

Università degli Studi di Salerno



Dipartimento di Informatica

Penetration Testing & Ethical Hacking

Enumerating Target e Port Scanning

Parte 3

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Outline

- Concetti Introduttivi
- Suite Protocollore TCP/IP
- Formato dei Messaggi TCP e UDP
- Active Enumeration
 - Network Scanner Nmap
 - Zenmap
 - Unicornscan
 - Masscan
- **Passive Enumeration**
 - Shodan
 - ZoomEye
 - FOFA
 - Censys

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Shodan

Caratteristiche

➤ <https://www.shodan.io/>

➤ **Motore di ricerca** che consente di trovare (tramite opportuni filtri) vari tipi di **dispositivi connessi ad Internet**

➤ **Noto per essere il più importante motore di ricerca per dispositivi connessi ad Internet**

➤ Computer

➤ Webcam

➤ Server

➤ Router

➤ Dispositivi IoT

➤ Etc

➤ Introdotto nel 2009 da John Matherly



Shodan

Caratteristiche

- <https://www.shodan.io/>
 - Anche definito come **motore di ricerca di *service banner***
 - Metadati che il Server invia al Client
 - I *service banner* potrebbero contenere informazioni interessanti sui software di rete in esecuzione sul Server
 - Messaggio di benvenuto
 - Eventuali opzioni supportate dal software
 - Qualsiasi altra cosa che potrebbe essere utile al Client prima di interagire con il Server

Shodan

Caratteristiche

- Shodan raccoglie dati su vari servizi di rete
 - *Web Server* (HTTP/HTTPS - porte 80, 8080, 443, 8443)
 - *FTP* (porta 21)
 - *SSH* (porta 22)
 - *Telnet* (porta 23)
 - *IMAP* (porte 143 o 993)
 - *SMTP* (porta 25)
 - *Real Time Streaming Protocol (RTSP)*, porta 554)
 - Utilizzato per accedere alle webcam ed ai loro flussi video
 - Etc

Shodan

Caratteristiche

- Shodan fornisce inoltre

- Una propria interfaccia a linea di comando (*Shodan Command-Line Interface*) utilizzabile online

- <https://cli.shodan.io/>

- API per scrivere programmi basati su Shodan

- Interessante strumento basato sulle API di Shodan

- *SearchDiggity*

- <https://www.bishopfox.com/resources/tools/google-hacking-diggity/attack-tools/>



Shodan

Caratteristiche

- Consente di scoprire
 - Quali sistemi sono presenti in un asset
 - I servizi attivi in tali sistemi (ed i relativi *banner*)
 - Le versioni di tali servizi
 - Etc
- **N.B.** Le **informazioni** fornite da tale strumento **potrebbero non essere allineate** con lo stato corrente dell'asset
 - Prima di proseguire con le fasi successive del processo di penetration testing sarebbe opportuno condurre anche una fase di *Active Target Enumeration*
 - Un host potrebbe non essere più attivo o lo stato/versione delle sue porte/servizi potrebbe essere cambiato

Shodan

Caratteristiche

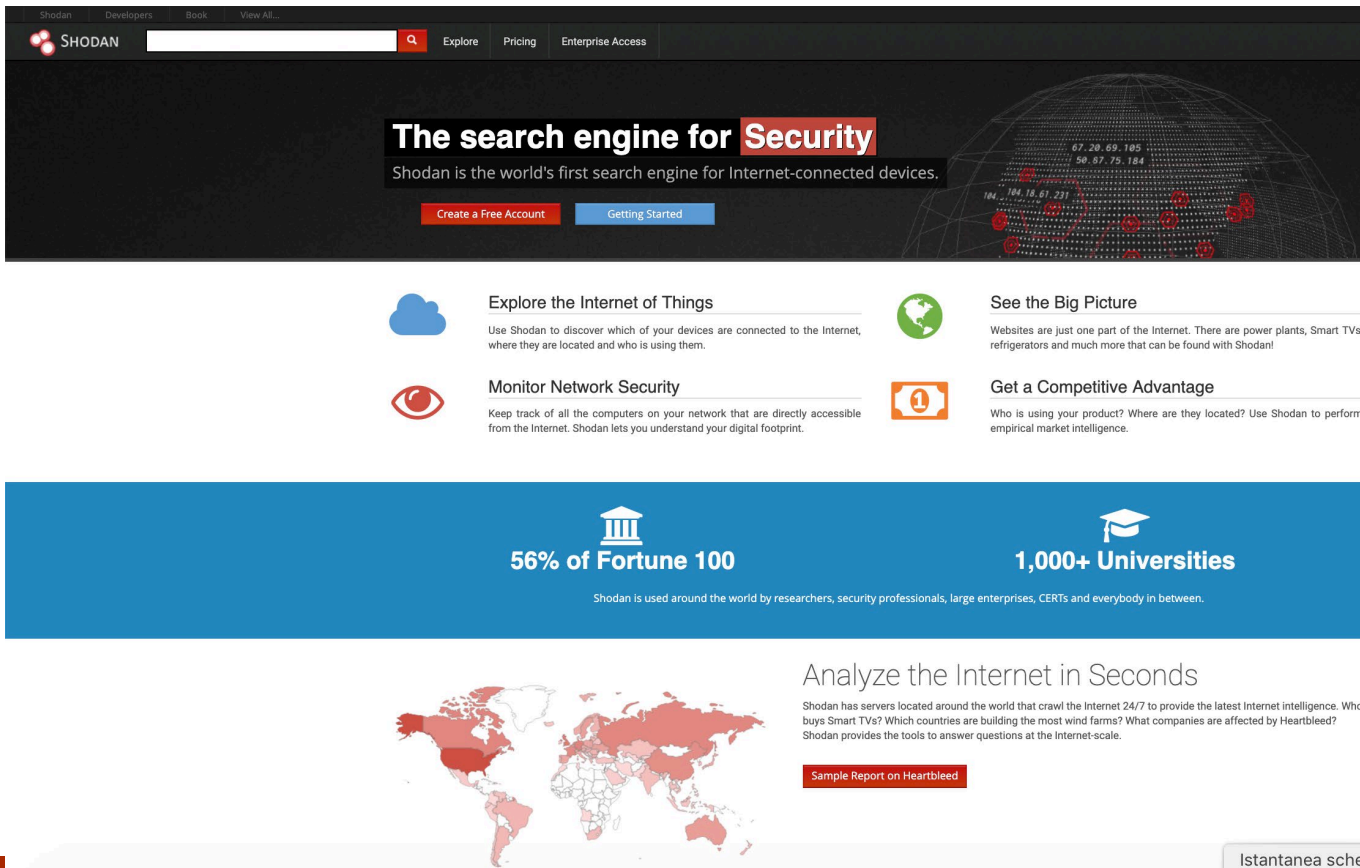
- Shodan è uno strumento utile per ottenere in maniera «discreta» ed affidabile numerose informazioni
- Il suo *timestamp* permette di avere un'idea su quanto siano recenti i risultati mostrati
- La versione «free» fornisce funzionalità molto limitate
 - Numero limitato di pagine come risultato per ciascuna query
 - Numero limitato di filtri da applicare
 - Etc



Shodan

Interfaccia

➤ <https://www.shodan.io/>



The screenshot displays the Shodan website's homepage. At the top, a navigation bar includes links for 'Shodan', 'Developers', 'Book', and 'View All...'. Below this is a search bar with the Shodan logo and a search icon, followed by links for 'Explore', 'Pricing', and 'Enterprise Access'. The main banner features the headline 'The search engine for Security' and the subtext 'Shodan is the world's first search engine for Internet-connected devices.' It includes two buttons: 'Create a Free Account' and 'Getting Started'. To the right of the banner is a graphic of a globe with IP addresses and red location markers. Below the banner, four feature sections are presented with icons and descriptions: 'Explore the Internet of Things' (cloud icon), 'Monitor Network Security' (eye icon), 'See the Big Picture' (globe icon), and 'Get a Competitive Advantage' (dollar sign icon). A blue section below these features highlights '56% of Fortune 100' and '1,000+ Universities' using Shodan, with a note that it is used by researchers, security professionals, large enterprises, CERTs, and everybody in between. The bottom section, titled 'Analyze the Internet in Seconds', describes Shodan's global server network and provides a 'Sample Report on Heartbleed' link. A world map with red location markers is shown on the left. A small 'Istantanea sche' label is visible in the bottom right corner of the screenshot.

Shodan

Developers

Book

View All...

SHODAN

Explore

Pricing

Enterprise Access

The search engine for Security

Shodan is the world's first search engine for Internet-connected devices.

Create a Free Account

Getting Started

Explore the Internet of Things

Use Shodan to discover which of your devices are connected to the Internet, where they are located and who is using them.

Monitor Network Security

Keep track of all the computers on your network that are directly accessible from the Internet. Shodan lets you understand your digital footprint.

See the Big Picture

Websites are just one part of the Internet. There are power plants, Smart TVs, refrigerators and much more that can be found with Shodan!

Get a Competitive Advantage

Who is using your product? Where are they located? Use Shodan to perform empirical market intelligence.

56% of Fortune 100

1,000+ Universities

Shodan is used around the world by researchers, security professionals, large enterprises, CERTs and everybody in between.

Analyze the Internet in Seconds

Shodan has servers located around the world that crawl the Internet 24/7 to provide the latest Internet intelligence. Who buys Smart TVs? Which countries are building the most wind farms? What companies are affected by Heartbleed? Shodan provides the tools to answer questions at the Internet-scale.

Sample Report on Heartbleed

Istantanea sche

Enumerating Target e Port Scanning

Shodan

Dorks – Query Shodan Predefinite

➤ <https://www.shodan.io/explore>

The screenshot displays the Shodan Explore interface. At the top, there's a grid of category tiles: Industrial Control Systems, Databases, Network Infrastructure, and Video Games. Below this is a 'RESEARCH' section with three featured items: 'Shodan 2000' (a retro-futuristic interface), 'Internet Observatory' (a high-level view of the Internet), and 'Favicon Map' (a map of common favicons). To the right of the research section is a 'BROWSE SEARCH DIRECTORY' section with a search bar, a list of popular tags (webcam, cam, camera, ip, router, scada, ftp, server, http, iot, test, password, cisco, web, default, login, ssh, it, nms, ipcam), and a 'What is the search directory?' section explaining the directory's purpose. On the far right, there's a vertical list of predefined search queries: 'Webcam' (best ip cam search), 'Cams' (admin admin), 'Netcam' (Netcam), 'default password' (Finds results with 'default password' in the ban.), and 'ufanet' ('80':8080). Each query tile shows the number of results and the query itself.

Explore

// CATEGORIES

Industrial Control Systems

Databases

Network Infrastructure

Video Games

// RESEARCH

Shodan 2000

Explore the Internet in style using an 80's retro-futuristic interface to synthwave music.

2000.SHODAN.IO

Internet Observatory

How exposed to the Internet is your country? What is the most common vulnerability? Get a high-level view of the Internet using our Observatory.

EXPOSURE.SHODAN.IO

Favicon Map

Favicons are the small icons you see in your browser tab at the top. See the most common favicons across the Internet.

FAVICONMAP.SHODAN.IO

// BROWSE SEARCH DIRECTORY

Search shared queries...

Popular Tags

webcam cam camera ip router scada ftp server http iot test password cisco web default login ssh it nms ipcam

What is the search directory?

Shodan lets users share their search queries with the community by saving them to the search directory. Shodan doesn't otherwise store or share your search queries. The queries in the search directory were explicitly shared by our users for the benefit of the community.

Note: The current Shodan website doesn't yet let you submit search queries to the directory.

Webcam

best ip cam search I have found yet.

12,519 webcam surveillance cams

Cams

admin admin

5,290 cam webcam

Netcam

Netcam

2,697 netcam

default password

Finds results with "default password" in the ban.

2,111 router default password

ufanet

'80':8080

1,413 ufanet

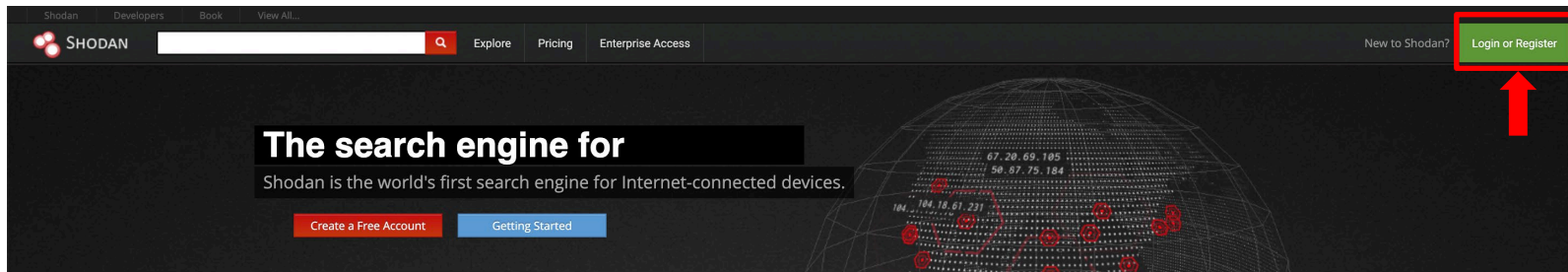
MORE

Enumerating Target e Port Scanning

Shodan

Registrazione

- **N.B.** Prima di utilizzare Shodan è **fortemente consigliata** la **registrazione** e l'**accesso** ad esso
 - **Altrimenti non possono essere usati i suoi filtri di ricerca**



Explore the Internet of Things

Use Shodan to discover which of your devices are connected to the Internet, where they are located and who is using them.



See the Big Picture

Websites are just one part of the Internet. There are power plants, Smart TVs, refrigerators and much more that can be found with Shodan!



Monitor Network Security

Keep track of all the computers on your network that are directly accessible from the Internet. Shodan lets you understand your digital footprint.



Get a Competitive Advantage

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1,000+ Universities

Enumerating Target e Port Scanning

Shodan

Registrazione

➤ <https://account.shodan.io/register>

Create Account

Username

Password

Confirm Password

Email

☐ Subscribe to the newsletter

By creating an account you are agreeing to our [Privacy Policy](#) and [Terms of Use](#)

CREATE



Shodan

Billing

➤ <https://account.shodan.io/billing>

Choose Your Plan

No contracts. No setup fees. Cancel anytime.

Freelancer

\$69/month

CHOOSE THIS PLAN

- ✓ Up to 1 million results per month *
- ✓ Scan up to 5,120 IPs per month
- ✓ Network Monitoring for 5,120 IPs

- ✓ Access to most filters
- ✓ Allows paging through search results
- ✓ Basic access to the Streaming API
- ✓ Commercial Use

- ✓ E-Mail support

Small Business

\$359/month

CHOOSE THIS PLAN

- ✓ Up to 20 million results per month *
- ✓ Scan up to 65,536 IPs per month
- ✓ Network Monitoring for 65,536 IPs

- ✓ Access to most filters
- ✓ Allows paging through search results
- ✓ Basic access to the Streaming API
- ✓ Commercial Use

- ✓ E-Mail support
- ✓ Vulnerability search filter

Corporate

\$1099/month

CHOOSE THIS PLAN

- ✓ **Unlimited** results per month *
- ✓ Scan up to 327,680 IPs per month
- ✓ Network Monitoring for 327,680 IPs

- ✓ Access to all filters
- ✓ Allows paging through search results
- ✓ Basic access to the Streaming API
- ✓ Commercial Use

- ✓ Premium Support
- ✓ Vulnerability search filter
- ✓ Batch IP Lookups
- ✓ Tag Search Filter
- ✓ Complementary Membership Upgrades

The logo for Shodan, featuring three red circles of varying sizes arranged in a triangular pattern to the left of the word "SHODAN" in a bold, black, sans-serif font. Below "SHODAN" is the text "Computer Search Engine" in a smaller, lighter font.

Enumerating Target e Port Scanning

Shodan

Filtri di Ricerca

- Shodan fornisce numerosi filtri di ricerca, raggruppati nelle seguenti categorie
 - **General**
 - **HTTP**
 - **SSL**
 - **Bitcoin**
 - **Restricted**
 - **NTP**
 - **Screenshots**
 - **SNMP**
 - **Telnet**
 - **Cloud**
 - **SSH**

Shodan

Filtri di Ricerca

- Alcuni tra i filtri di Shodan più utilizzati sono
 - **city**: cerca i dispositivi in una determinata città
 - **country**: cerca i dispositivi in un determinato paese
 - **geo**: cerca i dispositivi in base alle coordinate geografiche
 - **hostname**: cerca i dispositivi che corrispondono al nome di host
 - **net**: ricerca basata su un IP o CIDR
 - **os**: ricerca basata sul Sistema Operativo
 - **port**: cerca dispositivi che hanno determinate porte aperte
 - **before/after**: cerca risultati appartenenti ad un determinato intervallo temporale

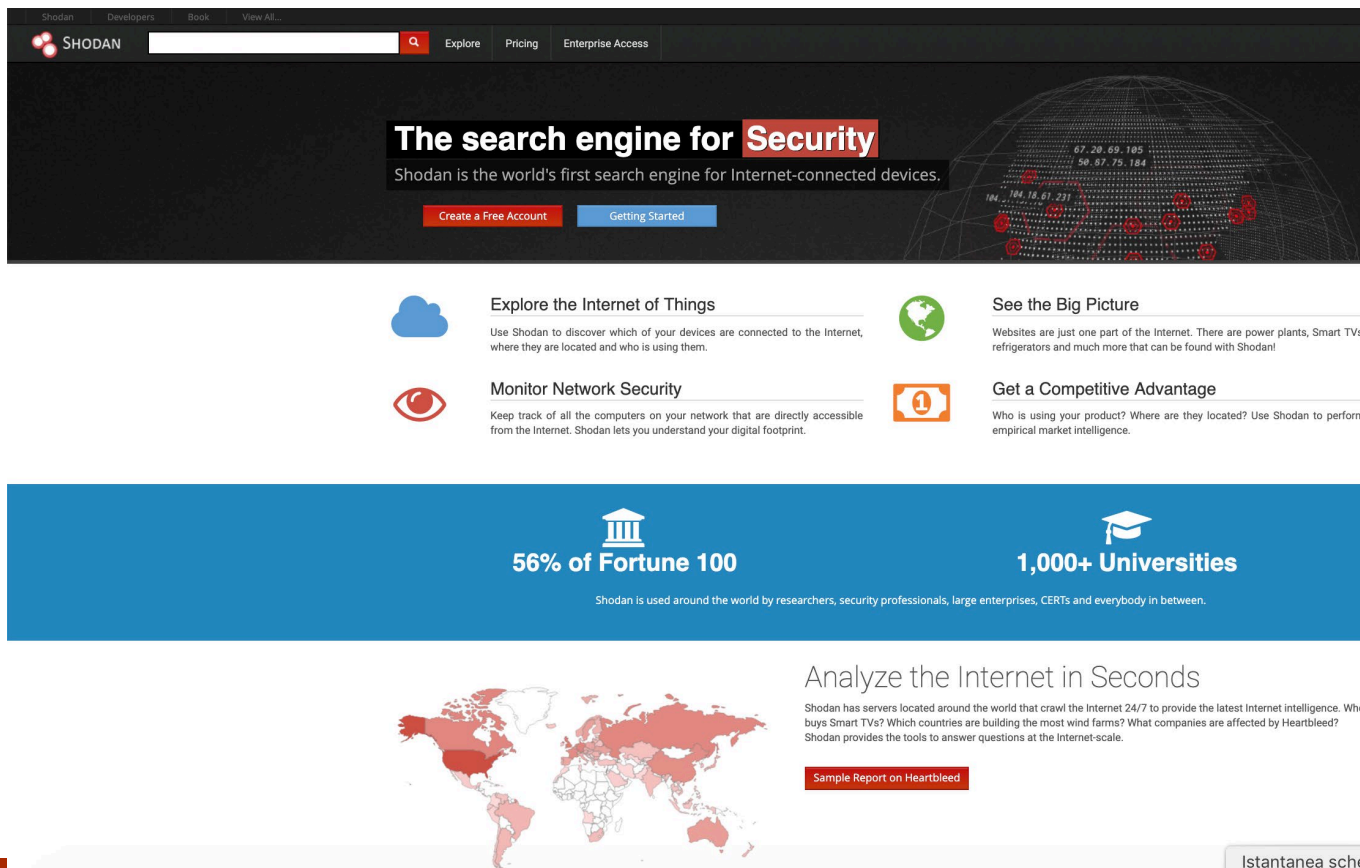
<https://www.shodan.io/search/filters>



Shodan

Esempio 1

➤ <https://www.shodan.io/>



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Shodan Developers Book View All...

SHODAN

Explore Pricing Enterprise Access

The search engine for Security

Shodan is the world's first search engine for Internet-connected devices.

Create a Free Account Getting Started

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Sample Report on Heartbleed

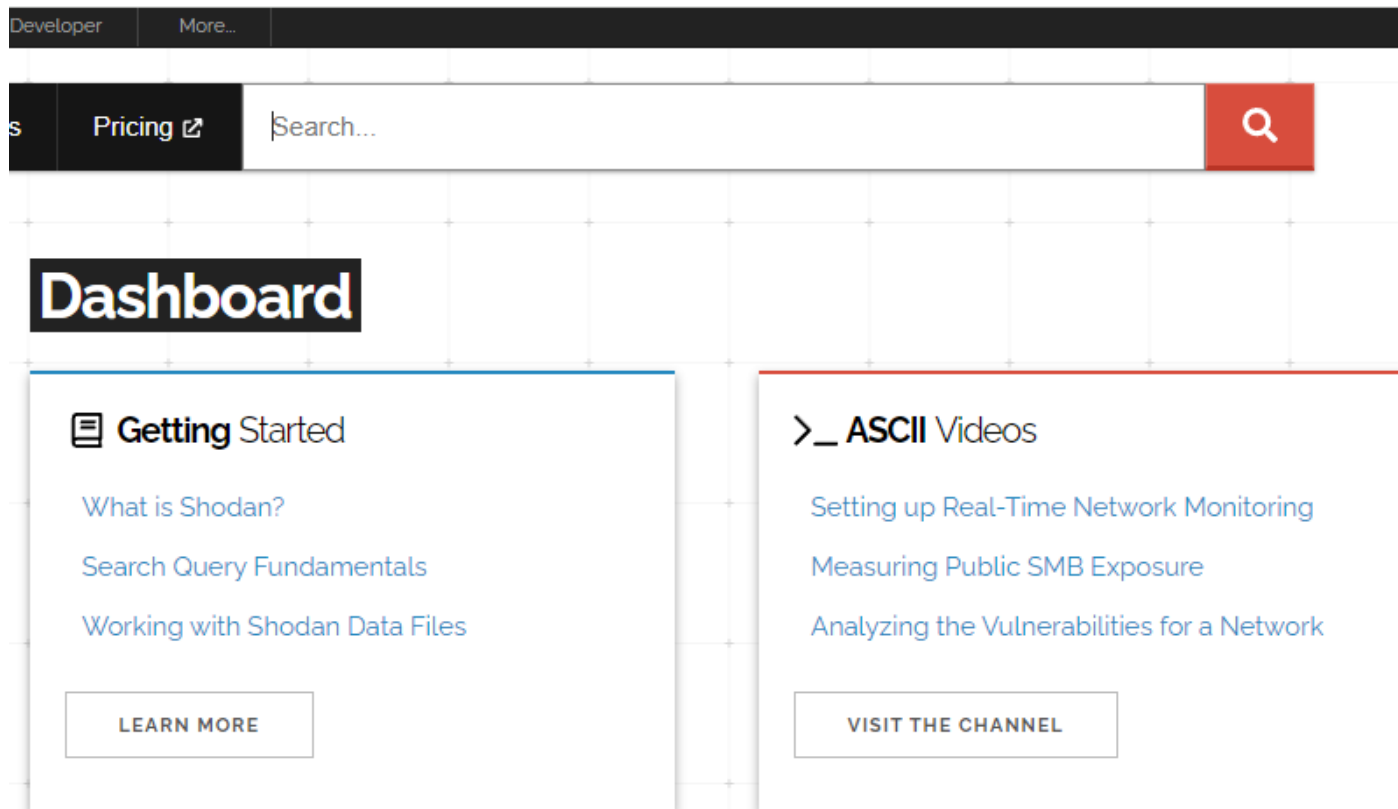
Enumerating Target e Port Scanning

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Shodan

Esempio 1

- Effettuiamo una ricerca per Autonomous System Network (**asn:as137**)



Shodan

Esempio 1

- Effettuiamo una ricerca per Autonomous System Network (**asn:as137**)

The screenshot shows the Shodan search results for the query 'asn:as137'. The interface includes a navigation bar with 'SHODAN', 'Explore', 'Downloads', and 'Pricing'. The search bar contains 'asn:as137'. The results section shows 'TOTAL RESULTS: 119,275'. Below this, 'TOP COUNTRIES' are listed with a world map highlighting Italy, Brazil, France, and Belgium. The top results are for IP addresses 157.138.174.18 and 193.205.148.175, both associated with Italy. The first result is for UNI-Venezia in Venice, and the second is for CNR - INO Firenze in Florence. The second result also shows an SSL Certificate issued by Let's Encrypt.

SHODAN Explore Downloads Pricing [asn:as137](#) 🔍

TOTAL RESULTS
119,275

TOP COUNTRIES

Italy 119,241
Brazil 13
France 9
Belgium 5

157.138.174.18
UNI-Venezia
Italy, Venice
No data returned

193.205.148.175
leonard.ino.it
CNR - INO Firenze
Italy, Florence

SSL Certificate
Issued By:
Common Name: R3
Organization: Let's Encrypt
Issued To:
Common Name:

SSL Error: TLSv1

Shodan

Esempio 1

- Effettuiamo una ricerca per Autonomous System Network (**asn:as137**)

Porte più utilizzate		TOP PORTS

		2000 18,456
		8008 17,868
		80 15,891
		443 14,235
		8010 10,134
		More...
Organizzazioni con il maggior numero di dispositivi censiti		TOP ORGANIZATIONS

		Universita' degli Studi di Udine 18,892
		UNI-Venezia 10,052
		INFN - LNF - Frascati 5,913
		UNI-Padova 4,385
		POLI-Torino 4,207
		More...

Shodan

Esempio 1

- Effettuiamo una ricerca per Autonomous System Network (**asn: as137**)

Software più utilizzati

TOP PRODUCTS

Apache httpd	12,474
OpenSSH	7,187
nginx	4,361
Fortinet FortiGate-600E	1,851
ciscoSystems	1,321
More...	

Sistemi Operativi
più utilizzati

TOP OPERATING SYSTEMS

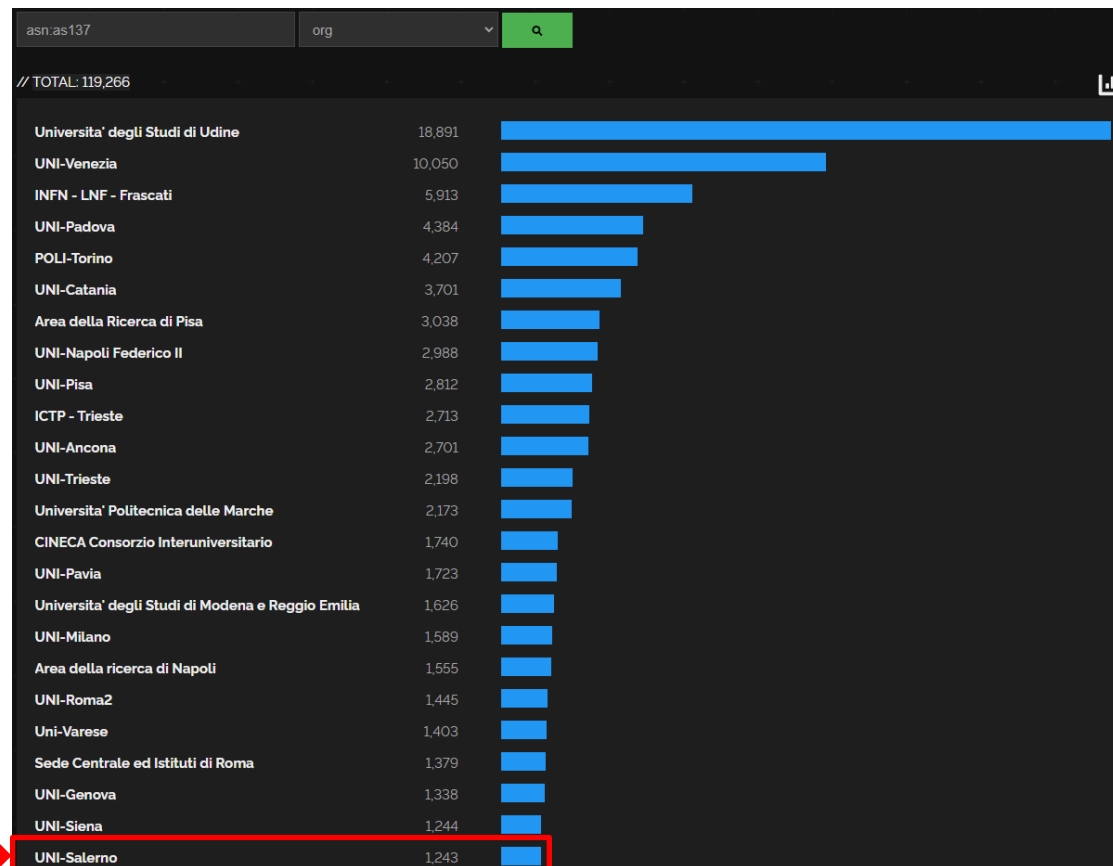
Windows	2,114
Ubuntu	1,895
Linux	940
Debian	474
Windows (build 10.0.19041)	201
More...	

Shodan

Esempio 1

➤ Selezioniamo l'organizzazione **UNI-Salerno**

TOP ORGANIZATIONS



Enumerating Target e Port Scanning

Shodan

Esempio 1

➤ Information freshness

The screenshot displays the Shodan search interface with three results. A red callout box labeled "Information freshness" points to the timestamp "2024-04-22T08:51:52.196107" associated with the first IP address.

IP Address	Organization	Location	Information Freshness
193.205.186.112	UNI-Salerno	Italy, Rome	2024-04-22T08:51:52.196107
193.205.162.176	UNI-Salerno	Italy, Lancusi-Penta-Bolano	2024-04-22T08:45:29.839106
193.205.184.221	dev2.unisa.it	Italy, Rome	2024-04-22T08:18:16.481011

193.205.186.112

UNI-Salerno
Italy, Rome

HTTP/1.1 400 Bad Request
Date: Mon, 22 Apr 2024 08:50:53 GMT
Server: Apache/2.4.46 (Unix) OpenSSL/1.1.1i PHP/8.0.2 mod_perl/2.0.11 Perl/v5.32.1
Vary: accept-language, accept-charset
Accept-Ranges: bytes
Connection: close
Content-Type: text/html; charset=utf-8
Content-Language: en
Expires: ...

193.205.162.176

UNI-Salerno
Italy, Lancusi-Penta-Bolano

No data returned

193.205.184.221

dev2.unisa.it
UNI-Salerno
Italy, Rome

No data returned

Shodan

Esempio 1

➤ Selezioniamo l'indirizzo IP 193.205.186.112

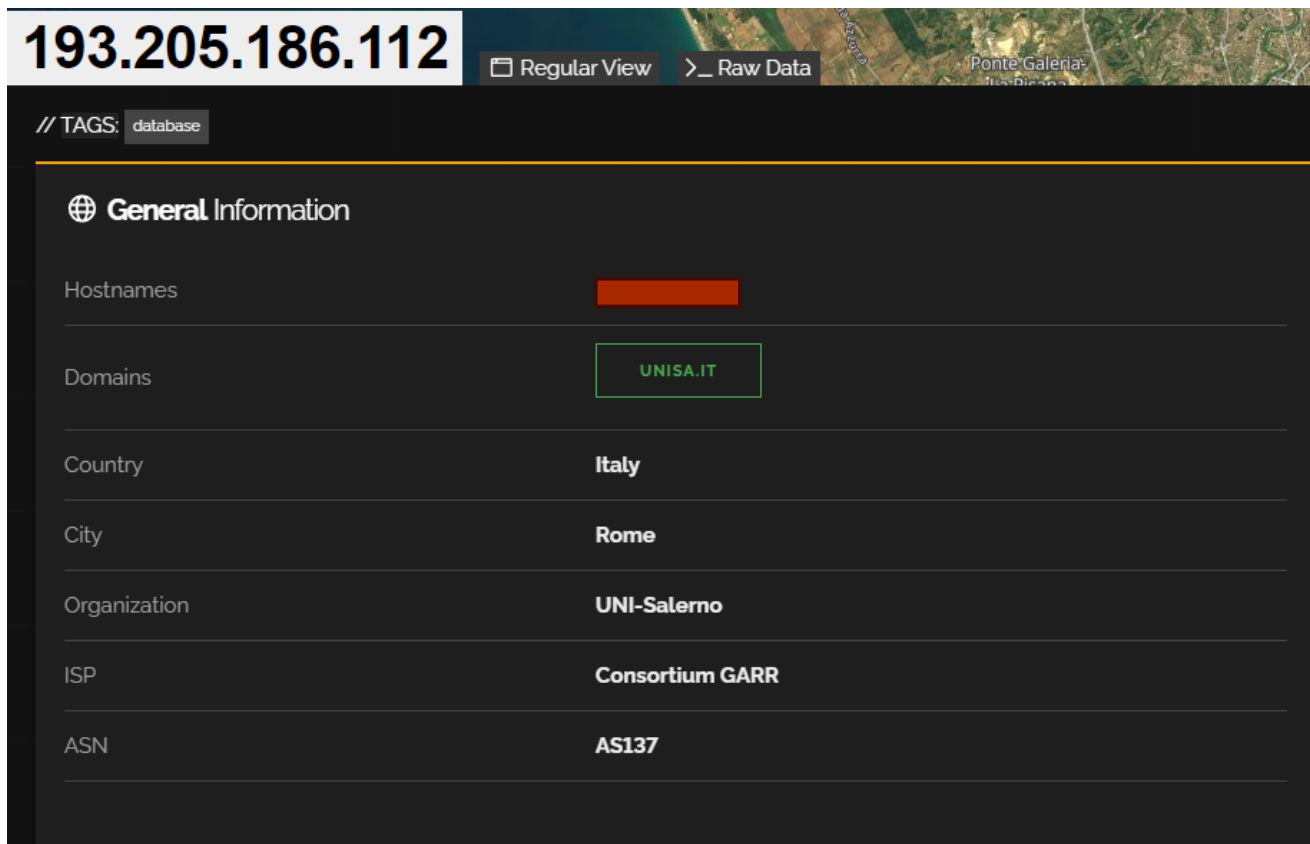
The screenshot shows the Shodan search results page. At the top, there are navigation links: "View Report", "Browse Images", and "View on Map". Below these is a "Partner Spotlight" banner for Gravwell. The main results list shows three entries. The first entry, for IP 193.205.186.112, is highlighted with a red box. It shows a "400 Bad Request" status and various HTTP headers. The second and third entries, for IP 193.205.162.176 and 193.205.184.221, both show "No data returned".

IP Address	Details	Timestamp
193.205.186.112	UNI-Salerno Italy, Rome HTTP/1.1 400 Bad Request Date: Mon, 22 Apr 2024 08:50:53 GMT Server: Apache/2.4.46 (Unix) OpenSSL/1.1.1i PHP/8.0.2 mod_perl/2.0.11 Perl/v5.32.1 Vary: accept-language,accept-charset Accept-Ranges: bytes Connection: close Content-Type: text/html; charset=utf-8 Content-Language: en Expires: ...	2024-04-22T08:51:52.196107
193.205.162.176	UNI-Salerno Italy, Lancusi-Penta-Bolano No data returned	2024-04-22T08:45:29.839106
193.205.184.221	dev2.unisa.it UNI-Salerno Italy, Rome No data returned	2024-04-22T08:18:16.481011

Shodan

Esempio 1

➤ Informazioni relative all'indirizzo IP 193.205.186.112 (Informazioni Generali)



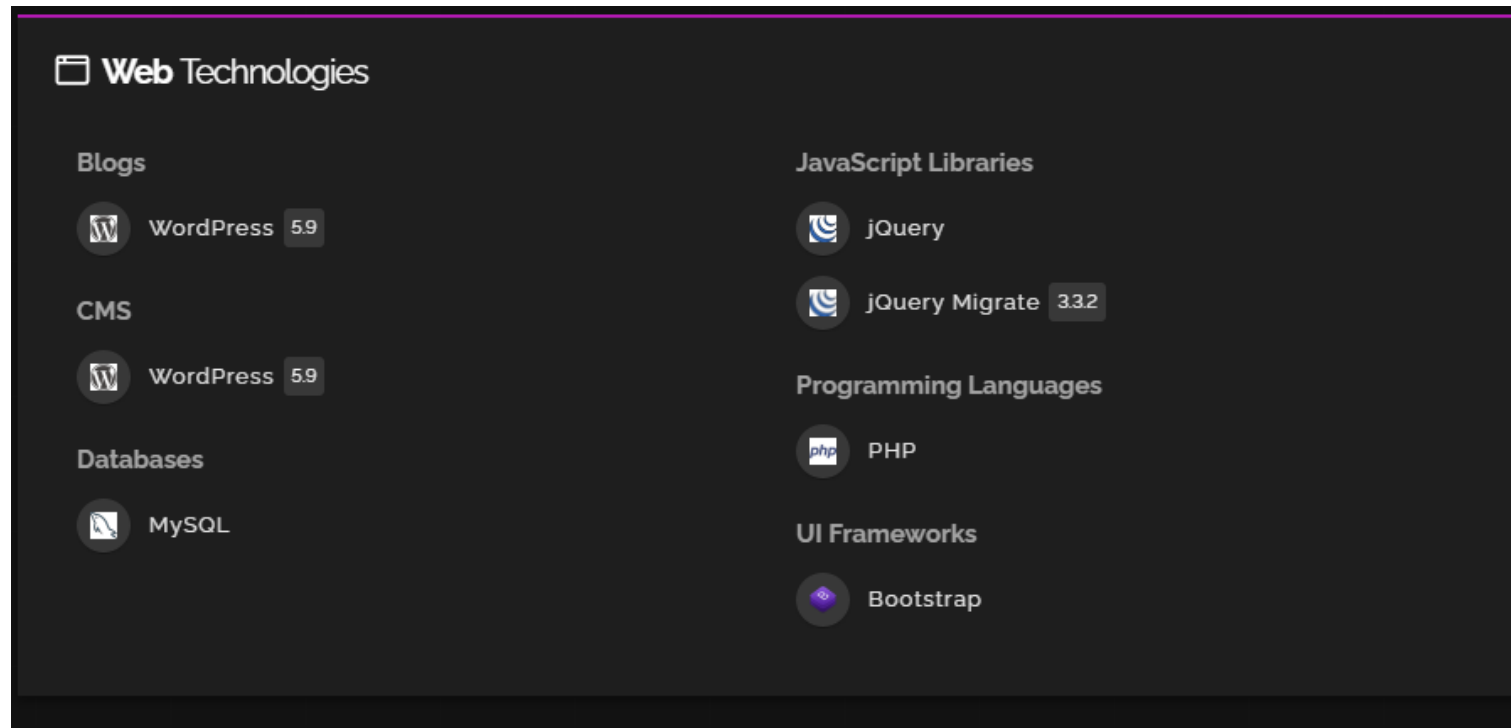
The screenshot shows the Shodan search results for the IP address 193.205.186.112. The interface is dark-themed. At the top, the IP address is displayed in large white text. Below it, there are two buttons: 'Regular View' and 'Raw Data'. To the right of these buttons is a small satellite map showing the location of the IP. Below the map, there is a section titled 'General Information' with a globe icon. This section contains a table of information about the IP address.

Field	Value
Hostnames	[REDACTED]
Domains	UNISA.IT
Country	Italy
City	Rome
Organization	UNI-Salerno
ISP	Consortium GARR
ASN	AS137

Shodan

Esempio 1

➤ Informazioni relative all'indirizzo IP 193.205.186.112 (Tecnologie Web)



Shodan

Esempio 1

➤ Informazioni relative all'indirizzo IP 193.205.186.112 (Porte Aperte)

The screenshot displays the Shodan interface for the IP address 193.205.186.112. At the top, under the 'Open Ports' section, a row of blue buttons lists the open ports: 21, 25, 80, 110, 137, 143, 443, 445, 2000, 3306, and 8010. Below this, the selected port 80 is shown with the protocol TCP. The main content area displays the banner for the Apache httpd 2.4.46 service. The banner text includes the HTTP status '200 OK', the date 'Wed, 03 Apr 2024 08:28:36 GMT', the server version 'Apache/2.4.46 (Unix) OpenSSL/1.1.1i PHP/8.0.2 mod_perl/2.0.11 Perl/v5.32.1', and various links and encoding information. A red rectangular box highlights the banner text. To the right of the banner, a red text overlay reads: 'Eventuali banner relativi alle porte aperte'.

Open Ports

21 25 80 110 137 143 443 445 2000 3306 8010

// 80 / TCP

1534238299 | 2024-04-03T08:29:28.496073

Apache httpd 2.4.46

HTTP/1.1 200 OK
Date: Wed, 03 Apr 2024 08:28:36 GMT
Server: Apache/2.4.46 (Unix) OpenSSL/1.1.1i PHP/8.0.2 mod_perl/2.0.11 Perl/v5.32.1
X-Powered-By: PHP/8.0.2
X-Pingback: http://www.biplab.unisa.it/home/xmlrpc.php
Link: <http://www.biplab.unisa.it/home/wp-json/>; rel="https://api.w.org/"
Link: <http://www.biplab.unisa.it/home/wp-json/wp/v2/pages/2>; rel="alternate"; type="application/json"
Link: <http://www.biplab.unisa.it/home/>; rel=shortlink
Transfer-Encoding: chunked
Content-Type: text/html; charset=UTF-8

Eventuali banner relativi alle porte aperte

Shodan

Esempio 1

➤ Informazioni relative all'indirizzo IP 193.205.186.112 (Vulnerabilità)

Vulnerabilities

Note: the device may not be impacted by all of these issues. The vulnerabilities are implied based on the software and version.

CVE-2024-0727

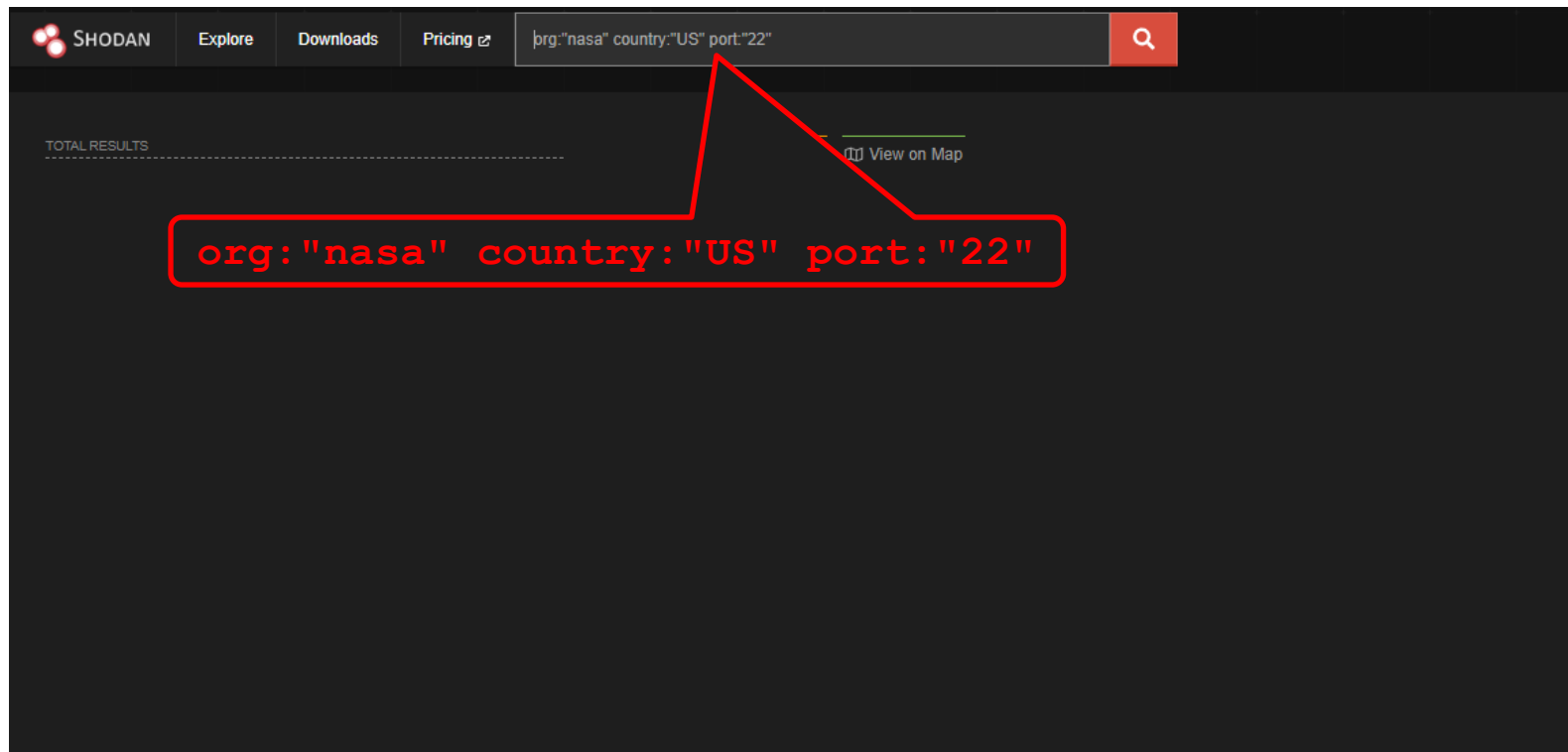
Issue summary: Processing a maliciously formatted PKCS12 file may lead OpenSSL to crash leading to a potential Denial of Service attack Impact summary: Applications loading files in the PKCS12 format from untrusted sources might terminate abruptly. A file in PKCS12 format can contain certificates and keys and may come from an untrusted source. The PKCS12 specification allows certain fields to be NULL, but OpenSSL does not correctly check for this case. This can lead to a NULL pointer dereference that results in OpenSSL crashing. If an application processes PKCS12 files from an untrusted source using the OpenSSL APIs then that application will be vulnerable to this issue. OpenSSL APIs that are vulnerable to this are: PKCS12_parse(), PKCS12_unpack_p7data(), PKCS12_unpack_p7encdata(), PKCS12_unpack_authsafes() and PKCS12_newpass(). We have also fixed a similar issue in SMIME_write_PKCS7(). However since this function is related to writing data we do not consider it security significant. The FIPS modules in 3.2, 3.1 and 3.0 are not affected by this issue.

CVE-2023-5678

Issue summary: Generating excessively long X9.42 DH keys or checking excessively long X9.42 DH keys or parameters may be very slow. Impact summary: Applications that use the functions

Shodan

Esempio 2



Shodan

Esempio 2

SHODAN Explore Downloads Pricing `prg:"nasa" country:"US" port:"22"` 🔍

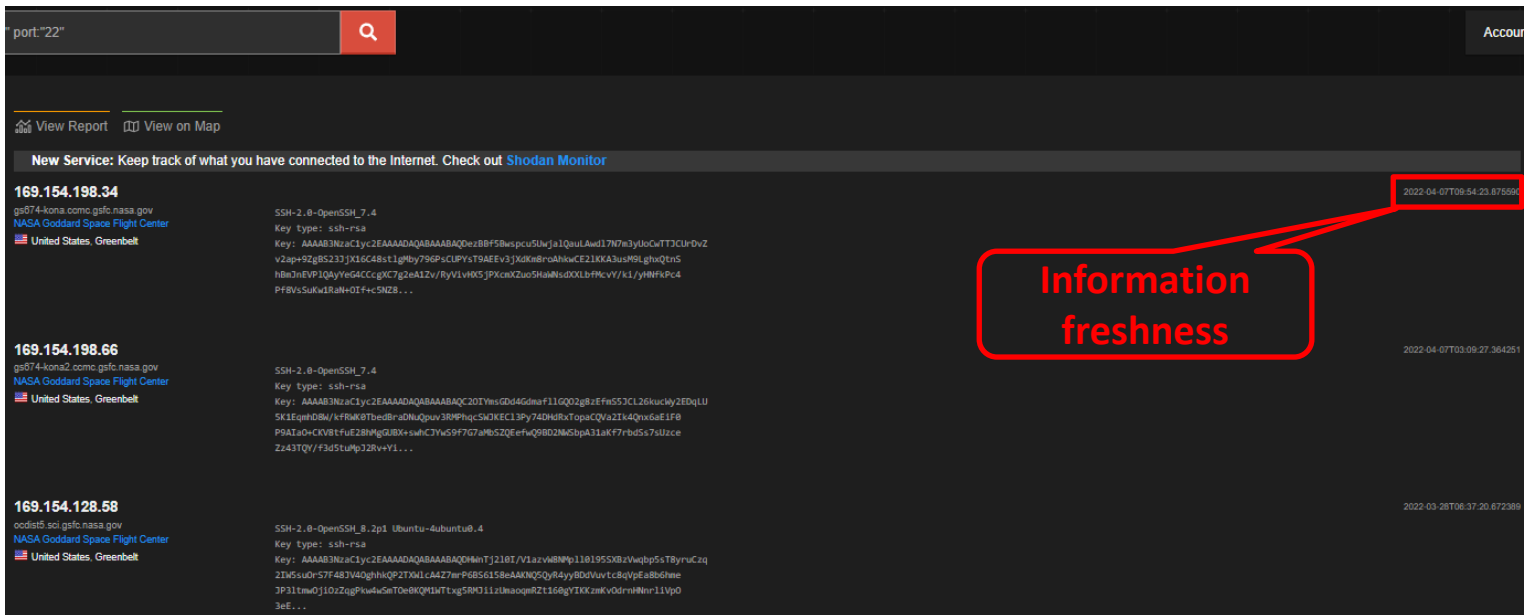
TOTAL RESULTS
3

[View Report](#) [View on Map](#)

New Service: Keep track of what you have connected to the Internet. Check out [Shodan Monitor](#)

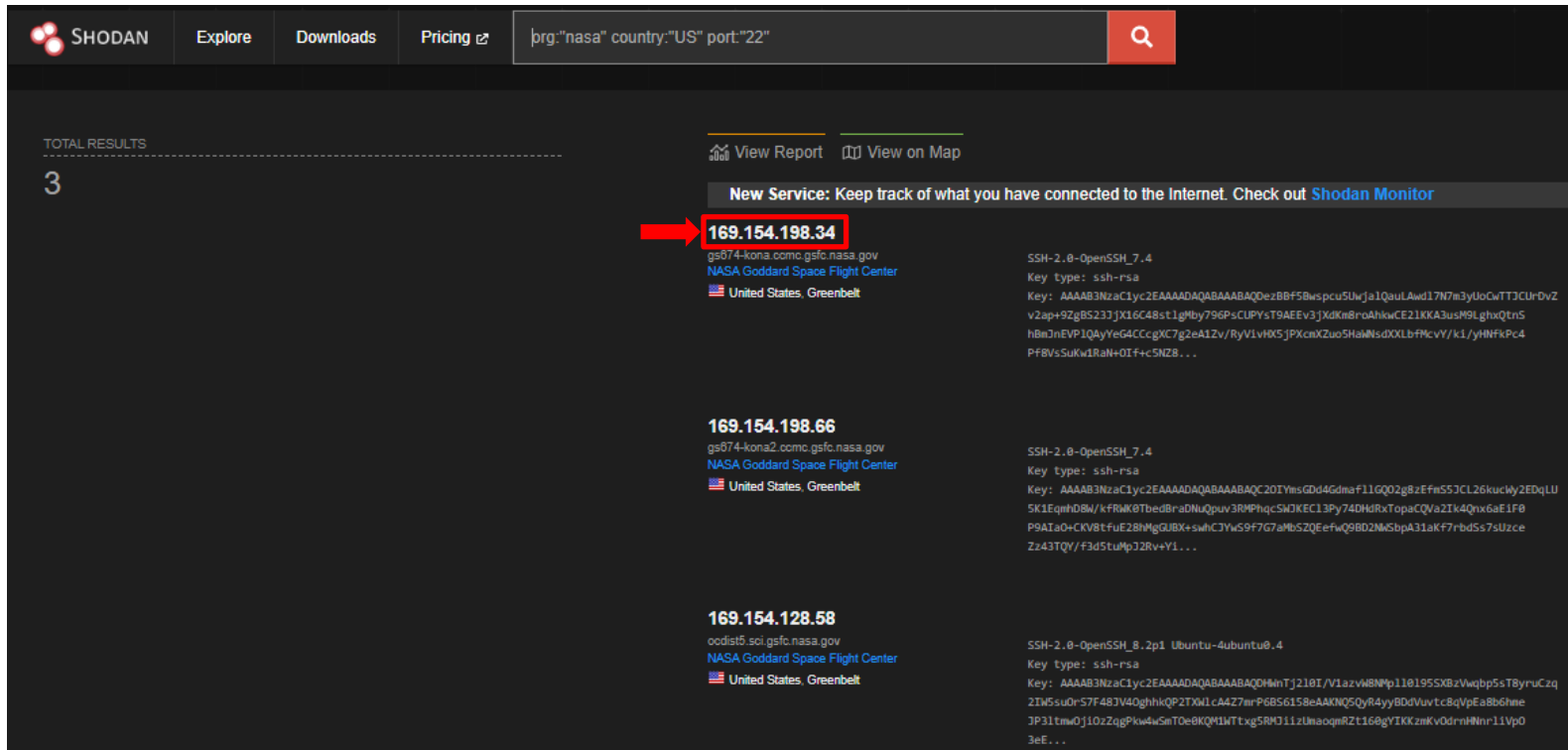
169.154.198.34 gs074-kona.cmc.gsfc.nasa.gov NASA Goddard Space Flight Center 🇺🇸 United States, Greenbelt	SSH-2.0-OpenSSH_7.4 Key type: ssh-rsa Key: AAAAB3NzaC1yc2EAAAADAQABAAQDezBBF5Bwspcu5Uwja1QauLawd17N7m3yUoCwTT3CLUrDvZv2ap+9ZgB5231jX16c48st1gMby796PsCUPYsT9AEV3jXdkmBroAhlwCE21KKA3usM9LghvQtnShBm3nEVP1QAYeG4CCcgXC7g2eA12v/RyV1vHKSjPXcmX2uoSHamNsDOLbFMcVY/k1/yHNFkPc4PF8VsSukw1RaH+OI+cSNZ8...
169.154.198.66 gs074-kona2.cmc.gsfc.nasa.gov NASA Goddard Space Flight Center 🇺🇸 United States, Greenbelt	SSH-2.0-OpenSSH_7.4 Key type: ssh-rsa Key: AAAAB3NzaC1yc2EAAAADAQABAAQDezBBF5Bwspcu5Uwja1QauLawd17N7m3yUoCwTT3CLUrDvZv2ap+9ZgB5231jX16c48st1gMby796PsCUPYsT9AEV3jXdkmBroAhlwCE21KKA3usM9LghvQtnShBm3nEVP1QAYeG4CCcgXC7g2eA12v/RyV1vHKSjPXcmX2uoSHamNsDOLbFMcVY/k1/yHNFkPc4PF8VsSukw1RaH+OI+cSNZ8...
169.154.128.58 oodist5.sci.gsfc.nasa.gov NASA Goddard Space Flight Center 🇺🇸 United States, Greenbelt	SSH-2.0-OpenSSH_8.2p1 Ubuntu-4ubuntu0.4 Key type: ssh-rsa Key: AAAAB3NzaC1yc2EAAAADAQABAAQDezBBF5Bwspcu5Uwja1QauLawd17N7m3yUoCwTT3CLUrDvZv2ap+9ZgB5231jX16c48st1gMby796PsCUPYsT9AEV3jXdkmBroAhlwCE21KKA3usM9LghvQtnShBm3nEVP1QAYeG4CCcgXC7g2eA12v/RyV1vHKSjPXcmX2uoSHamNsDOLbFMcVY/k1/yHNFkPc4PF8VsSukw1RaH+OI+cSNZ8...

Esempio 2



Shodan

Esempio 2



SHODAN Explore Downloads Pricing `prg:"nasa" country:"US" port:"22"` 🔍

TOTAL RESULTS
3

View Report View on Map

New Service: Keep track of what you have connected to the Internet. Check out [Shodan Monitor](#)

169.154.198.34
gs074-kona.ccmc.gsfc.nasa.gov
NASA Goddard Space Flight Center
🇺🇸 United States, Greenbelt

SSH-2.0-OpenSSH_7.4
Key type: ssh-rsa
Key: AAAAB3NzaC1yc2EAAAADAQABAAQDezBBF5Bwspcu5Uwja1QauLawd17N7m3yUoCwTT3CLUrDvZv2ap+9ZgB5231jX16c48st1gMby796PsCUPYsT9AEV3jXdkmBroAhlwCE21KKA3usM9LghvQtnShBm3nEVP1QAYeG4CCcgXC7g2eA12v/RyV1vHXSjPXcmX2uoSHamNsdxLbFMcVY/k1/yHNFkPc4PF8VsSukw1RaH+OI+cSNZB...

169.154.198.66
gs074-kona2.ccmc.gsfc.nasa.gov
NASA Goddard Space Flight Center
🇺🇸 United States, Greenbelt

SSH-2.0-OpenSSH_7.4
Key type: ssh-rsa
Key: AAAAB3NzaC1yc2EAAAADAQABAAQDezBBF5Bwspcu5Uwja1QauLawd17N7m3yUoCwTT3CLUrDvZv2ap+9ZgB5231jX16c48st1gMby796PsCUPYsT9AEV3jXdkmBroAhlwCE21KKA3usM9LghvQtnShBm3nEVP1QAYeG4CCcgXC7g2eA12v/RyV1vHXSjPXcmX2uoSHamNsdxLbFMcVY/k1/yHNFkPc4PF8VsSukw1RaH+OI+cSNZB...

169.154.128.58
oodist5.sci.gsfc.nasa.gov
NASA Goddard Space Flight Center
🇺🇸 United States, Greenbelt

SSH-2.0-OpenSSH_8.2p1 Ubuntu-4ubuntu0.4
Key type: ssh-rsa
Key: AAAAB3NzaC1yc2EAAAADAQABAAQDezBBF5Bwspcu5Uwja1QauLawd17N7m3yUoCwTT3CLUrDvZv2ap+9ZgB5231jX16c48st1gMby796PsCUPYsT9AEV3jXdkmBroAhlwCE21KKA3usM9LghvQtnShBm3nEVP1QAYeG4CCcgXC7g2eA12v/RyV1vHXSjPXcmX2uoSHamNsdxLbFMcVY/k1/yHNFkPc4PF8VsSukw1RaH+OI+cSNZB...

Shodan

Esempio 2

The screenshot displays the Shodan search engine interface. At the top, there's a navigation bar with 'SHODAN', 'Explore', 'Downloads', and 'Pricing'. A search bar is on the right. Below the navigation bar, a map shows the location of the selected host, 169.154.198.34, in Greenbelt, Maryland. The map includes labels for nearby areas like Silver Spring, Takoma Park, and College Park. Below the map, the host's IP address is displayed, along with options for 'Regular View', 'Raw Data', and 'History'. The main content area is divided into two sections: 'General Information' and 'Vulnerabilities'. The 'General Information' section lists details about the host, including its hostname, domains, country, city, organization, ISP, and ASN. The 'Vulnerabilities' section lists known vulnerabilities, including CVE-2018-15919 and CVE-2017-15906.

Geolocalizzazione dell'host selezionato

General Information	
Hostnames	gs674-kona.ccmc.gsfc.nasa.gov
Domains	NASA.GOV
Country	United States
City	Greenbelt
Organization	NASA Goddard Space Flight Center
ISP	NASA Goddard Space Flight Center
ASN	AS7847

Vulnerabilities	
CVE-2018-15919	Remotely observable behaviour in auth-gss2.c in OpenSSH through 7.8 could be used by remote attackers to detect existence of users on a target system when GSS2 is in use. NOTE: the discoverer states "We understand that the OpenSSH developers do not want to treat such a username enumeration (or "oracle") as a vulnerability".
CVE-2017-15906	The process_open function in sftp-server.c in OpenSSH before 7.6 does not properly prevent write operations in readonly mode, which allows attackers to create zero-length files.

Shodan

Esempio 2

The screenshot displays the Shodan search engine interface. At the top, there is a navigation bar with links for SHODAN, Explore, Downloads, and Pricing, along with a search bar. Below the navigation bar, a map shows the location of the selected IP address, 169.154.198.34, in Silver Spring, Maryland. The main content area is divided into two sections: General Information and Vulnerabilities.

General Information

Hostnames	gs674-kona.ccmc.gsfc.nasa.gov
Domains	NASA.GOV
Country	United States
City	Greenbelt
Organization	NASA Goddard Space Flight Center
ISP	NASA Goddard Space Flight Center
ASN	AS7847

Vulnerabilities

Note: the device may not be impacted by all of these issues. The vulnerabilities are implied based on the software and version.

- CVE-2018-15919** Remotely observable behaviour in auth-gss2.c in OpenSSH through 7.8 could be used by remote attackers to detect existence of users on a target system when GSS2 is in use. NOTE: the discoverer states "We understand that the OpenSSH developers do not want to treat such a username enumeration (or "oracle") as a vulnerability".
- CVE-2017-15906** The process_open function in sftp-server.c in OpenSSH before 7.6 does not properly prevent write operations in readonly mode, which allows attackers to create zero-length files.

Shodan

Esempio 2

The screenshot shows the Shodan search engine interface. At the top, there is a navigation bar with links for SHODAN, Explore, Downloads, and Pricing. A search bar is located on the right. Below the navigation bar, a map displays the location of the selected IP address, 169.154.198.34, which is highlighted in a white box. The map shows the area around Silver Spring, Maryland, with labels for various locations like Sterling, Great Falls, Potomac, and Greenbelt. Below the map, the IP address 169.154.198.34 is displayed in a large white font. To the right of the IP address, there are links for Regular View, Raw Data, and History. The main content area is divided into two sections: General Information and Vulnerabilities. The General Information section lists details about the host, including Hostnames, Domains, Country, City, Organization, ISP, and ASN. The Vulnerabilities section lists known vulnerabilities associated with the host, including CVE-2018-15919 and CVE-2017-15906. A red speech bubble points to the Vulnerabilities section with the text "Eventuali vulnerabilità dell'host selezionato".

SHODAN Explore Downloads Pricing Search...

169.154.198.34 Regular View Raw Data History

General Information

Hostnames	gs674-kona.ccmc.gsfc.nasa.gov
Domains	NASA.GOV
Country	United States
City	Greenbelt
Organization	NASA Goddard Space Flight Center
ISP	NASA Goddard Space Flight Center
ASN	AS7847

Vulnerabilities

Note: the device may not be impacted by all of these issues. The vulnerabilities are implied based on the software and version.

CVE-2018-15919 Remotely observable behaviour in auth-gss2.c in OpenSSH through 7.8 could be used by remote attackers to detect existence of users on a target system when GSS2 is in use. NOTE: the discoverer states "We understand that the OpenSSH developers do not want to treat such a username enumeration (or "oracle") as a vulnerability".

CVE-2017-15906 The process_open function in sftp-server.c in OpenSSH before 7.6 does not properly prevent write operations in readonly mode, which allows attackers to create zero-length files.

Eventuali vulnerabilità dell'host selezionato

Shodan

Esempio 2

```
Open Ports
22

// 22 / TCP
2004830606 | 2022-04-08T22:10:59.661749

OpenSSH 7.4

SSH-2.0-OpenSSH 7.4
Key type: ssh-rsa
Key: AAAAB3NzaC1yc2EAAAADAQABAAQDQzBBfS9wpcu5Uwja1QoutAed17N7e3yUoCwIT3CjUrDvZ
v2ap+9Zg852J3jX16CA8stlgMy79GpsCUPVt9A8Ev3jKd8m8roAbhwC1KKA3uMlghwQtn5
h8e3nEVp1QayYnG4CCgK72oA12v/RyYivK5jPXcmZusSh8Wu8d0LbfMcvV/ki/yHfMfPc4
PF8VssuwaR8W0if+c5N2B8stlgb/YT63Nwz1QqPvptDv0NB/23WYluRAC/x6w14Hg0let
5uzMy8YDrgvIdCuEHM/1Bc15Yxidlua8bch5wu0aelH1Nfiv6B7za10a30ZrHP180qv
Fingerprint: fc:e9:36:25:00:ae:d6:df:5c:42:9c:b8:11:89:b7:b6

Kex Algorithms:
curve25519-sha256
curve25519-sha256@libssh.org
ecdh-sha2-nistp256
ecdh-sha2-nistp384
ecdh-sha2-nistp521
diffie-hellman-group-exchange-sha256
diffie-hellman-group16-sha512
diffie-hellman-group18-sha512
diffie-hellman-group-exchange-sha1
diffie-hellman-group14-sha256
diffie-hellman-group14-sha1
diffie-hellman-group1-sha1

Server Host Key Algorithms:
ssh-rsa
rsa-sha2-512
rsa-sha2-256
ecdsa-sha2-nistp256

Encryption Algorithms:
aes128-ctr
aes192-ctr
aes256-ctr
aes128-gcm@openssh.com
aes256-gcm@openssh.com

MAC Algorithms:
hmac-sha2-256
hmac-sha2-512

Compression Algorithms:
none
zlib@openssh.com
```

Ulteriori informazioni (banner)
sui servizi erogati dall'host

Shodan

Esempio 3

SHODAN Explore Downloads Pricing [country:"IT" port:"445"](#) 🔍

TOTAL RESULTS
17,867

TOP CITIES

City	Count
Milan	3,820
Rome	2,488
Turin	640
Naples	517
Bologna	434

[More...](#)

TOP ORGANIZATIONS

Organization	Count
Fastweb SpA	3,413
Telecom Italia S.p.A.	3,204
Vodafone Italia S.p.A.	1,575
Telecom Italia S.p.A. TIN EASY LITE	1,488
Tiscali Italia S.P.A.	685

[More...](#)

TOP PRODUCTS

Product	Count
Samba	5,995
snmbx	375
NQE 2.0	300
NQ 6.2	6
YNQ 1.4.5	5

[More...](#)

TOP OPERATING SYSTEMS

OS	Count
Windows 6.1	2,334
Unix	2,316
QTS	1,353
Windows Server 2012 R2 Standard 9600	555
Darwin	375

[More...](#)

View Results **View on Map**

country: "IT" port: "445"

Italy Milano

93.42.154.234
93.42.154.234.p87.fastwebnet.it
Fastweb SpA

Italy Milan

185.251.124.23
host:185-251-124-23.ipv4.infrastel.com
Linguanti Claudia trading as Micro Servizi

Italy Gramsciele

31.198.27.98
host:31-198-27-98.business.telecomitalia.it
TROTTA BUS SERVICES SPA

Italy Milan

87.26.8.236
host:87-26-8-236.business.telecomitalia.it
Telecom Italia S.p.A. IPTV

Italy Palazzolo sull'Oglio

System Information:

- SMB Version: 1
- OS: Windows 6.1
- Software: Samba 4.7.1
- Capabilities: dfs, extended-security, infolevel-passthru, large-files, large-reads, large-writes, level2-oplocks, lock-and-read, nt-find, nt-smb, nt-status, raw-mode, rpc-remote-api, unicode, unix

Shares:

Name	Type	Comments
print\$	Disk	Printer Drivers
GCORE_Windows_Server_2016	Disk	
IPC\$	IPC	IPC Service (Samba 4.7.1)
ISPysystem_Windows_Server_2016	Disk	

SMB Status:

- Authentication: enabled
- SMB Version: 1
- OS: Windows Server 2016 Standard 14393
- Software: Windows Server 2016 Standard 6.3
- Capabilities: extended-security, infolevel-passthru, large-files, large-reads, large-writes, level2-oplocks, lock-and-read, lwa, nt-find, nt-smb, nt-status, rpc-remote-api, unicode

Enumerating Target e Port Scanning

Shodan

Esempio 3

SHODAN Explore Downloads Pricing country:"IT" port:"445" 🔍

TOTAL RESULTS
17,867

View Report Browse Images View on Map

New Service: Keep track of what you have connected to the Internet. Check out [Shodan Monitor](#)

92.38.174.4
do-m2.gommba.com
G Core Labs S.A.
Italy Milano

SMB Status:
Authentication: disabled
SMB Version: 1
OS: Windows 6.1
Software: Samba 4.7.1
Capabilities: dfs, extended-security, infolevel-passthru, large-files, large-reads, large-writes, level2-oplocks, lock-and-read, nt-find, nt-smb, nt-status, raw-mode, rpc-remote-api, unicode, unix

Shares

Name	Type	Comments
print\$	Disk	Printer Drivers
GCORE_Windows_Server_2016	Disk	
IPC\$	IPC	IPC Service (Samba 4.7.1)
ISPysystem_Windows_Server_2016	Disk	

93.42.154.234
93-42-154-234.p87.fastwebnet.it
Fastweb SpA
Italy Milan

SMB Status:
Authentication: enabled
SMB Version: 1
OS: Windows Server 2016 Standard 14393
Software: Windows Server 2016 Standard 6.3
Capabilities: extended-security, infolevel-passthru, large-files, large-reads, large-writes, level2-oplocks, lock-and-read, lwa, nt-find, nt-smb, nt-status, rpc-remote-api, unicode

185.251.124.23
host:185-251-124-23.ipv4.infrastel.com
Linguanti Claudia trading as Micro Servizi
Italy Gramsciele

SMB Status:
Authentication: enabled
SMB Version: 1
OS: Windows Server 2016 Datacenter 14393
Software: Windows Server 2016 Datacenter 6.3
Capabilities: extended-security, infolevel-passthru, large-files, large-reads, large-writes, level2-oplocks, lock-and-read, lwa, nt-find, nt-smb, nt-status, rpc-remote-api, unicode

31.198.27.98
host:31-198-27-98.business.telecomitalia.it
TROTTA BUS SERVICES SPA
Italy Milan

SMB Status:
Authentication: enabled
SMB Version: 1
OS: Windows 6.1
Software: Samba 4.3.11-Ubuntu
Capabilities: dfs, extended-security, infolevel-passthru, large-files, large-reads, large-writes, level2-oplocks, lock-and-read, nt-find, nt-smb, nt-status, rpc-remote-api, unicode, unix

87.26.8.236
host:87-26-8-236.business.telecomitalia.it
Telecom Italia S.p.A. IPTV
Italy Palazzo aurOglio

SMB Status:
Authentication: enabled
SMB Version: 1
OS: QOS
Software: Samba 4.11.17
Capabilities: dfs, extended-security, infolevel-passthru, large-files, large-reads, large-writes, level2-oplocks, lock-and-read, nt-find, nt-smb, nt-status, raw-mode, rpc-remote-api, unicode

TOP CITIES

Milan	3,820
Rome	2,488
Turin	640
Naples	517
Bologna	434
More...	

TOP ORGANIZATIONS

Fastweb SpA	3,413
Telecom Italia S.p.A.	3,204
Vodafone Italia S.p.A.	1,575
Telecom Italia S.p.A. TIN EASY LITE	1,488
Tiscali Italia S.P.A.	685
More...	

TOP PRODUCTS

Samba	5,995
smbrx	375
NQE 2.0	300
NQ 6.2	6
YNQ 1.4.6	5
More...	

TOP OPERATING SYSTEMS

Windows 6.1	2,334
Unix	2,316
QTS	1,353
Windows Server 2012 R2 Standard 9600	555
Darwin	375
More...	

Enumerating Target e Port Scanning

Shodan

Esempio 3

Informazioni sulle altre
porte aperte presenti
sull'host selezionato

The screenshot displays the Shodan search engine interface. The top navigation bar includes links for SHODAN, Explore, Downloads, and Pricing, along with a search bar. The main header shows the IP address 92.38.174.4 and a map of the location in Milan, Italy. Below the header, the 'General Information' section provides details about the host, including its hostnames, domains, country, city, organization, ISP, ASN, and operating system. The 'Web Technologies' section shows the use of jQuery. The 'Open Ports' section, highlighted with a red box, lists the open ports and their associated programs and protocols.

General Information

Field	Value
Hostnames	dci-mi2.gcorelabs.com, gcorelabs.com
Domains	G-CORELABS.COM
Country	Italy
City	Milano
Organization	G-Core Labs S.A.
ISP	G-Core Labs S.A.
ASN	AS199524
Operating System	Windows 6.1

Web Technologies

- JQUERY

Open Ports

80 111 443 445

// 80 / TCP

92.38.174.4 | 2022-04-09T04:14:44.109Z

HTTP/1.1 200 Moved Permanently
Content-Length: 0
Connection: close
Location: https://92.38.174.4/
Date: Sat, 09 Apr 2022 04:14:44 GMT

// 111 / TCP

2931177911 | 2022-04-09T08:07:21.155650

Portmap	Program	Version	Protocol	Port
portmapper	4	tcp	111	
portmapper	3	tcp	111	
portmapper	2	tcp	111	
portmapper	4	udp	111	
portmapper	3	udp	111	
portmapper	2	udp	111	
status	1	udp	15207	
status	1	tcp	50978	
mountd	1	udp	20048	
mountd	1	tcp	20048	
mountd	2	udp	20048	
mountd	2	tcp	20048	
mountd	3	udp	20048	
mountd	3	tcp	20048	
nfs	3	tcp	2049	
nfs	4	tcp	2049	
100227	3	tcp	2049	
nfs	3	udp	2049	
nfs	4	udp	2049	
100227	3	udp	2049	
nlockmgr	1	udp	33489	
nlockmgr	3	udp	33489	
nlockmgr	4	udp	33489	
nlockmgr	1	tcp	33009	
nlockmgr	3	tcp	33009	
nlockmgr	4	tcp	33009	

// 443 / TCP

1011000510 | 2022-04-09T08:11:46.802722

Enumerating Target e Port Scanning

Shodan

Esempio 4

SHODAN Explore Downloads Pricing `/VSFTPD 2.3.4 country:"US" port:"21"` 🔍

TOTAL RESULTS: 393

View Report

New Service: [Check out Shodan Monitor](#)

60.95.203.400

VSFTPD 2.3.4 country:"US" port:"21"

TOP CITIES

New York City	137
Newark	15
Ashburn	11
San Jose	11
Los Angeles	10
More...	

TOP ORGANIZATIONS

Techie Hosting, Inc.	137
Comcast Cable Communications, LLC	27
T-Mobile USA, Inc.	19
Charter Communications Inc	17
Charter Communications	10
More...	

TOP PRODUCTS

vsftpd	1
vsftpd (before 2.0.8) or WU-FTP	1

66.211.130.34

66-211-130-34 static.firslight.net
First Light Fiber
United States, Portland

228 (vsftpd 2.3.4)
SMB Login Incorrect.
SMB Please login with USER and PASS.
211-Features:
EPRT
EPSV
MDTM
PASV
REST STREAM
SIZE
TVFS
UTF8
211 End

208.71.131.16

Techie Hosting, Inc.
United States, New York City

228 (vsftpd 2.3.4)
SMB Login Incorrect.
SMB Please login with USER and PASS.
211-Features:
EPRT
EPSV
MDTM
PASV
REST STREAM
SIZE
TVFS
UTF8
211 End

208.71.130.44

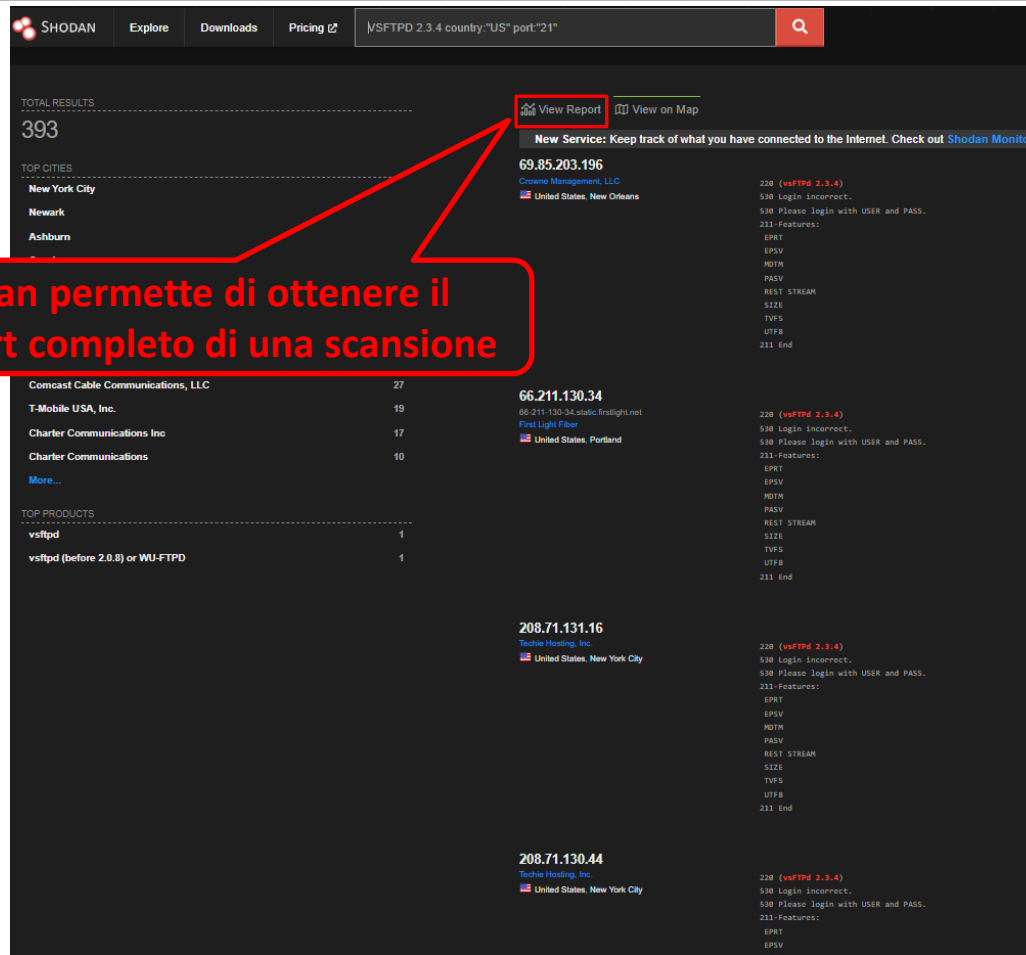
Techie Hosting, Inc.
United States, New York City

228 (vsftpd 2.3.4)
SMB Login Incorrect.
SMB Please login with USER and PASS.
211-Features:
EPRT
EPSV
MDTM

Enumerating Target e Port Scanning

Shodan

Esempio 4



SHODAN Explore Downloads Pricing | vsftpd 2.3.4 country:"US" port:"21" 🔍

TOTAL RESULTS
393

TOP CITIES
New York City
Newark
Ashburn

TOP PRODUCTS
vsftpd 1
vsftpd (before 2.0.8) or WU-FTPD 1

View Report View on Map

New Service: Keep track of what you have connected to the Internet. Check out [Shodan Monitor](#)

69.85.203.196
Crowe Management, LLC
United States, New Orleans

228 (vsftpd 2.3.4)
SMB Login Incorrect.
SMB Please login with USER and PASS.
211-Features:
EPRT
EPSV
MDTM
PASV
REST STREAM
SIZE
TVFS
UTF8
211 End

66.211.130.34
66-211-130-34.static.firstlight.net
First Light Fiber
United States, Portland

228 (vsftpd 2.3.4)
SMB Login Incorrect.
SMB Please login with USER and PASS.
211-Features:
EPRT
EPSV
MDTM
PASV
REST STREAM
SIZE
TVFS
UTF8
211 End

208.71.131.16
Techie Hosting, Inc.
United States, New York City

228 (vsftpd 2.3.4)
SMB Login Incorrect.
SMB Please login with USER and PASS.
211-Features:
EPRT
EPSV
MDTM
PASV
REST STREAM
SIZE
TVFS
UTF8
211 End

208.71.130.44
Techie Hosting, Inc.
United States, New York City

228 (vsftpd 2.3.4)
SMB Login Incorrect.
SMB Please login with USER and PASS.
211-Features:
EPRT
EPSV
MDTM

➤ Shodan permette di ottenere il report completo di una scansione

Shodan

Esempio 5

The screenshot displays the Shodan search engine interface. The search bar at the top contains the query "rfb 003.007 authentication country:ru". The search results are categorized into three sections: TOP CITIES, TOP PORTS, and TOP ORGANIZATIONS. A red box highlights the search query in the search bar.

Search Query: rfb 003.007 authentication country: "ru"

TOTAL RESULTS: 157

TOP CITIES:

City	Count
Moscow	29
Saint Petersburg	28
Yekaterinburg	17
Tula	13
Kazan	6

TOP PORTS:

Port	Count
5900	112
5901	37
5800	2
443	1
2323	1

TOP ORGANIZATIONS:

Organization	Count
PJSC Vimpelcom	11
CJSC ER-Telecom Holding Tula branch	10
Selectel Network	10
PJSC Rostelecom	6
JSC ER-Telecom Holding	4

Search Results Details:

79.172.36.66
79.172.36.66.ural.ru
JSC ER-Telecom Holding
Russian Federation, Yekaterinburg

87.251.183.194
Fenster
Russian Federation, Yekaterinburg

188.235.160.190
dynamicip-188-235-160-190.pppoe.sarazov.ertelecom.ru
CJSC ER-Telecom Holding Saratov branch
Russian Federation, Saratov

83.171.73.101
PJSC Rostelecom
Russian Federation, Krasnogorskiy

RFB 003.007
Authentication disabled
VNC:
Protocol Version: 3.7
Security Types:
1: None
18: TLS

Enumerating Target e Port Scanning

Shodan

Esempio 5

```
lMonb3oBaTenb: Cit
Hlo6po noxkanoBatTb B i2651-trueconf-01
17:15 05
MAS. NH.
21:1Yal7 pM pf-Tofe
Sy hhh hhh bbb bbb hhh bbb hh
;
Cals
EN
CapsLock Bpikn.
VNC:
Protocol Version: 3.7
Security Types:
  1: None
  18: TLS
Server Name: cit@i2651-trueconf-0...
```



Shodan

Esempio 6

The screenshot displays the Shodan search engine interface with the search term 'netcam' entered in the top navigation bar. A red box highlights the word 'netcam' in the search bar. The interface shows a total of 3,963 results. On the left, there are filters for 'TOP COUNTRIES' (United States, Germany, Hong Kong, Russian Federation, France) and 'TOP PORTS' (80, 81, 82, 8080, 8000). Below these are 'TOP ORGANIZATIONS' (Vultr Holdings, LLC, Deutsche Telekom AG, etc.) and 'TOP PRODUCTS' (Teradici PCoIP Management Console, mDNS, etc.). The main results area lists several IP addresses with associated details:

- 113.253.233.239**: HGC Global Communications Limited, Hong Kong. HTTP/1.1 481 Unauthorized. Content-Type: text/html. Connection: keep-alive. WWW-Authenticate: Basic realm="netcam". Content-Length: 17.
- 193.252.194.239**: Orange S.A., France. HTTP/1.1 481 Unauthorized. Content-Type: text/html. Connection: keep-alive. WWW-Authenticate: Basic realm="netcam". Content-Length: 17.
- 149.28.168.149**: Vultr Holdings, LLC, Australia, Sydney. HTTP/1.1 200 OK. Server: MiniUPnPd/1.4. Pragma: no-cache. Cache-Control: no-cache. Content-Type: text/html; charset=utf-8. Content-Length: 55868.
- 129.28.171.180**: Tencent Cloud Computing (Beijing) Co., Ltd, China, Guangzhou. HTTP/1.1 200 OK. Server: MiniUPnPd/1.4. Pragma: no-cache. Cache-Control: no-cache. Content-Type: text/html; charset=utf-8. Content-Length: 55868.
- 37.143.130.93**: Onaprovider.com - Madrid Infrastructure, Spain, Madrid. HTTP/1.1 200 OK. Server: MiniUPnPd/1.4. Pragma: no-cache. Cache-Control: no-cache. Content-Type: text/html; charset=utf-8. Content-Length: 55868.

Enumerating Target e Port Scanning

Shodan

Esempio 7

The screenshot displays the Shodan search engine interface. At the top, the navigation bar includes 'SHODAN', 'Explore', 'Downloads', 'Pricing', and a search bar containing 'Axis M1013'. A red box highlights the search term 'Axis M1013' in the search bar. Below the navigation bar, the 'TOTAL RESULTS' section shows '48'. The 'TOP COUNTRIES' section features a world map and a list of countries: Mexico (20), Russian Federation (9), United States (8), Nicaragua (3), and Brazil (2). The 'TOP ORGANIZATIONS' section lists: Gestión de direccionamiento UniNet (12), Uninet S.A. de C.V. (8), EQUIPOS Y SISTEMAS S.A. (3), Novgorod Datacom (3), and Purdue University (3). The main content area displays search results for 'Axis M1013'. Each result includes an IP address, a host name, a location, and a list of open ports. The results are as follows:

IP Address	Host Name	Location	Open Ports
189.190.91.166	dsl-189-190-91-166-dyn.prod-infinilum.com.mx	Gestión de direccionamiento UniNet, Mexico, Puebla	220 AXIS M1013 Fixed Network Camera 5.51.7.3 (2020) ready., 530 Login incorrect., 214-The following commands are implemented., USER QUIT PASS SYST HELP PORT PASV LIST, NLST RETR STOR TYPE MKD RMD DELE PWD, CMD SITE CDUP RNFR RNTD NOOP...
128.46.190.92	shincam-me38.eon.purdue.edu	Purdue University, United States, West Lafayette	220 AXIS M1013 Fixed Network Camera 5.50.5.1 (2014) ready., 530 Login incorrect., 214-The following commands are implemented., USER QUIT PASS SYST HELP PORT PASV LIST, NLST RETR STOR TYPE MKD RMD DELE PWD, CMD SITE CDUP RNFR RNTD NOOP...
67.215.107.37	shincam-me38.eon.purdue.edu	Puerto Rico, San Juan	220 AXIS M1013 Fixed Network Camera 5.40.5.3 (2013) ready., 530 Login incorrect., 214-The following commands are implemented., USER QUIT PASS SYST HELP PORT PASV LIST, NLST RETR STOR TYPE MKD RMD DELE PWD, CMD SITE CDUP RNFR RNTD NOOP...
200.145.110.201	host110201.fca.unesp.br	UNIVERSIDADE ESTADUAL PAULISTA, Brazil, Jaboticabal	220 AXIS M1013 Fixed Network Camera 5.40.5.2 (2013) ready., 530 Login incorrect., 214-The following commands are implemented., USER QUIT PASS SYST HELP PORT PASV LIST, NLST RETR STOR TYPE MKD RMD DELE PWD, CMD SITE CDUP RNFR RNTD NOOP...
84.242.195.42	Novgorod Datacom	Russian Federation, Velikiy Novgorod	220 AXIS M1013 Fixed Network Camera 5.40.5.3 (2013) ready., 530 Login incorrect., 214-The following commands are implemented., USER QUIT PASS SYST HELP PORT PASV LIST, NLST RETR STOR TYPE MKD RMD DELE PWD, CMD SITE CDUP RNFR RNTD NOOP...

Enumerating Target e Port Scanning

Shodan

Esempio 8

The screenshot displays the Shodan search engine interface with the search query "default password" entered in the top search bar. The interface is divided into several sections:

- TOTAL RESULTS:** 897
- TOP COUNTRIES:** A world map showing the distribution of results by country. The United States has the highest count at 154, followed by Spain (104), Russian Federation (82), Argentina (47), and Ukraine (46).
- TOP PORTS:** A list of ports with their respective counts. The top ports are 10081 (520), 50000 (87), 5984 (50), 8081 (42), and 8080 (36).
- TOP ORGANIZATIONS:** A list of organizations. The top ones are OLE COMUNICACION S.L. (76), Chunghwa Telecom Co., Ltd. (26), Linode (17), South Central Rural Telecommunications Cooperative Inc. (17), and TOV NEO-TELECOM (16).
- TOP PRODUCTS:** A list of products. The top ones are LAP (129), Teradici PCoIP Management Console (85), LAP-HP (74), LMS (70), and NBS (48).

The main content area displays three search results, each with a red box highlighting the phrase "default password":

- 85.207.247.18**
static.ralskoni.cz
RADIOKOMUNIKACE a.s.
Czechia, Stráž pod Ralskem
compromised
Ubiquiti Networks Device:
IP Address: 85.207.247.18
MAC Address: 08:15:6D:89:11:09
Alternate IP Address: 169.254.17.213
Alternate MAC Address: 08:15:6D:8A:11:05
Hostname: HACKED-ROUTER-HELP-505-HAD-DEFAULT-PASSWORD
Product: NSS
Version: X55.ar2113.v4.0.4.5074.150734.1344
- 88.8.205.34**
34.red-88-8-205.dynasticp.rima-ide.net
Telefonica de Espana SAU
Spain, Terrassa
HTTP/1.1 200 OK
Date: Sun, 10 Apr 2022 19:42:41 GMT
Server: Apache
Set-Cookie: adw=deleted; expires=Thu, 01-Jan-1970 00:00:01 GMT; Max-Age=0
Set-Cookie: adw=deleted; expires=Thu, 01-Jan-1970 00:00:01 GMT; Max-Age=0
Set-Cookie: cookie_sib=deleted; expires=Thu, 01-Jan-1970 00:00:01 GMT; Max...
- 37.114.229.53**
Tosoh Fanavai Ertebatat Pasargad Arian Co.
PAE
Iran, Islamic Republic of, Pasargad
compromised
Ubiquiti Networks Device:
IP Address: 37.114.229.53
MAC Address: 44:09:E7:CE:F8:8F
Alternate IP Address: 192.168.1.1
Alternate MAC Address: 04:09:E7:CF:F8:8F
Hostname: HACKED-ROUTER-HELP-505-HAD-DEFAULT-PASSWORD
Product: LMS
Version: XM.ar934x.v5.6.2.licensed.28182.198885.1456

Enumerating Target e Port Scanning

Shodan

Esempio 9

The screenshot displays the Shodan search engine interface. At the top, the search bar contains the query "Server: SQ-WEBCAM". A red box highlights the search results, with a red arrow pointing to the search bar. The search results are organized into several sections:

- TOTAL RESULTS:** 320
- TOP COUNTRIES:** A world map showing the distribution of results by country. The top countries listed are Russian Federation (84), Ukraine (43), United Kingdom (35), United States (35), and India (24).
- TOP PORTS:** A list of the most common ports scanned. The top ports are 80 (30), 81 (29), 52869 (29), 7547 (28), and 37215 (27).
- TOP ORGANIZATIONS:** A list of the most common organizations. The top organizations are Softline Trade JSC (84), Linode (80), 139.162.0.0/16 (26), LTD HOSTPRO LAB (24), and Linode, LLC (22).

The main search results are displayed in a table with the following columns:

- IP Address:** 172.104.47.64, 139.59.56.96, 139.162.119.9, 93.90.222.17, 139.162.119.9, 93.90.222.28
- Organization:** Linode, DigitalOcean, LLC, Softline Trade JSC, Russian Federation, Moscow
- Location:** Singapore, Singapore, India, Doddaballapura, Japan, Tokyo
- Cloud Provider:** cloud, honeybot
- HTTP Status:** HTTP/1.1 200 OK
- Server:** 368 web server, 792/71644 HTTP Server version 2.0 - TELDAT S.A., A18M5/1.00, ADB Broadband HTTP Server, ADH-Web, AR, ASUSTeK UPnP/1.0 MiniUPnPd/1.4

Enumerating Target e Port Scanning

Shodan

Esempio 10

The screenshot displays the Shodan search engine interface. The search bar at the top contains the query "linux upnp avtech". The results are categorized into several sections:

- TOTAL RESULTS:** 231,376
- TOP COUNTRIES:** A world map showing the distribution of results by country. The top countries listed are Malaysia (48,428), Thailand (39,224), Indonesia (16,746), Taiwan (14,441), and China (12,371).
- TOP PORTS:** A list of ports and their associated result counts: 80 (18,695), 88 (5,297), 8080 (4,081), 81 (3,839), and 8000 (2,773).
- TOP ORGANIZATIONS:** A list of organizations and their result counts: Triple T Broadband Public Company Limited (24,563), TMNST (23,773), Maxis Broadband Sdn Bhd (12,862), Internet Service Provider in Sri Lanka (10,910), and Chunghwa Telecom Co., Ltd. (10,714).
- TOP PRODUCTS:** A list of products and their result counts: Avtech AVN801 network camera (58,476), lighttpd (74), Apache Httpd (63), Teradici PCoIP Management Console (59), and CherryPy httpd (34).

The main content area displays detailed information for the first three search results, each including a "View Report" link, the IP address, the organization, the location, the date, the server, the connection status, the last modified date, the content type, the ETag, and the content length.

- Result 1:** IP: 123.195.224.17, Organization: Kbro CO. Ltd., Location: Taiwan, Taoyuan City, Date: Mon, 11 Apr 2022 04:05:12 GMT, Server: Linux/2.x Uhm/1.0 Artech/1.0, Connection: close, Last-Modified: Tue, 26 Nov 2019 07:36:01 GMT, Content-Type: text/html, ETag: 230-54373-1514731761, Content-Length: 54171.
- Result 2:** IP: 218.111.101.201, Organization: STREAMYX-BIZ-CENTRAL, Location: Malaysia, George Town, Date: Mon, 11 Apr 2022 04:05:03 GMT, Server: Linux/2.x Uhm/1.0 Artech/1.0, Connection: close, Last-Modified: Wed, 27 Jun 2018 03:19:35 GMT, Content-Type: text/html, ETag: 388-15850-153009575, Content-Length: 15850.
- Result 3:** IP: 115.32.1.244, Organization: TMNST, Location: Malaysia, Bukit Mertajam, Date: Mon, 11 Apr 2022 04:04:42 GMT, Server: Linux/2.x Uhm/1.0 Artech/1.0, Connection: close, Last-Modified: Thu, 30 Mar 2017 06:47:43 GMT, Content-Type: text/html, ETag: 268-16695-149856463, Content-Length: 16695.

The fourth result is partially visible, showing the IP address 87.153.124.44 and the organization Deutsche Telekom AG.

Enumerating Target e Port Scanning

Outline

- Concetti Introductivi
- Suite Protocollore TCP/IP
- Formato dei Messaggi TCP e UDP
- Active Enumeration
 - Network Scanner Nmap
 - Zenmap
 - Unicornscan
 - Masscan
- Passive Enumeration
 - Shodan
 - **ZoomEye**
 - FOFA
 - Censys

ZoomEye

Caratteristiche

- Motore di ricerca che permette di ottenere informazioni su
 - Dispositivi
 - Siti Web
 - Servizi di rete
 - Componenti di rete
 - Etc
- Prima di utilizzare ZoomEye è fortemente consigliata la registrazione
- Talvolta per l'accesso a ZoomEye potrebbe essere necessario l'utilizzo di VPN
 - A causa di filtri su indirizzi IP provenienti dall'Unione Europea



ZoomEye

Pricing

<https://www.zoomeye.ai/pricing>

Features & Price		Registered User	Personal (Security enthusiasts) \$700 \$190/Year 73% OFF Buy Now	Professional (Security teams and analysts) \$1,900 \$1,090/Year 43% OFF Buy Now	Business (Medium to large enterprises) \$19,100 \$10,990/Year 43% OFF Buy Now	Corporate (Tailored enterprise solutions) Contact Now
New Explore+	ZoomEye Hub NEW	×	✓	✓	✓	✓
	ZoomEyeGPT(AI Search) NEW	5 results/Day	30 results/Day	100 results/Day	1000 results/Day	Custom
Primary	Monthly Maximum Results	3,000	100,000	800,000	10,000,000	Custom
	Maximum Result Pages	5	50	50	100	Custom
	API Access Available	Use ZoomEye-Point	✓	✓	✓	✓
	API Request Rate Limit (requests/second)	0.5	1	1	2	Custom
	Number of Result Fields Available	46	71	74	78	78
Ability	Web-based Results Download	Use ZoomEye-Point	✓	✓	✓	✓
	Honeypot Filtering	50 Points	10 Points	10 Points	✓	✓
	Aggregated Results Data	20	50	50	100	100
	Historical Data Access	×	×	×	✓	✓



ZoomEye

Interfaccia

➤ <https://www.zoomeye.ai/>

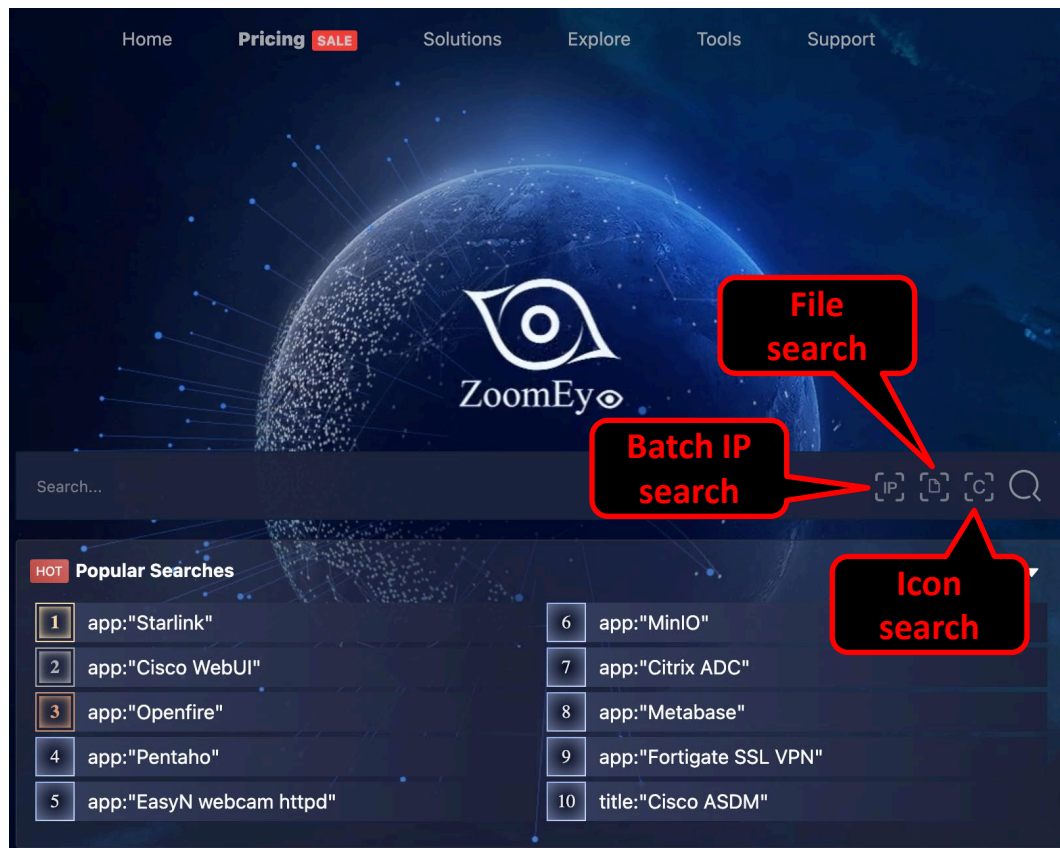


Enumerating Target e Port Scanning

ZoomEye

Interfaccia

➤ <https://www.zoomeye.ai/>



Enumerating Target e Port Scanning

ZoomEye

API

- ZoomEye offre agli utenti la possibilità di utilizzare API
- <https://github.com/knownsec/ZoomEye>

ZoomEye API SDK <https://www.zoomeye.org/api>

zoomeye zoomeye-api zoomeye-sdk

 23 commits

 2 branches

 1 release

 5 contributors

 GPL-2.0

Branch: master ▾

New pull request

Find file

Clone or download ▾

 hysia Merge pull request #4 from knownsec/feature/tests ... Latest commit e06afc5 on 29 Nov 2018

 tests	add unittest script	3 years ago
 .gitignore	Initial commit	4 years ago
 LICENSE	Initial commit	4 years ago
 MANIFEST.in	add setup.py	3 years ago
 README.md	Update Readme.md	11 months ago
 run_tests.sh	run a tests shell script	3 years ago
 setup.py	remove requires - getpass	3 years ago
 zoomeye.py	python3 (raw_input -> input)	3 years ago

Enumerating Target e Port Scanning

ZoomEye

Topics

➤ <https://www.zoomeye.ai/topics>



Power automation topics



Git platform topics



Blockchain topics



Industrial topics



Certificate topics



Firewall topics



Router topics



Printer topics



WAF topics



DNS topics



Camera topics



Network storage topics

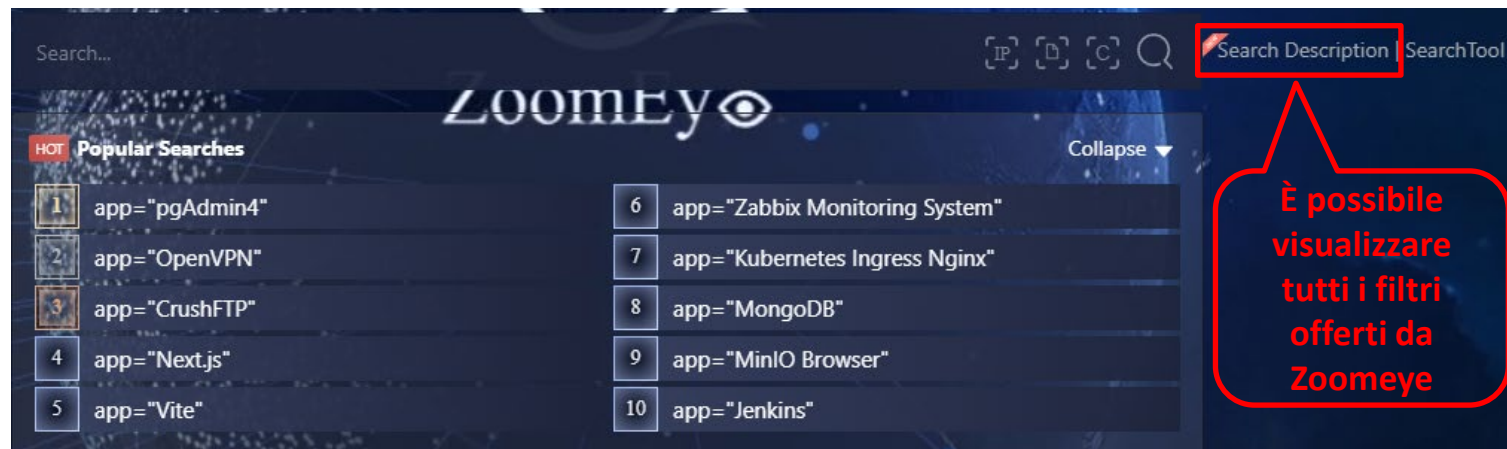


Database topics

ZoomEye

Filtri di Ricerca

➤ <https://www.zoomeye.ai/>



ZoomEye

Filtri di Ricerca

➤ <https://www.zoomeye.ai/>

Search Description

- Search scope covers devices (IPv4, IPv6) and websites (domain names)
- When entering a search string, the system will match the keywords in "global" mode, covering content from various protocols such as HTTP, SSH, FTP, etc. (e.g., HTTP/HTTPS protocol headers, body, SSL, title, and other protocol banners)
- The search string is case-insensitive and will be matched after segmentation (the search results page provides a "segmentation" test function). Use == for precise matching and strict restriction of search syntax case sensitivity.
- Please use quotation marks for search strings (e.g., "Cisco System" or 'Cisco System'). If there are quotation marks in the search string, use \ for escape, e.g., "a\"b". If there are brackets in the search string, use \ for escape, e.g., portinfo\(\)

Search Logic Operations

SearchLogic	Description	Example
=	Search for assets containing keywords	<code>title="knownsec"</code> Search for websites with titles containing Knowsec's assets
	Accurate search, indicating a	<code>title=="knownsec"</code> Precise search, which means exact match of

ZoomEye

Filtri di Ricerca

➤ Alcuni tra i principali filtri di ricerca

- **app**: nome dell'applicazione
- **country**: country code (ad es., **UK**, **IT**, **ES**, **FR**, **CN**, **JP**)
- **city**: nome della città
- **port**: numero di porta
- **os**: nome del Sistema Operativo (ad es., **os:linux**)
- **service**: nome del servizio
- **hostname**: hostname (ad es., **hostname:google.com**)
- **ip**: indirizzo IP (ad ed., **ip:8.8.8.8**)
- **cidr**: segmento CIDR (ad es., **cidr:8.8.8.0/24**)

ZoomEye

➤ Operatori Logici

Simbolo	Descrizione
=	Ricerca asset che abbiano una corrispondenza rispetto alle keyword
==	Ricerca accurata, che richiede una corrispondenza esatta rispetto alle keyword (case sensitive)
	OR Logico
&&	AND Logico
!=	NOT Logico
()	Forniscono prioritizzazione delle operazioni
*	Fuzzy search, use * for search

ZoomEye

Dorks

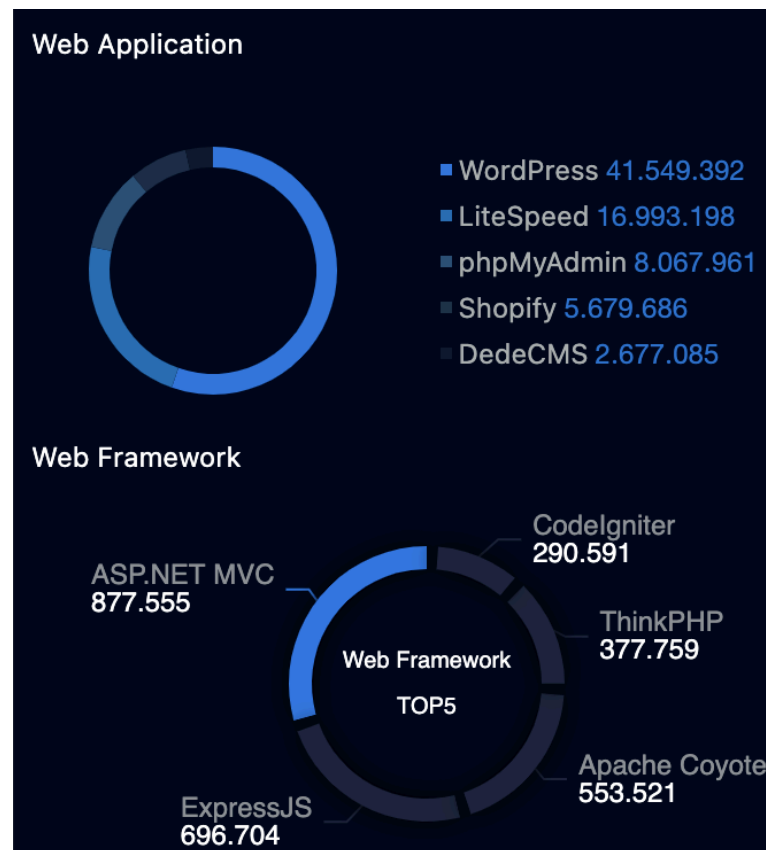
➤ <https://www.zoomeye.ai/>



ZoomEye

Statistiche

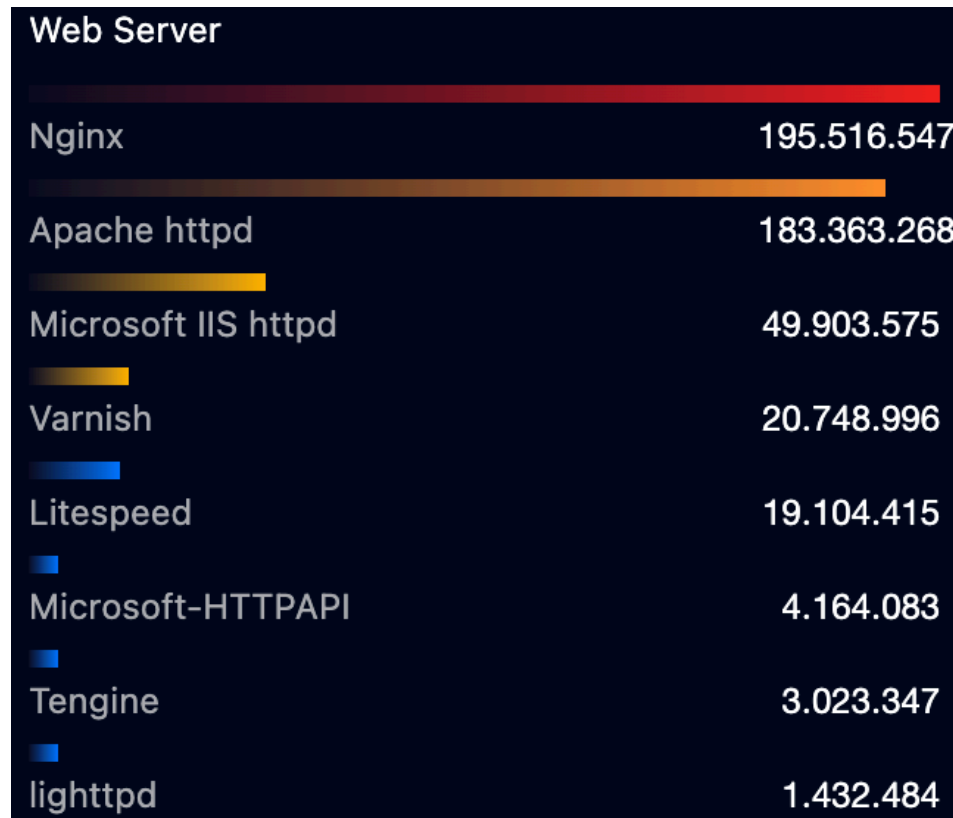
➤ <https://www.zoomeye.ai/statistics>



ZoomEye

Statistiche

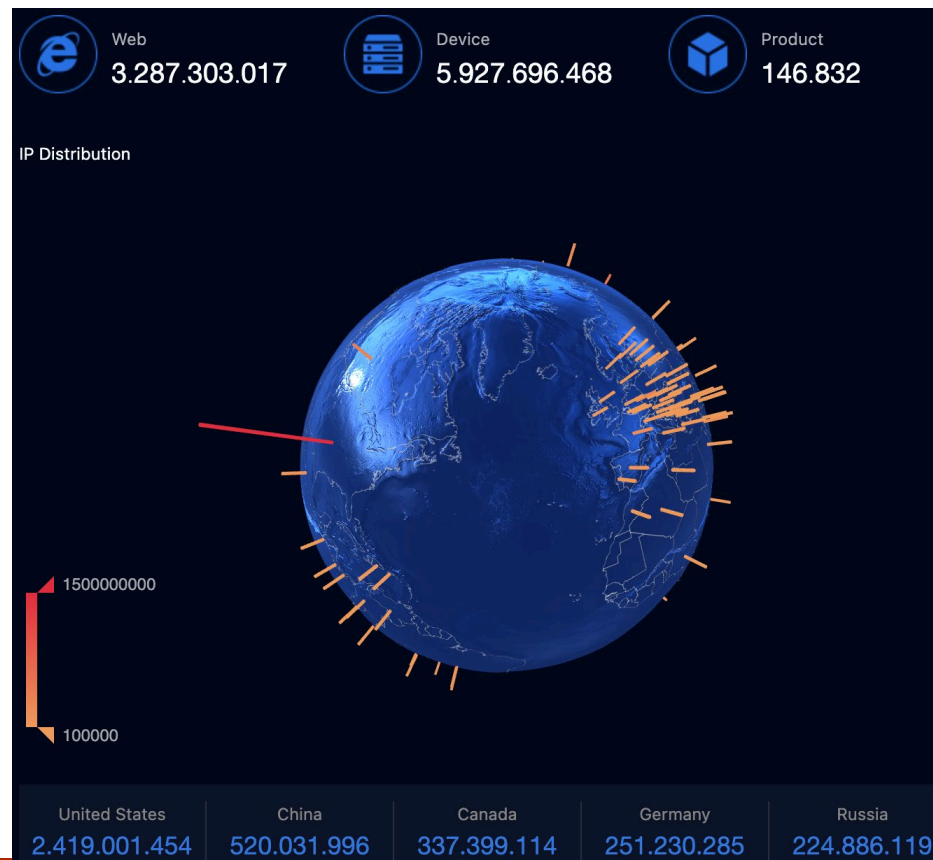
➤ <https://www.zoomeye.ai/statistics>



ZoomEye

Statistiche

➤ <https://www.zoomeye.ai/statistics>

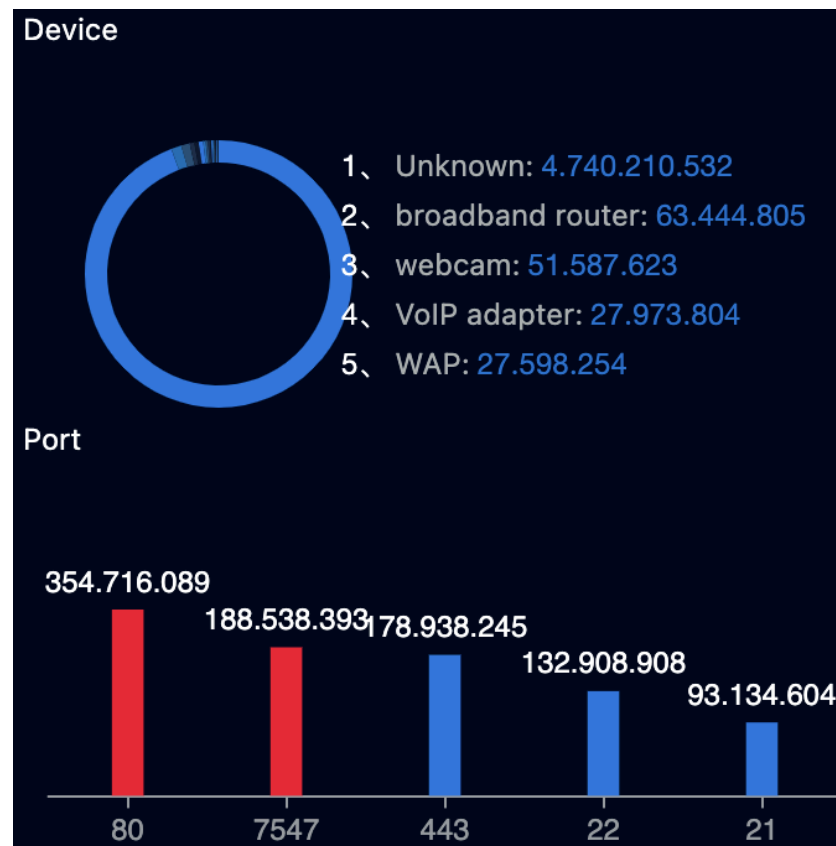


Enumerating Target e Port Scanning

ZoomEye

Statistiche

➤ <https://www.zoomeye.ai/statistics>



Enumerating Target e Port Scanning

ZoomEye

Statistiche

➤ <https://www.zoomeye.ai/statistics>

Product		
Unknown		1.820.634.872
Incapsula CDN httpd		1.425.864.267
Apache httpd		388.064.595
nginx		131.620.602
OpenSSH		93.855.070
Service		
ftp	96.546.135	
unknown	71.565.963	

ZoomEye

ZoomEyeGPT Beta

➤ <https://www.zoomeye.ai/gpt>

ZoomEyeGPT Beta

ZoomEyeGPT is an AI-powered tool specifically designed to enhance cyberspace asset searches. It addresses the challenges users face when navigating ZoomEye's search syntax. Whether you're a system administrator, an academic researcher, a network technology enthusiast, or a cybersecurity beginner, ZoomEyeGPT provides tailored syntax recommendations based on your needs. By reducing the learning curve, streamlining complex operations, and simplifying the search process, it makes powerful network exploration accessible to all.

ZoomEyeGPT Search Query:

Enter your target asset, and we'll auto-generate the search syntax. For example: Find Nginx assets in Los Angeles, USA.

(Daily limit: 5)



Examples

Search for Starlink devices in Ukraine.

Search for publicly accessible security cameras in Russia.

Search for data on port 22 or port 3306 in the United States.

Search for assets containing Arabic payment keywords.

Please help me convert the Censys query 'services.http.response.body: "index of"' into a ZoomEye query.

ZoomEye

Componenti

➤ <https://www.zoomeye.ai/component>

Components

ZoomEye component search navigation, let you find target assets quickly and accurately

Equipment type

- Gateway(395)
- PBX(391)
- Manager Platform(505)
- Industrial Equipment(79)
- CDN(43)
- Big Data(13)
- CMS(100)
- Web Framework(70)
- IP Camera(395)
- Video Recorder(156)
- IDS(15)
- Software Platfoem(61)
- Office Anywhere System(55)
- Management Software(113)



Gateway

Cisco SIP Gateway	Cisco Expressway E	zyxel AAM1212	Cisco TANDBERG ...	Cisco Application ...
▼				

PBX

Samsung PBX teln...	Asterisk PBX 11.20...	Asterisk PBX 1.6.2....	Asterisk PBX 1.8.1...	Asterisk PBX 1.6.2...
▼				

Enumerating Target e Port Scanning

ZoomEye

Esempio 1

- Effettuiamo una ricerca per Autonomous System Network (**asn=as137**)

The screenshot shows the ZoomEye search interface. At the top, the search bar contains the query `asn=as137`. Below the search bar, there are tabs for 'Result', 'Report', and 'Maps'. The 'Report' tab is selected and highlighted with a red box. A red arrow points from a red box containing the text `asn=as137` to the search bar. The search results show a list of IP addresses and their associated information. The first result is for the IP address `150.145.38.62:8787`. The details for this IP are shown in a table with columns for 'Banner' and 'Hash'. The 'Banner' column contains the following information:

Banner
HTTP/1.1 200 OK Content-Type: text/html X-Content-Type-Options: nosniff Server: RStudio Set-Cookie: rs-csrf-token=d5dc1d03-733f-415b-bb83-2c55d0afd790; path=/; csrf-token=d5dc1d03-733f-415b-bb83-2c55d0afd790; path=/; HttpOnly Pragma: no-cache Cache-Control: no-cache, no-store, max-age=0, must-revalidate Date: Mon, 05 May 2025 14:34:00 GMT X-Frame-Options: DENY

The 'Hash' column is empty. Below the table, there is a link to 'See All Text'. The second result is for the IP address `infophd.disi.unitn.it:80`. The details for this IP are shown in a table with columns for 'Header', 'Body', and 'Hash'. The 'Header' column contains the following information:

Header
HTTP/1.1 302 Found

The 'Body' and 'Hash' columns are empty. On the right side of the interface, there is a 'FILTER' section with a checkbox for 'Hide Honeypot' and a 'SEARCH TYPE' section with a list of search types and their counts:

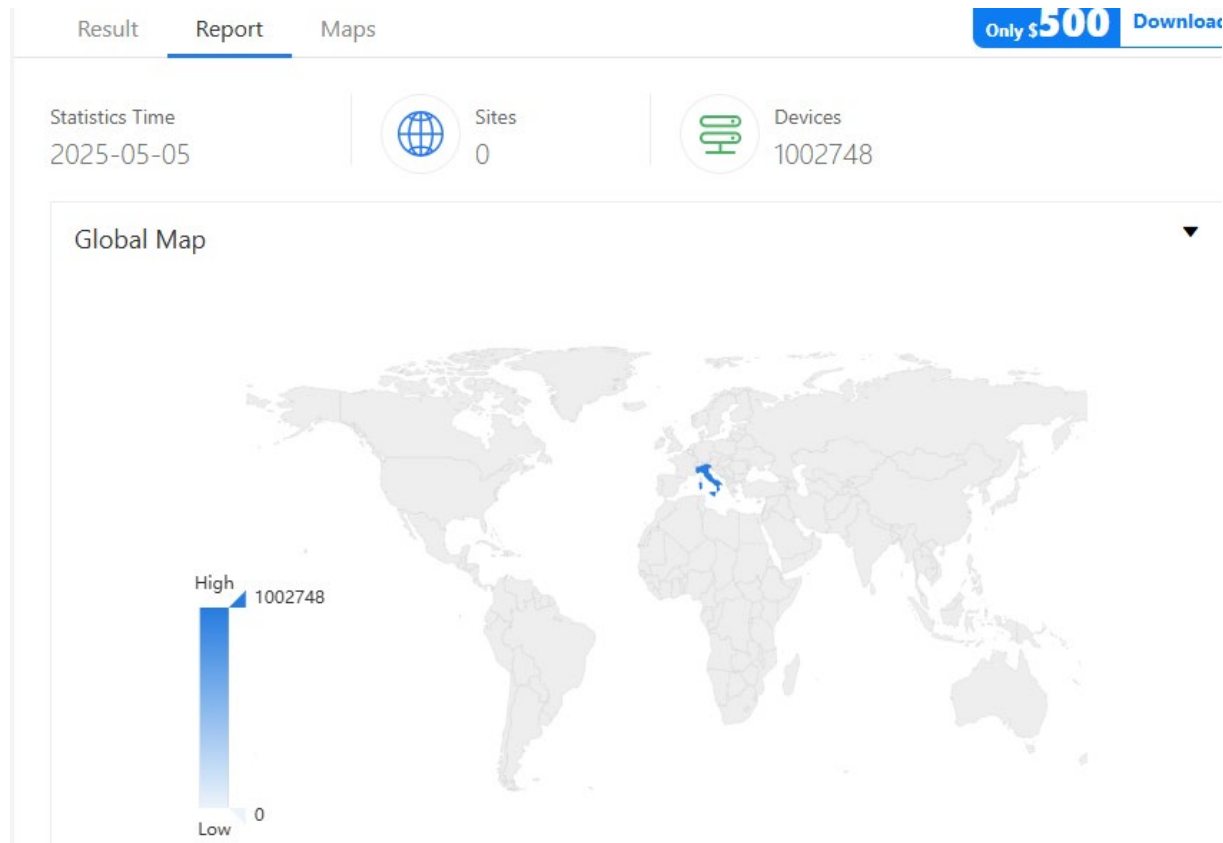
SEARCH TYPE	Count
Devices	1.003.004
Ipv4	1.002.748
Ipv6	256
Websites	22.457

Enumerating Target e Port Scanning

ZoomEye

Esempio 1

- Possiamo visualizzare numerose statistiche relative all'AS AS137

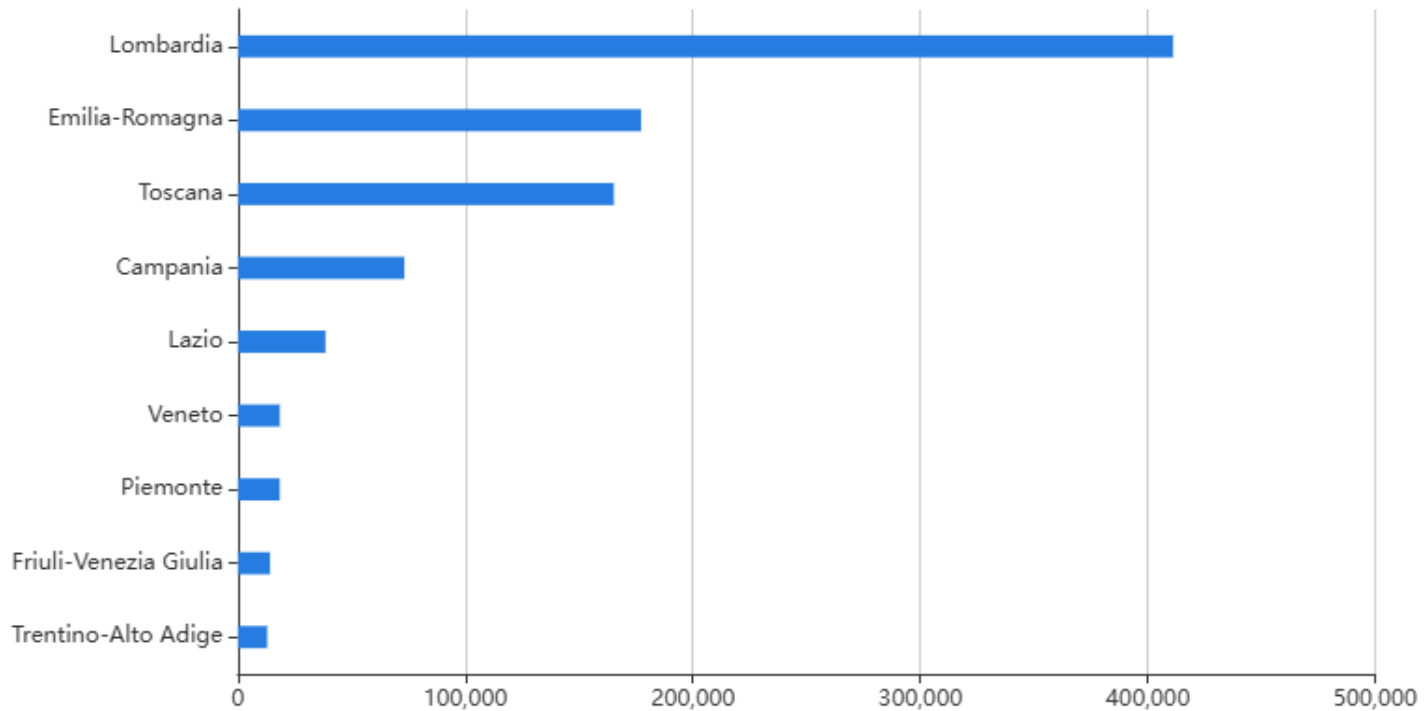


ZoomEye

Esempio 1



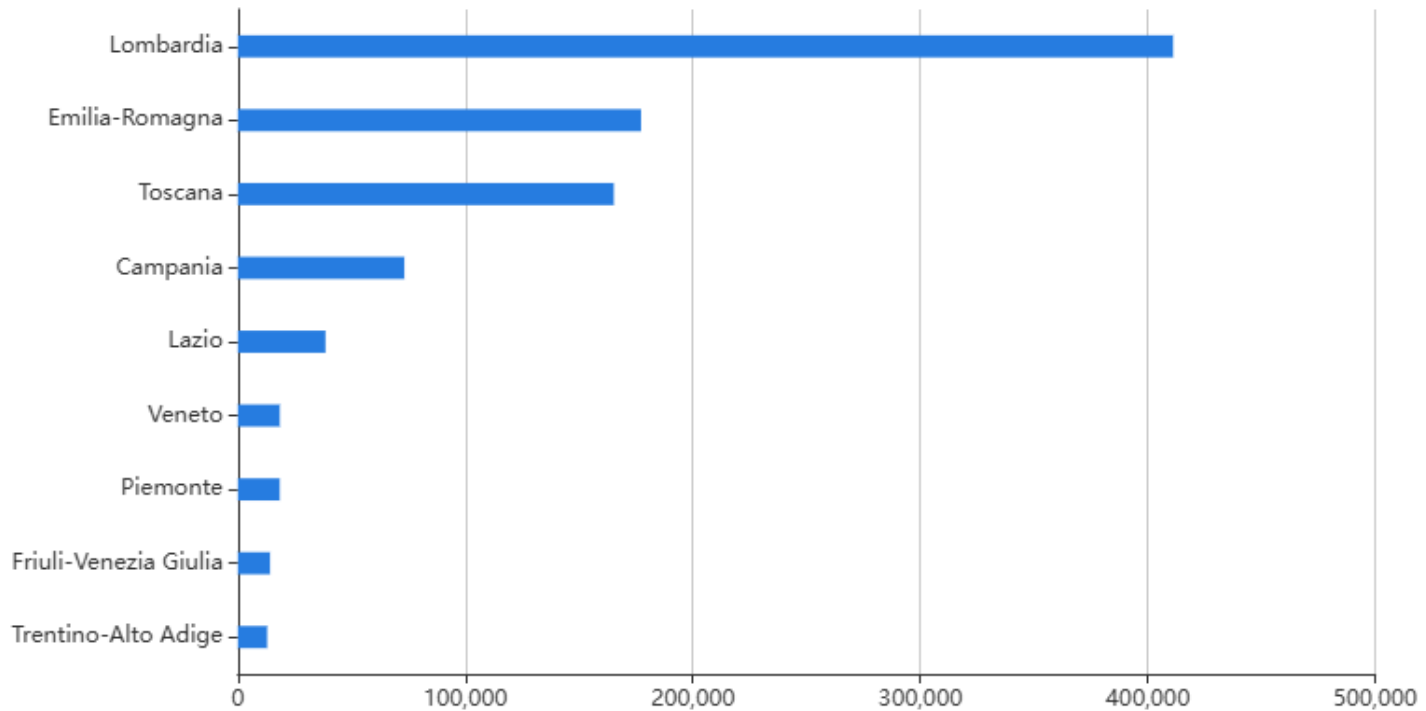
Regional Distribution



ZoomEye

Esempio 1

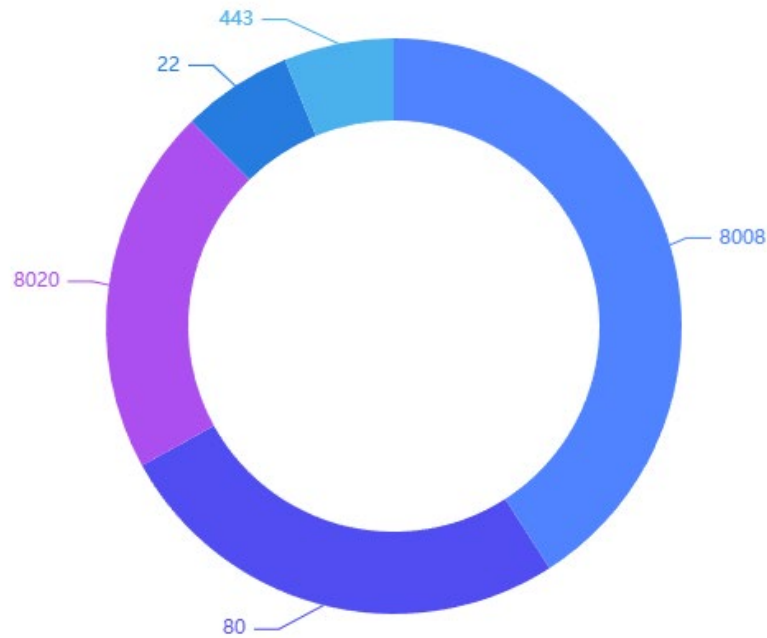
Regional Distribution



ZoomEye

Esempio 1

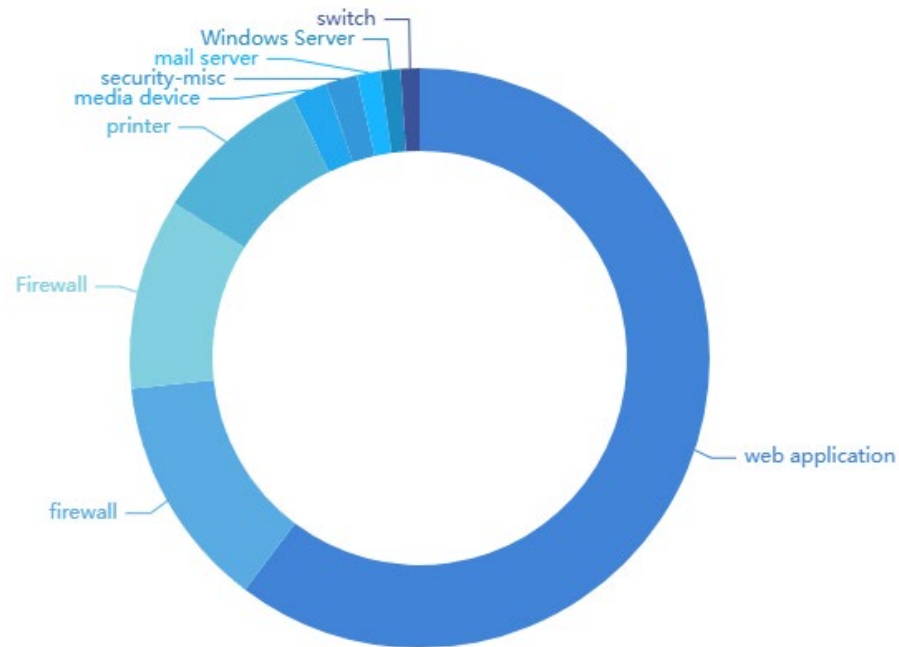
Port Statistics



ZoomEye

Esempio 1

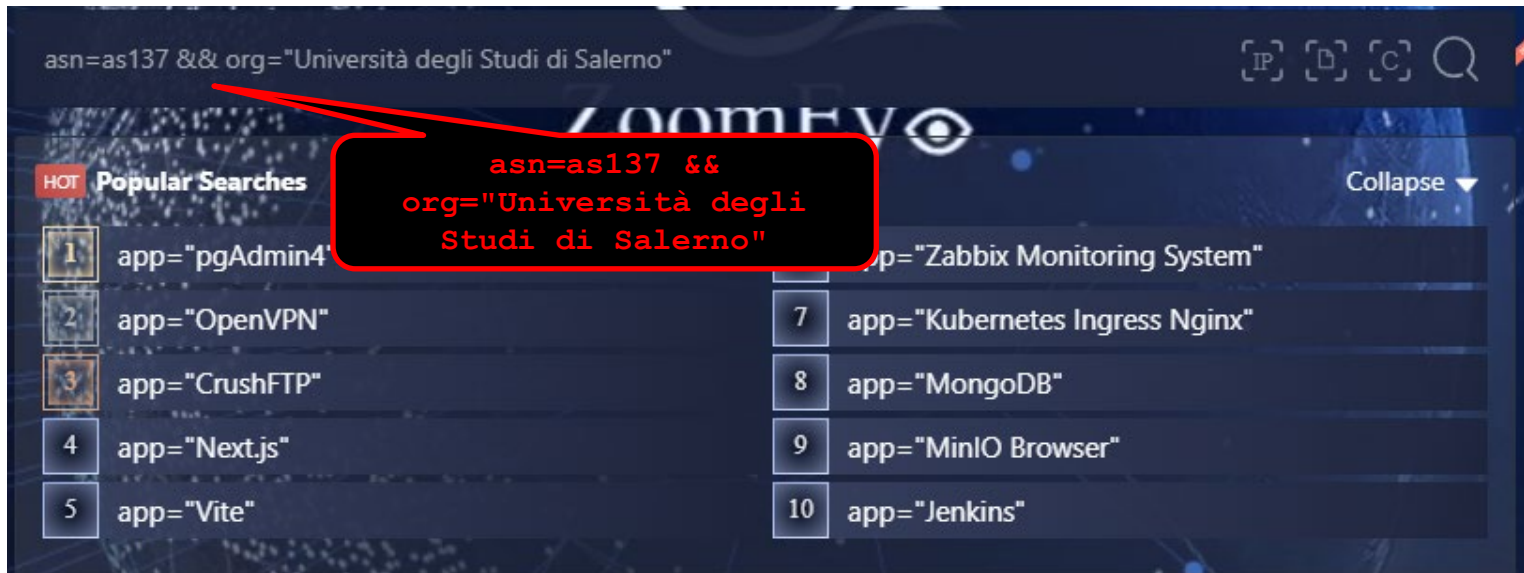
Device



ZoomEye

Esempio 2

- Specifichiamo l'organizzazione «Università degli Studi di Salerno»



ZoomEye

Esempio 2

➤ Vengono mostrati tutti i dispositivi appartenenti all'organizzazione

193.205.161.7:9000

9000 http

193.205.161.7

Data update

Italy, Campania, Salerno

Organization: Università degli Stu...

ASN: AS137

2025-05-04 03:57

Banner

Hash

HTTP/1.1 303 See Other
Set-Cookie: PLAY_SESSION=; Max-Age=0; Expires=Thu, 01 Jan 1970 00:00:00
X-Frame-Options: SAMEORIGIN
X-Xss-Protection: 1; mode=block
X-Content-Type-Options: nosniff
Location: /jatos/login
X-Permitted-Cross-Domain-Policies: master-only
Date: Sat, 03 May 2025 21:13:16 GMT
Content-Length: 0
Referrer-Policy: origin-when-cross-origin, strict-origin-when-cross-orig
See All Text

193.205.161.136:9000

9000 http

193.205.161.136

Data update

Italy, Campania, Salerno

Hostname: music.di.unisa.it

Organization: Università degli Stu...

ASN: AS137

Title: Page not found at /

2025-05-04 03:57

Banner

Hash

HTTP/1.1 404 Not Found
Date: Sat, 03 May 2025 20:19:17 GMT
Server: WSGIServer/0.2 CPython/3.11.0
Content-Type: text/html; charset=utf-8
X-Frame-Options: DENY
Content-Length: 4851
X-Content-Type-Options: nosniff
Referrer-Policy: same-origin
Cross-Origin-Opener-Policy: same-origin
Vary: Origin
See All Text



FILTER

☐ Hide Honeypot

SEARCH TYPE

Devices

1.993

Ipv4

1.993

Ipv6

0

Websites

17

YEAR

2025

75

2024

50

2023

304

More

ITALY

Campania

1.085

ZoomEye

Esempio 2

➤ Possiamo controllare i dettagli relativi a ciascun dispositivo

193.205.161.7:9000 9000 http

➡ 193.205.161.7 Data update

Italy, Campania, Salerno
Organization: Università degli Stu...
ASN: AS137
🕒 2025-05-04 03:57

Banner Hash

HTTP/1.1 303 See Other
Set-Cookie: PLAY_SESSION=; Max-Age=0; Expires=Thu, 01 Jan 1970 00:00:00
X-Frame-Options: SAMEORIGIN
X-Xss-Protection: 1; mode=block
X-Content-Type-Options: nosniff
Location: /jatos/login
X-Permitted-Cross-Domain-Policies: master-only
Date: Sat, 03 May 2025 21:13:16 GMT
Content-Length: 0
Referrer-Policy: origin-when-cross-origin, strict-origin-when-cross-origin

See All Text

193.205.161.136:9000 9000 http

193.205.161.136 Data update

Italy, Campania, Salerno
Hostname: music.di.unisa.it
Organization: Università degli Stu...
ASN: AS137
Title: Page not found at /
🕒 2025-05-04 03:57

Banner Hash

HTTP/1.1 404 Not Found
Date: Sat, 03 May 2025 20:19:17 GMT
Server: WSGIServer/0.2 CPython/3.11.0
Content-Type: text/html; charset=utf-8
X-Frame-Options: DENY
Content-Length: 4851
X-Content-Type-Options: nosniff
Referrer-Policy: same-origin
Cross-Origin-Opener-Policy: same-origin
Vary: Origin

See All Text



FILTER

☐ Hide Honeypot

SEARCH TYPE

Devices	1.993 ▾
Ipv4	1.993
Ipv6	0
Websites	17

YEAR

2025	75
2024	50
2023	304
	More

ITALY

Campania	1.085
----------	-------

ZoomEye

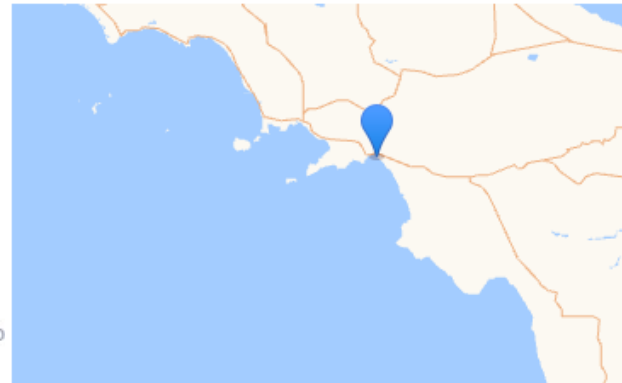
Esempio 2

➤ Informazioni di Base

Basic Information

Component details

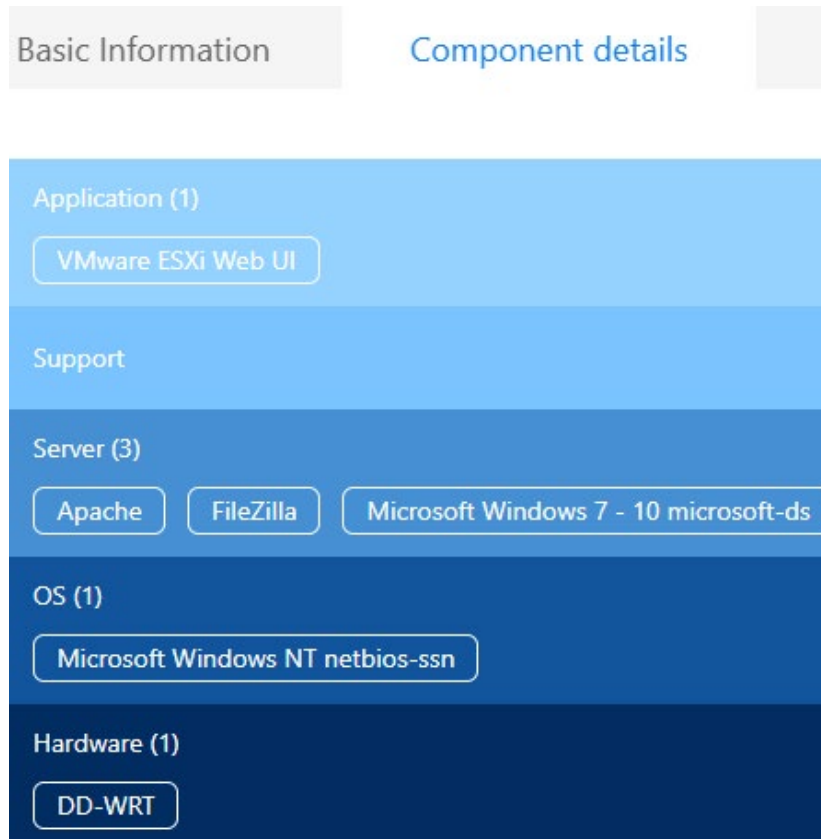
IP Address	193.205.161.7
City	Salerno
Province / State	Campania
Country	 Italy
Location	40.673361, 14.769705
Organization	Universitxe0 degli Studi di Salerno
ISP	
Hostnames	ISIS-PC
ASN	AS137



ZoomEye

Esempio 2

➤ Dettagli relativi al dispositivo



The screenshot displays the 'Component details' tab of a ZoomEye search result. It lists various components categorized by type and count:

- Basic Information** (selected tab)
- Component details**
- Application (1)**: VMware ESXi Web UI
- Support**
- Server (3)**: Apache, FileZilla, Microsoft Windows 7 - 10 microsoft-ds
- OS (1)**: Microsoft Windows NT netbios-ssn
- Hardware (1)**: DD-WRT

ZoomEye

Esempio 2

➤ Informazioni sulle porte aperte ed i relativi servizi e banner

Ports / Service 11 Whois DNS Analysis 2 User Tags 0

21/ftp 22/ssh 80/http 137/netbios-ns 443/https 445/microsoft-ds 3389/ms-wbt-server 8008/http 8080/http

9000/http 9080/http

21/ftp

Device	
Product	FileZilla ftpd
Version	0.9.44
Service	ftp
ProtType	TCP
OS	
Updated	2020-02-02 03:14

Banner

22/ssh

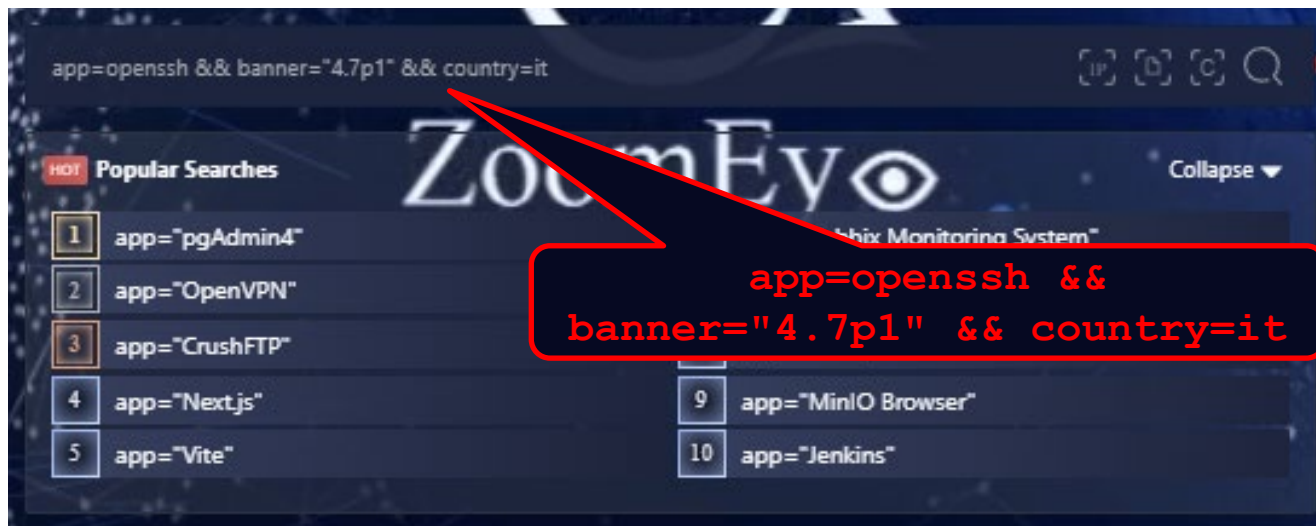
Device	
Product	dropbear
Version	0.52
Service	ssh
ProtType	TCP

Banner

ZoomEye

Esempio 3

➤ <https://www.zoomeye.ai/>



ZoomEye

Esempio 3

➤ <https://www.zoomeye.ai/>

app = "openssh" × && banner = "4.7p1" × && country = "it" × Not satisfied with the search, try [ZoomEyeGPT](#)

About 1.775 results (Nearly year: 35 results) 0.584 seconds

Result Report Maps Only \$10 Download All [Subscribe](#) [Tokenizer](#) [Collection](#)

188.14.169.244:2222 

2222 ssh

188.14.169.244  [Data update](#)

 Italy, Lombardia, Milan

OS: Linux

Hostname: host-188-14-169-244...

Organization: Telecom Italia S.p.A.

ASN: AS3269

[IDC](#)

🕒 2025-05-05 13:07

Banner Hash

SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu1.2

213.212.143.63:2223 

2223 ssh

213.212.143.63  [Data update](#)

 Italy, Lombardia, Milan

OS: Linux

Hostname: 63.143.212.213.static.a...

Organization: Aries Srl

ISP: IT.Gate S.p.A.

Banner Hash

SSH-2.0-OpenSSH_4.7p1 Debian-8ubuntu3

FILTER

☐ Hide Honeypot

SEARCH TYPE

Devices	1.775 ▾
Ipv4	1.774
Ipv6	1

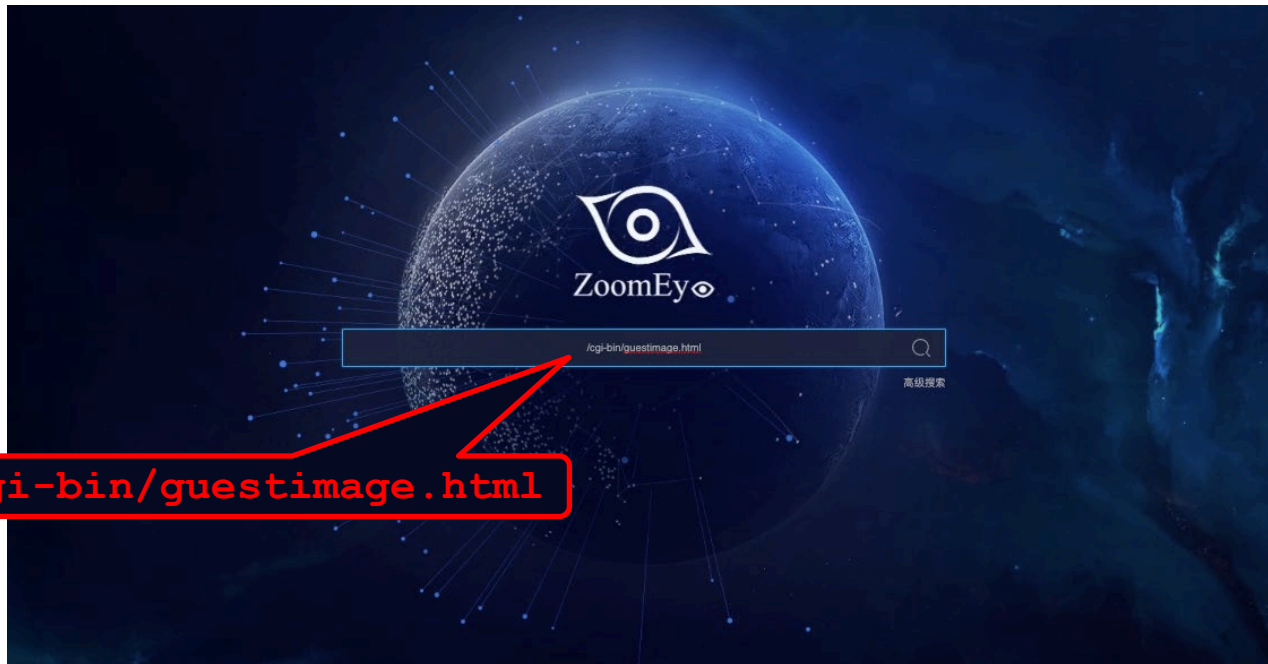
YEAR

2025	4
2024	42

ZoomEye

Esempio 4

➤ <https://www.zoomeye.ai/>



ZoomEye

Esempio 4

➤ <https://www.zoomeye.ai/>

82.226.203.114 ⓘ ble59-3-82-226-203-114.fbx.pro... 3000/http 🇫🇷 France, Lille 🕒 2019-11-02 06:29	HTTP/1.0 302 Found Location: /cgi-bin/guestimage.html Content-type: text/html; charset=utf-8 Cache-Control: no-cache <!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"> <html> <head>p <title> Redirect to guestimage: /cgi-bin/guestimage.html </title> <meta http-equiv="Content-Type" content="text/html; charset=utf-8"; </head> <body> </>	Singapore 1.701 ▲ Australia 1.696 ▲ Austria 1.451 ▲ Spain 1.407 ▲ Switzerland 1.405 ▲ Belgium 913 ▲
50.199.221.101 ⓘ mail.masslaborers.org 8888/http 🇺🇸 United States, Milford 🕒 2019-11-02 06:17	HTTP/1.0 302 Found Location: /cgi-bin/guestimage.html Content-type: text/html; charset=ISO-8859-1 Cache-Control: no-cache <!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"> <html> <head> <title> Redirect to guestimage: /cgi-bin/guestimage.html </title> <meta http-equiv="Content-Type" content="text/html; charset=ISO-885 </head> <body> </>	组件 MOBOTIX webcam... 21.380 Mobotix M10 PRIS... 10.576 thttpd 3.182 Unknown 1.487 Mobotix Camera ht... 1.038 Apache httpd 40 nginx 2 LANDesk remote ... 1 QNAP NAS storage... 1 服务 http 36.270

ZoomEye

Esempio 4 (Caso 1)

➤ <https://www.zoomeye.ai/>

82.226.203.114

ble59-3-82-226-203-114.fbx.pro...

3000/http

France, Lille

2019-11-02 06:29

HTTP/1.0 302 Found

Location: /cgi-bin/guestimage.html

Content-type: text/html; charset=utf-8

Cache-Control: no-cache

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN">

<html>

<head>

<title>

Redirect to guestimage: /cgi-bin/guestimage.html

</title>

<meta http-equiv="Content-Type" content="text/html; charset=utf-8";

</head>

<body>

</body>

50.199.221.101

mail.masslaborers.org

8888/http

United States, Milford

2019-11-02 06:17

HTTP/1.0 302 Found

Location: /cgi-bin/guestimage.html

Content-type: text/html; charset=ISO-8859-1

Cache-Control: no-cache

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN">

<html>

<head>

<title>

Redirect to guestimage: /cgi-bin/guestimage.html

</title>

<meta http-equiv="Content-Type" content="text/html; charset=ISO-8859-1";

</head>

<body>

</body>

Singapore

1.701 ▲

Australia

1.696 ▲

Austria

1.451 ▲

Spain

1.407 ▲

Switzerland

1.405 ▲

Belgium

913 ▲

组件

MOBOTIX webcam...

21.380

Mobotix M10 PRIS...

10.576

thttpd

3.182

Unknown

1.487

Mobotix Camera ht...

1.038

Apache httpd

40

nginx

2

LANDesk remote ...

1

QNAP NAS storage...

1

服务

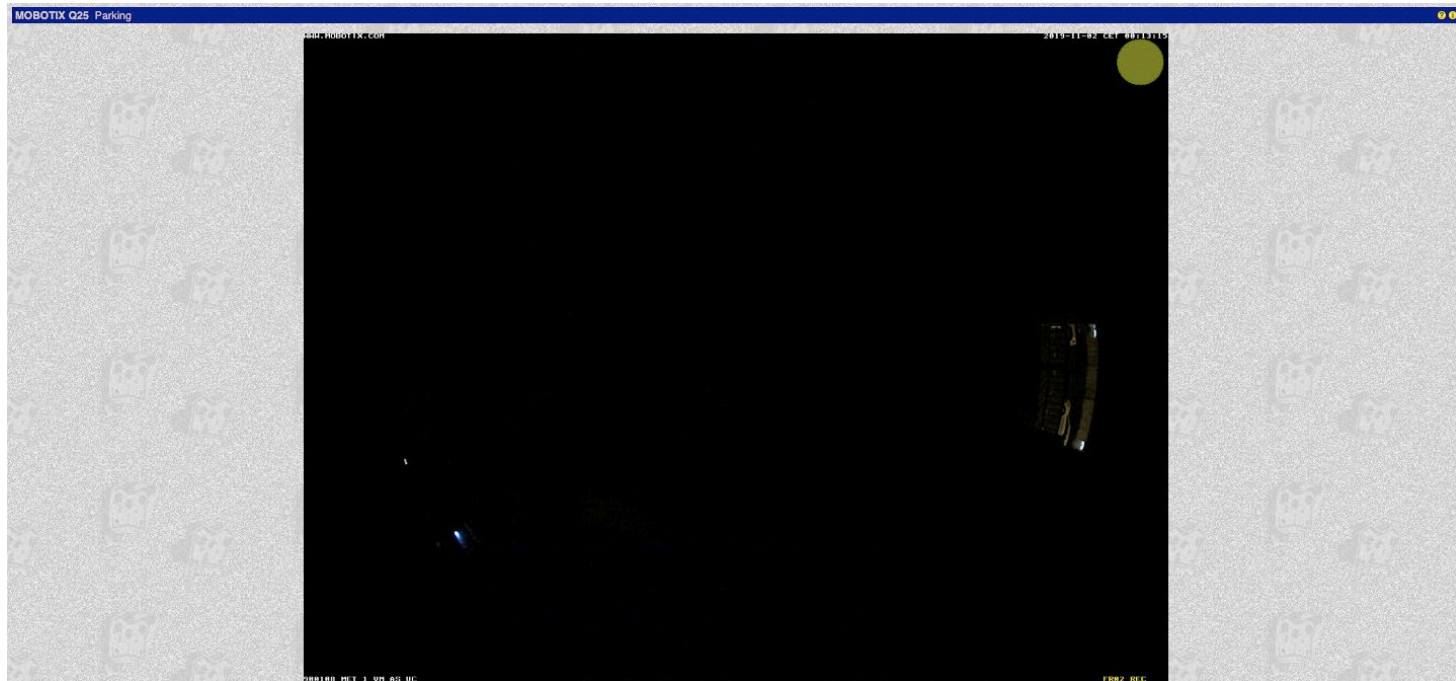
http

36.270

ZoomEye

Esempio 4 (Caso 1)

➤ <https://www.zoomeye.ai/>



Enumerating Target e Port Scanning

ZoomEye

Esempio 4 (Caso 2)

➤ <https://www.zoomeye.ai/>

82.226.203.114 ⓘ
ble59-3-82-226-203-114.fbx.pro...
3000/http
🇫🇷 France, Lille
🕒 2019-11-02 06:29

HTTP/1.0 302 Found
Location: /cgi-bin/guestimage.html
Content-type: text/html; charset=utf-8
Cache-Control: no-cache

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN">
<html>
<head>
<title>
Redirect to guestimage: /cgi-bin/guestimage.html
</title>
<meta http-equiv="Content-Type" content="text/html; charset=utf-8">
</head>
<body>
</body>
</html>

50.199.221.101 ⓘ
mail.masslaborers.org
8888/http
🇺🇸 United States, Milford
🕒 2019-11-02 06:17

HTTP/1.0 302 Found
Location: /cgi-bin/guestimage.html
Content-type: text/html; charset=ISO-8859-1
Cache-Control: no-cache

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN">
<html>
<head>
<title>
Redirect to guestimage: /cgi-bin/guestimage.html
</title>
<meta http-equiv="Content-Type" content="text/html; charset=ISO-8859-1">
</head>
<body>
</body>
</html>

Singapore 1.701 ▲
Australia 1.696 ▲
Austria 1.451 ▲
Spain 1.407 ▲
Switzerland 1.405 ▲
Belgium 913 ▲

组件

MOBOTIX webcam... 21.380
Mobotix M10 PRIS... 10.576
thttpd 3.182
Unknown 1.487
Mobotix Camera ht... 1.038
Apache httpd 40
nginx 2
LANDesk remote ... 1
QNAP NAS storage... 1

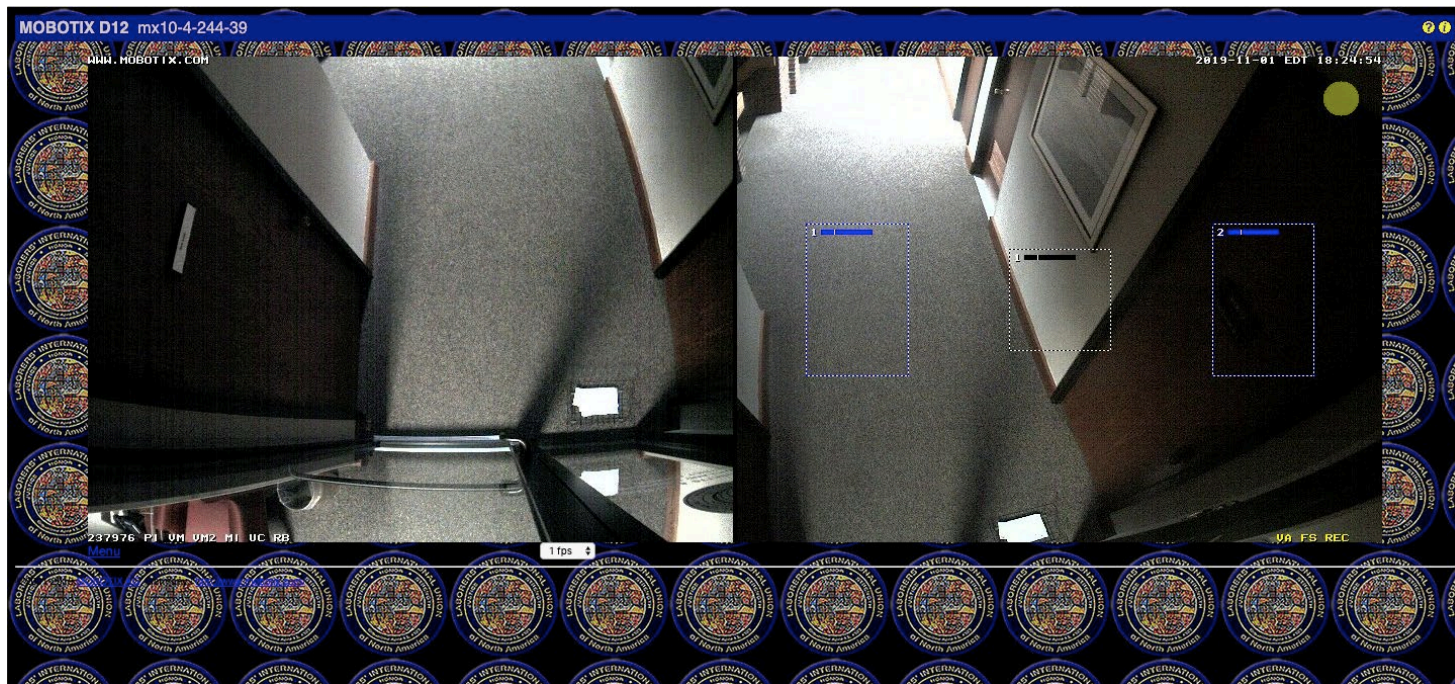
服务

http 36.270

ZoomEye

Esempio 4 (Caso 2)

➤ <https://www.zoomeye.ai/>

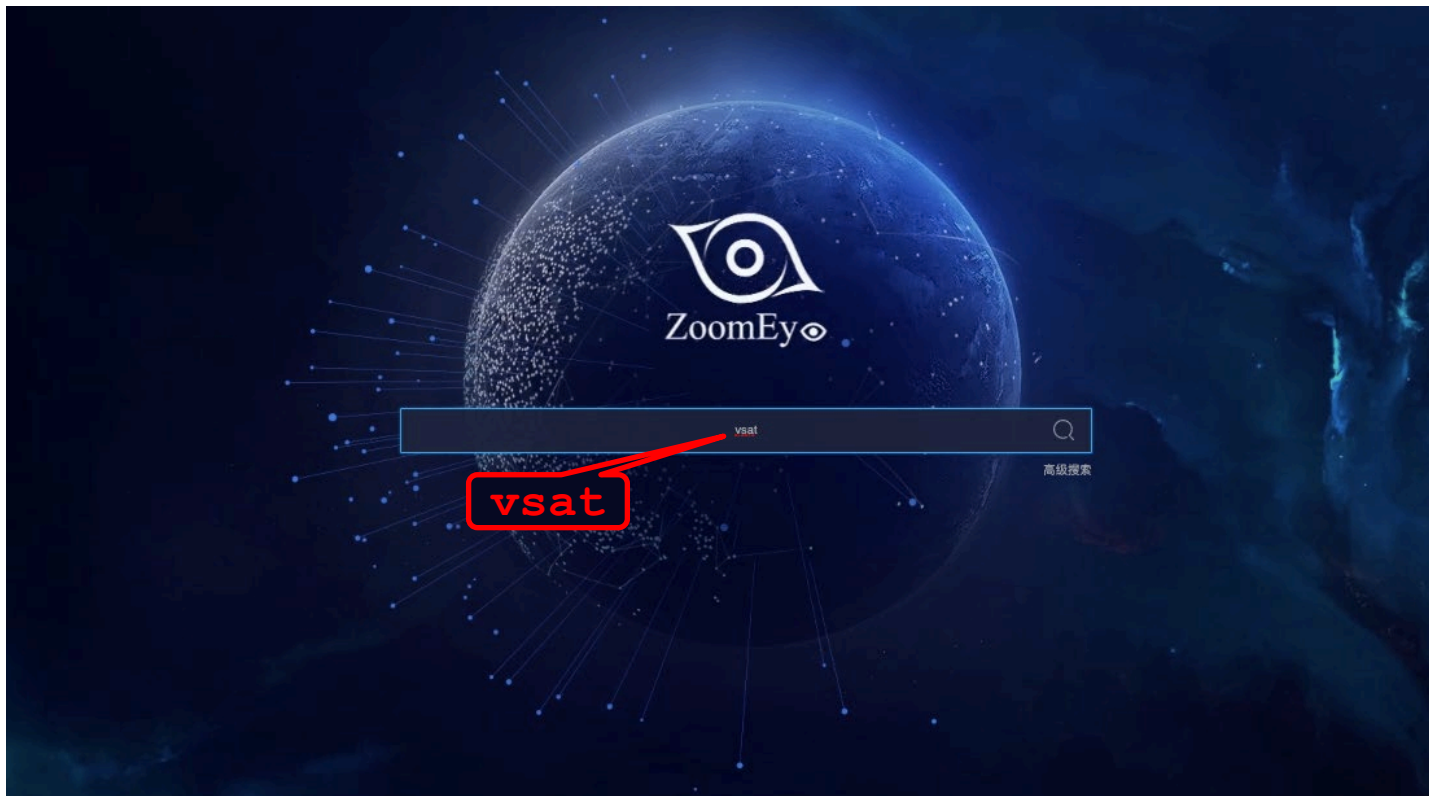


Enumerating Target e Port Scanning

ZoomEye

Esempio 5

➤ <https://www.zoomeye.ai/>



ZoomEye

Esempio 5

➤ <https://www.zoomeye.ai/>



ZoomEye

Esempio 5

➤ <https://www.zoomeye.ai/>

Result Report Maps Vulnerability

About 137.887 results (Nearly year: 57.164 results) 2.196 seconds

Value ranking

vsat X

avisho.r...

avisho.rozblog.com

Unknown, Unknown

2023-04-20 00:20

Cloudflare, Inc. cloudflare.com

ASN: AS13335

Banner SSL

Data update

HTTP/1.1 200 OK
Date: Wed, 19 Apr 2023 16:08:13 GMT
Content-Type: text/html; charset=utf-8
Transfer-Encoding: chunked
Connection: keep-alive
expires: Thu, 19 Nov 1981 08:52:00 GMT
cache-control: no-store, no-cache, must-revalidate
pragma: no-cache
content-language: fa
vary: Accept-Encoding, User-Agent
strict-transport-security: max-age=0;
x-turbo-charged-by: LiteSpeed
CF-Cache-Status: DYNAMIC
Report-To: {"endpoints":[{"url":"https://a.nel.cloudflare.com/report/"}]}

Universi...

www.gobisons.ca

Windows

Microsoft IIS Httpd:10.0

United States, Dallas

2023-04-20 00:12

Rackspace Hosting

rackspace.com

ASN: AS33070

TITLE: University of Manitoba Athle...

Banner SSL

Data update

HTTP/1.1 301 Moved Permanently
Location: https://gobisons.ca/
Server: Microsoft-IIS/10.0
Strict-Transport-Security: max-age=0;
X-Content-Type-Options: nosniff
X-XSS-Protection: 1; mode=block
Referrer-Policy: no-referrer-when-downgrade
Date: Wed, 19 Apr 2023 16:07:06 GMT
Content-Length: 0

<!doctype html>
<html id="ct100_html" lang="en" class=" index homepage">

SEARCH TYPE

Devices	113.778
IPv4	113.749
IPv6	29
Websites	24.108

YEAR

2023	37.407
2022	23.992
2021	21.718

More

COUNTRY

United States	29.697
China	19.653
Viet Nam	17.682
Indonesia	6.566
Ivory Coast	6.565

More

Enumerating Target e Port Scanning

ZoomEye

Esempio 5

➤ <https://www.zoomeye.ai/>

The screenshot displays the ZoomEye search results page. At the top, there are tabs for 'Result', 'Report', 'Maps', and 'Vulnerability'. A search bar contains 'vsat'. The results show 'About 137.887 results (Nearly year: 57.164 results) 2.196 seconds'. A 'Value ranking' button is visible. The first result is for 'avisho.r...' with details like 'avisho.rozblog.com', 'Unknown, Unknown', and '2023-04-20 00:20'. The second result is for 'Universi...' with details like 'www.gobisons.ca', 'Windows', 'Microsoft IIS Httpd:10.0', 'United States, Dallas', '2023-04-20 00:12', 'Rackspace Hosting', 'rackpace.com', 'ASN: AS33070', and 'TITLE: University of Manitoba Athle...'. The sidebar on the right includes a world map, 'SEARCH TYPE' (Devices: 113,778, IPv4: 113,749, IPv6: 29, Websites: 24,108), 'YEAR' (2023: 37,407, 2022: 23,992, 2021: 21,718), and 'COUNTRY' (United States: 29,697, China: 19,653, Viet Nam: 17,682, Indonesia: 6,566, Ivory Coast: 6,565).

Result Report Maps Vulnerability

Subscribe Collection download share tokenizer

About 137.887 results (Nearly year: 57.164 results) 2.196 seconds

Value ranking

vsat X

avisho.r...

avisho.rozblog.com

Unknown, Unknown

2023-04-20 00:20

Cloudflare, Inc. cloudflare.com

ASN: AS13335

Banner SSL

Data update

HTTP/1.1 200 OK

Date: Wed, 19 Apr 2023 16:08:13 GMT

Content-Type: text/html; charset=utf-8

Transfer-Encoding: chunked

Connection: keep-alive

expires: Thu, 19 Nov 1981 08:52:00 GMT

cache-control: no-store, no-cache, must-revalidate

pragma: no-cache

content-language: fa

vary: Accept-Encoding, User-Agent

strict-transport-security: max-age=0;

x-turbo-charged-by: LiteSpeed

CF-Cache-Status: DYNAMIC

Report-To: {"endpoints":[{"url":"https://a.nel.cloudflare.com/report/...

Universi...

www.gobisons.ca

Windows

Microsoft IIS Httpd:10.0

United States, Dallas

2023-04-20 00:12

Rackspace Hosting

rackpace.com

ASN: AS33070

TITLE: University of Manitoba Athle...

Banner SSL

Data update

HTTP/1.1 301 Moved Permanently

Location: https://gobisons.ca/

Server: Microsoft-IIS/10.0

Strict-Transport-Security: max-age=0;

X-Content-Type-Options: nosniff

X-XSS-Protection: 1; mode=block

Referrer-Policy: no-referrer-when-downgrade

Date: Wed, 19 Apr 2023 16:07:06 GMT

Content-Length: 0

<!doctype html>

<html id="ct100_html" lang="en" class=" index homepage">

SEARCH TYPE

Devices 113,778

IPv4 113,749

IPv6 29

Websites 24,108

YEAR

2023 37,407

2022 23,992

2021 21,718

More

COUNTRY

United States 29,697

China 19,653

Viet Nam 17,682

Indonesia 6,566

Ivory Coast 6,565

More

Enumerating Target e Port Scanning

ZoomEye

Esempio 5

➤ <https://www.zoomeye.ai/>

The official athletics website for the University of Manitoba Bisons

Basic Information

Component details

Website	www.gobisons.ca
IP Address	174.143.104.95
City	Dallas
Province / State	Texas
Country	 United States
Location	32.78183, -96.79586
Organization	Rackspace Hosting
ISP	rackspace.com
ASN	AS33070



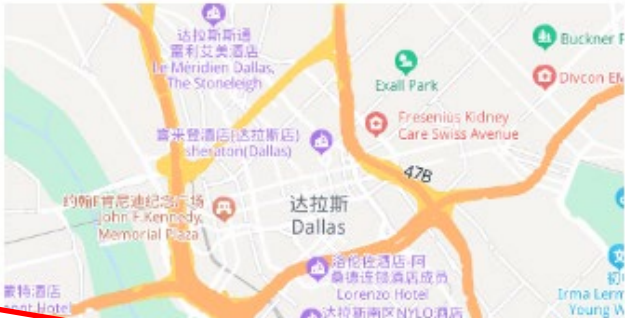
ZoomEye

Esempio 5

➤ <https://www.zoomeye.ai/>

The official athletics website for the University of Manitoba Bisons

Basic Information	Component details
Website	www.gobisons.ca
IP Address	174.143.104.95
City	Dallas
Province / State	Texas
Country	United States
Location	32.78183, -96.79586
Organization	Rackspace Hosting
ISP	rackspace.com
ASN	AS33070



➤ Cercando le seguenti coordinate tramite Google Maps possiamo geolocalizzare l'imbarcazione

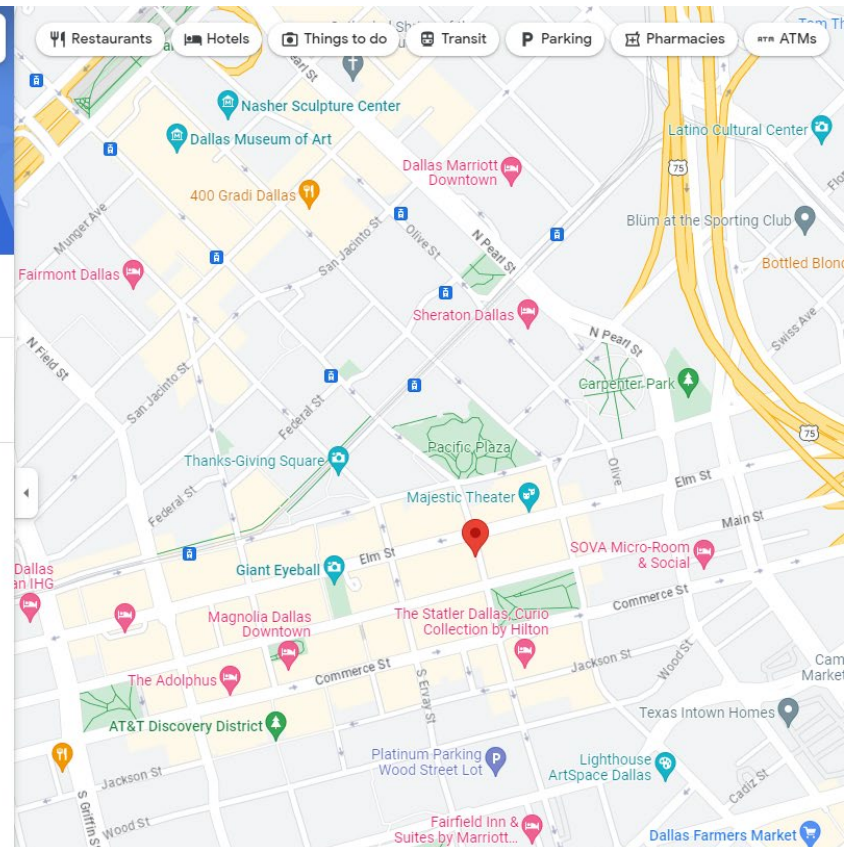
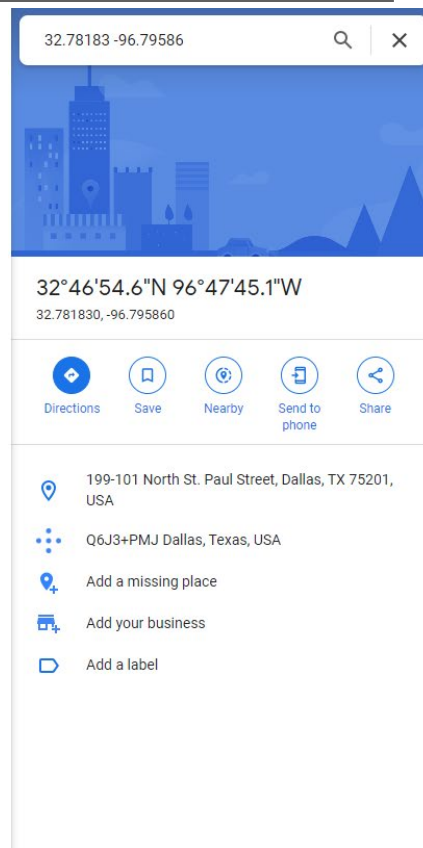
➤ 32.78183, -96.79586

ZoomEye

Esempio 5

➤ <https://www.zoomeye.ai/>

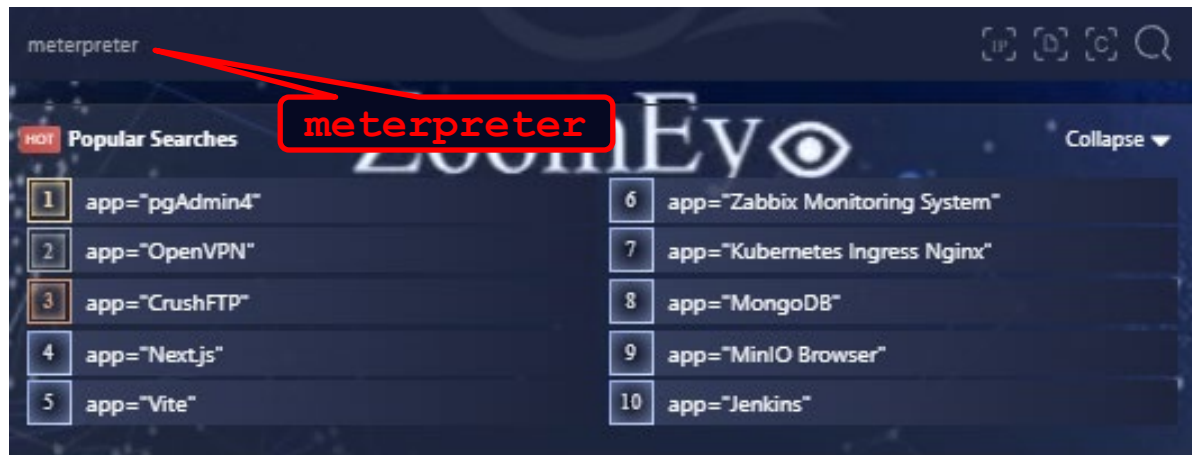
**Possiamo inserire le
coordinate
geografiche su
Google Maps**



ZoomEye

Esempio 6

➤ <https://www.zoomeye.ai/>



- **Meterpreter**
 - Payload avanzato, appartenente alla suite Metasploit
 - Consente l'accesso remoto ad una macchina target

ZoomEye

Esempio 6

➤ <https://www.zoomeye.ai/>

Result Report Maps

Only \$10 Download All Subscribe Tokenizer Collection

charge.elastic.co:80

charge.elastic... Data update

United States
Organization: Google LLC
ASN: AS15169
Title: Billing Information for Elastic...
2025-05-05 20:14

Header	Body	Hash
HTTP/1.1 308 Permanent Redirect Cache-Control: private Location: https://charge.elastic.co:443/ Content-Length: 0 Date: Mon, 05 May 2025 11:59:50 GMT Content-Type: text/html; charset=UTF-8		

prelert-info.elastic.co:80

prelert-info.el... Data update

United States
Organization: Google LLC
ASN: AS15169
Title: Machine Learning for Elastic...
2025-05-05 20:00

Header	Body	Hash
HTTP/1.1 308 Permanent Redirect Cache-Control: private Location: https://prelert-info.elastic.co:443/ Content-Length: 0 Date: Mon, 05 May 2025 11:44:12 GMT Content-Type: text/html; charset=UTF-8		

artifacts.elastic.co:80

artifacts.elasti... Data update

United States
Organization: Google LLC
ASN: AS390982
Title: Download and provision Elastic...
2025-05-05 19:51

Header	Body	Hash
HTTP/1.1 308 Permanent Redirect Cache-Control: private Location: https://artifacts.elastic.co:443/ Content-Length: 0 Date: Mon, 05 May 2025 11:36:24 GMT Content-Type: text/html; charset=UTF-8		

Map

FILTER

☐ Hide HoneyPot

SEARCH TYPE

Devices	2.860
Ipv4	2.860
Ipv6	0
Websites	71

YEAR

2025	121
2024	195
2023	183

[More](#)

COUNTRY

United States	923
China	353
Germany	260
The Netherlands	163
India	129

Enumerating Target e Port Scanning

ZoomEye

Esempio 6

➤ <https://www.zoomeye.ai/>

The screenshot displays the ZoomEye search results interface. At the top, there are tabs for 'Result', 'Report', and 'Maps'. A blue button indicates 'Only \$10' and a 'Download All' link. On the right, there are links for 'Subscribe', 'Tokenizer', and 'Collection'. The main content area shows three search results for IP addresses associated with Elastic.co:80. Each result includes a table with 'Header', 'Body', and 'Hash' columns. The first result is for 'charge.elastic.co:80', the second for 'prelert-info.elastic.co:80', and the third for 'artifacts.elastic.co:80'. All three results show a 'Permanent Redirect' status. To the right of the results, there is a world map and a 'FILTER' section with options for 'Hide Honeypot', 'SEARCH TYPE' (Devices, Ipv4, Ipv6, Websites), 'YEAR' (2025, 2024, 2023), and 'COUNTRY' (United States, China, India).

Result	Report	Maps
charge.elastic.co:80		
charge.elastic.co:80	Data update	Header Body Hash
United States	HTTP/1.1 308 Permanent Redirect	
Organization: Google LLC	Cache-Control: private	
ASN: AS15169	Location: https://charge.elastic.co:443/	
Title: Billing Information for Elastic...	Content-Length: 0	
2025-05-05 20:14	Date: Mon, 05 May 2025 11:59:50 GMT	
	Content-Type: text/html; charset=UTF-8	
prelert-info.elastic.co:80		
prelert-info.elastic.co:80	Data update	Header Body Hash
United States	HTTP/1.1 308 Permanent Redirect	
Organization: Google LLC	Cache-Control: private	
ASN: AS15169	Location: https://prelert-info.elastic.co:443/	
Title: Machine Learning for Elastic...	Content-Length: 0	
2025-05-05 20:00	Date: Mon, 05 May 2025 11:44:12 GMT	
	Content-Type: text/html; charset=UTF-8	
artifacts.elastic.co:80		
artifacts.elastic.co:80	Data update	Header Body Hash
United States	HTTP/1.1 308 Permanent Redirect	
Organization: Google LLC	Cache-Control: private	
ASN: AS15169	Location: https://artifacts.elastic.co:443/	
Title: Machine Learning for Elastic...	Content-Length: 0	
2025-05-05 20:00	Date: Mon, 05 May 2025 11:44:12 GMT	
	Content-Type: text/html; charset=UTF-8	

FILTER

☐ Hide Honeypot

SEARCH TYPE

Devices	2,860
Ipv4	2,860
Ipv6	0
Websites	71

YEAR

2025	121
2024	195
2023	183

[More](#)

COUNTRY

United States	923
China	353
India	129

La suite Meterpreter verrà mostrata in dettaglio durante le fasi di **Exploitation** e **Post Exploitation**

Enumerating Target e Port Scanning

Outline

- Concetti Introduttivi
- Suite Protocollore TCP/IP
- Formato dei Messaggi TCP e UDP
- Active Enumeration
 - Network Scanner Nmap
 - Zenmap
 - Unicornscan
 - Masscan
- Passive Enumeration
 - Shodan
 - ZoomEye
 - **FOFA**
 - Censys

FOFA

- Motore di ricerca cinese per dispositivi e servizi esposti su Internet, simile a Shodan o ZoomEye
- Effettua ricerche passive su IP, domini, certificati SSL, header HTTP ed altri metadati di rete
- Usato in ambito:
 - OSINT
 - Security assessment
 - Threat hunting
 - Bug bounty e pentest

FOFA

Pricing

➤ <https://en.fofa.info/vip>

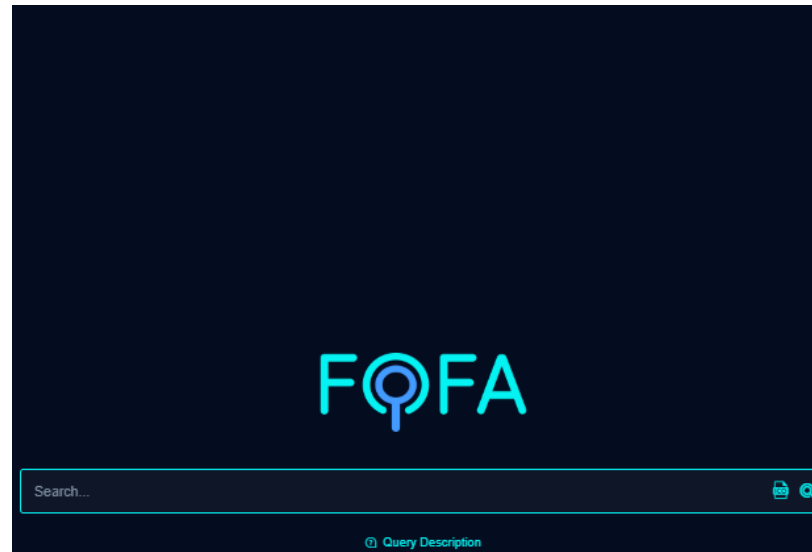
Registered User (Non-commercial only)	Personal	Professional	Business	Corporate
	\$588	\$4,788	\$47,988	\$64,800 /year
Free to Use	Subscribe	Subscribe	Subscribe	Contact us
Purchase Activation Code	Purchase Activation Code	Purchase Activation Code	Purchase Activation Code	
✓ Query Credits 300/month ✓ 3k Results/month ✓ Access 58 Syntaxes ② ✓ Web Download Feature ✓ 1 User	✓ Query Credits 10,000/month ✓ Up to 100k Results/month ✓ Access 68 Syntaxes ② ✓ Web Download Feature ✓ 1 User	✓ Query Credits 80,000/month ✓ Up to 800k Results/month ✓ Access 78 Syntaxes ② ✓ Web Download Feature ✓ 1 User	✓ Query Credits 900,000/month ✓ Up to 9million Results/month ✓ Access All Syntaxes ② ✓ Web Download Feature ✓ 5 Users ②	✓ Query Credits Custom ✓ Custom Results/month ✓ Access All Syntaxes ② ✓ Web Download Feature ✓ Custom Users ②
✓ FOFA Fingerprint Rule ✓ Website Feature ID (FID) ✓ Cloud Assets Tag	✓ Access 36 fields in Query API ✓ 1 Request/second in Query API ✓ HOST Aggregation API	✓ Access 41 fields in Query API ✓ 1 Request/second in Query API ✓ HOST Aggregation API ✓ Statistic Aggregation API	✓ Access 47 fields in Query API ✓ 2 Request/second in Query API ✓ HOST Aggregation API ✓ Statistic Aggregation API	✓ Access all fields in Query API ✓ 5 Request/second in Query API ✓ HOST Aggregation API ✓ Statistic Aggregation API
	✓ FOFA Fingerprint Rule ✓ Website Feature ID (FID) ✓ Cloud Assets Tag ✓ Fuzzy Search ②	✓ FOFA Fingerprint Rule ✓ Website Feature ID (FID) ✓ Cloud Assets Tag ✓ Fuzzy Search ② ✓ Product Tag ✓ Category Tag	✓ FOFA Fingerprint Rule ✓ Website Feature ID (FID) ✓ Cloud Assets Tag ✓ Fuzzy Search ② ✓ Product Tag ✓ Category Tag ✓ Third Party SDK Tag	✓ FOFA Fingerprint Rule ✓ Website Feature ID (FID) ✓ Cloud Assets Tag ✓ Fuzzy Search ② ✓ Product Tag ✓ Category Tag ✓ Third Party SDK Tag ✓ Asset Refresh ✓ Statistical Visualization

Enumerating Target e Port Scanning

FOFA

Interfaccia

➤ <https://en.fofa.info/>



FOFA

Query Syntax

- Offre funzionalità molto simili a quelle fornite da Shodan e ZoomEye, accessibili mediante opportuni «filtri»



FOFA

Query Description

➤ <https://en.fofa.info/>

Output Parziale

Query Description

If the query keyword has no syntax or filter, it will default to search from HTML, HTTP header and URL;
If the query syntax has multiple AND & OR relationships, try including it with ();
Add "==" to complete the match, such as finding all hosts of qq.com. It can be domain=="qq.com".

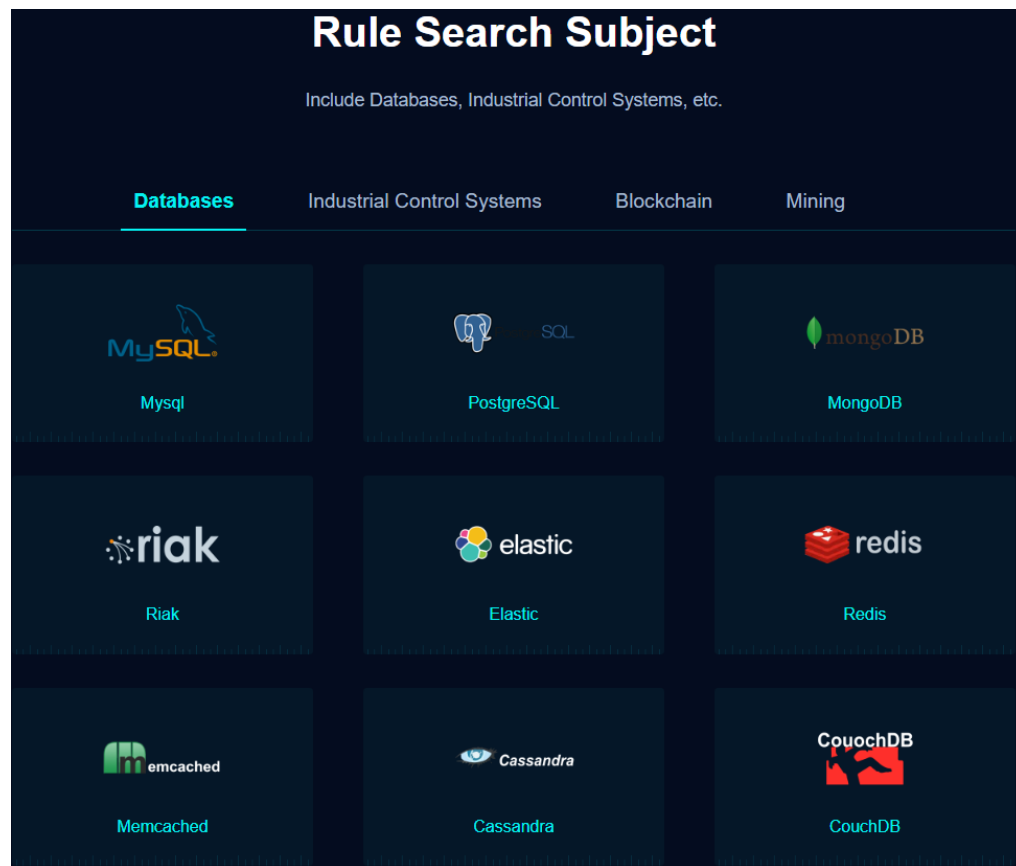
+ Advanced Search

Filter Example (Click to search)	Description	Tips
title="bing"	Query the "bing" from the title.	-
header="elastic"	Query "elastic" from the header.	-
body="google"	Query the "google" assets from HTML.	-
fid="sSXXGNUO2FefBTcCLIT/2Q=="	Search the same fingerprint results by fid.	Type is subdomain.
domain="bing.com"	Query the "bing.com" by the domain name.	-
icp="京ICP证030173号"	Query the website record number.	Type is subdomain.
js_name="js/jquery.js"	Query " js/jquery.js" from the website body.	Type is subdomain.
js_md5="82ac3f14327a8b7ba49baa208d4eaa15"	Query the related results by js_md5.	-
cname="ap21.inst.siteforce.com"	Query "ap21.inst.siteforce.com" assets from Cname	-

FOFA

Featured Categories

➤ <https://en.fofa.info/subject>



Enumerating Target e Port Scanning

FOFA

API

- FOFA offre delle proprie API
- <https://en.fofa.info/api>

FOFA

FOFA API Reference

API Introduction

Limits

Request Structure

SDK

Query Interface

▼ Aggregation & Statistics

Statistic Aggregation

HOST Aggregation

▼ Basic Interface

Account Information

Query Interface

It provides a way to search for hosts and get detailed information to make development easier.

GET <https://fofa.info/api/v1/search/all>

No.	Field	Necessary	Type	Description	Example
1	qbase64	Yes	string	The query syntax after base 64 encoding is your query content.	aXA0iJEwMy4zNS4xNjguMzgi
2	fields	No	string	FOFA API can select the fields; the default is host, IP, and port. See Appendix 1 for more details.	host,ip,port
3	page	No	int	Turn the page; the default is page 1, sorted by update time.	1
4	size	No	int	The number of results per page; the default size is 100, and the maximum size per page is 10,000.	100
5	full	No	boolean	Results within one year are displayed. Specify true to search all results.	false

Enumerating Target e Port Scanning

FOFA

Registrazione

- Prima di utilizzare FOFA è opportuno registrarsi su
- https://i.nosec.org/register?locale=en&service=https://en.fofa.info/f_login

Register

Email address, need activate account. 

Password 

Confirm your password 

Username 

Click the image to change. 

☒ I agree to the [Service&User Agreement](#)

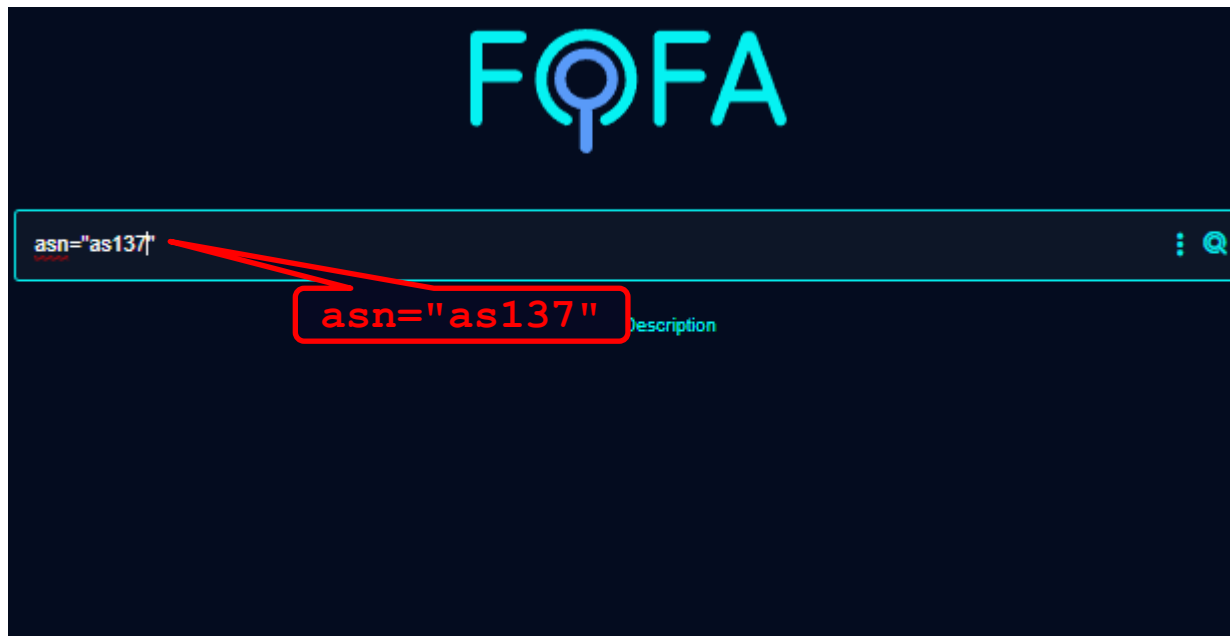
Register

[Already have an account?](#) [Didn't receive the email?](#)

FOFA

Esempio 1

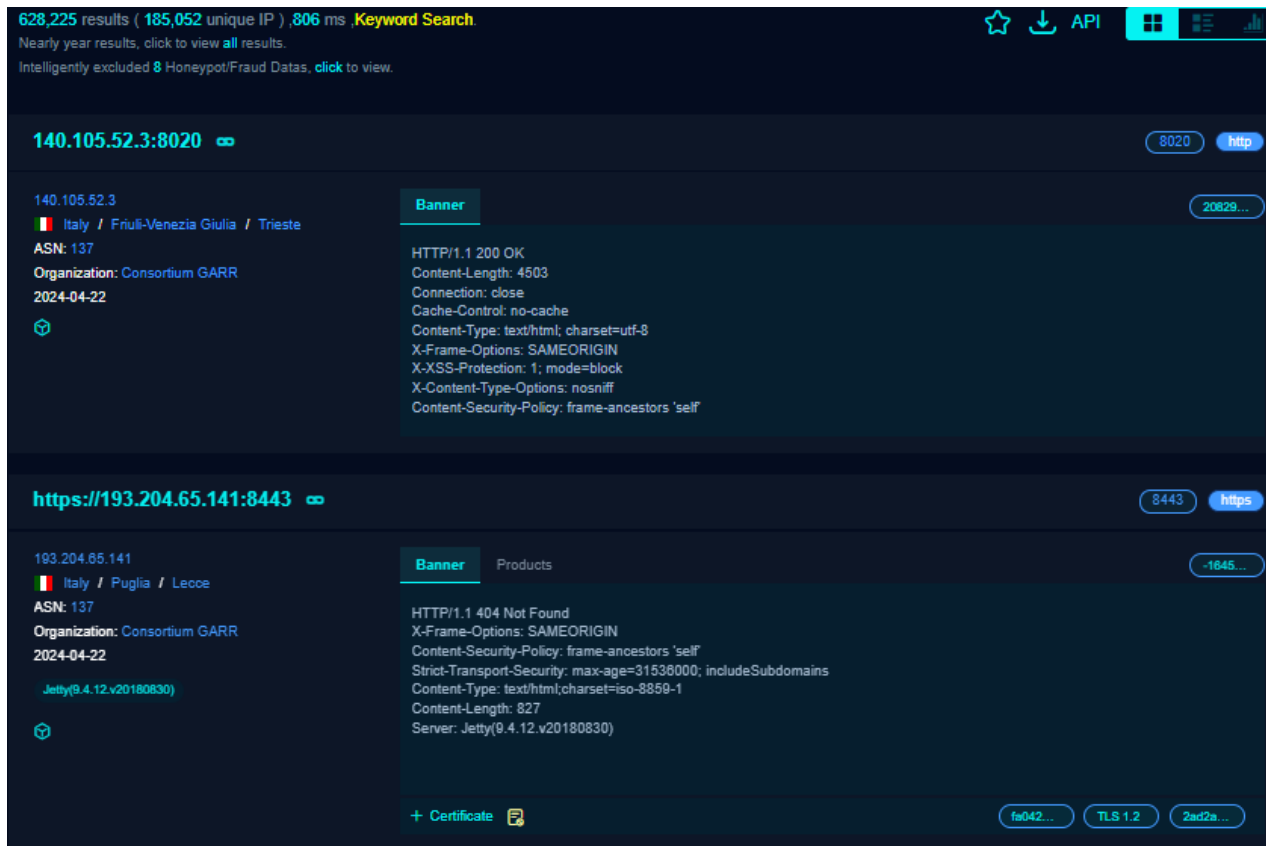
- Effettuiamo una ricerca per Autonomous System Network (**asn=as137**)



FOFA

Esempio 1

- Effettuiamo una ricerca per Autonomous System Network (**asn=as137**)



The screenshot displays the FOFA search interface with the query **asn=as137**. The top bar shows 628,225 results (185,052 unique IP), a response time of 806 ms, and a 'Keyword Search' filter. Below the search bar, two results are shown:

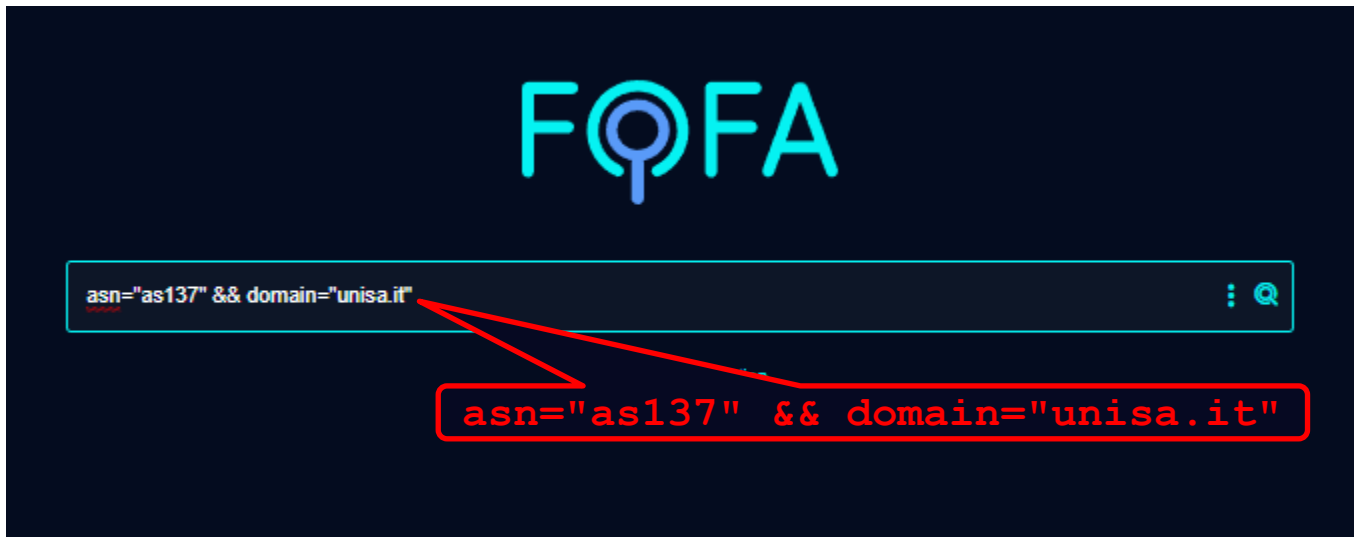
- 140.105.52.3:8020** (http):
 - Location: Italy / Friuli-Venezia Giulia / Trieste
 - ASN: 137
 - Organization: Consortium GARR
 - Date: 2024-04-22
 - Banner: HTTP/1.1 200 OK, Content-Length: 4503, Connection: close, Cache-Control: no-cache, Content-Type: text/html; charset=utf-8, X-Frame-Options: SAMEORIGIN, X-XSS-Protection: 1; mode=block, X-Content-Type-Options: nosniff, Content-Security-Policy: frame-ancestors 'self'
- https://193.204.65.141:8443** (https):
 - Location: Italy / Puglia / Lecce
 - ASN: 137
 - Organization: Consortium GARR
 - Date: 2024-04-22
 - Server: Jetty(9.4.12.v20180830)
 - Banner: HTTP/1.1 404 Not Found, X-Frame-Options: SAMEORIGIN, Content-Security-Policy: frame-ancestors 'self', Strict-Transport-Security: max-age=31536000; includeSubdomains, Content-Type: text/html; charset=iso-8859-1, Content-Length: 827

Enumerating Target e Port Scanning

FOFA

Esempio 2

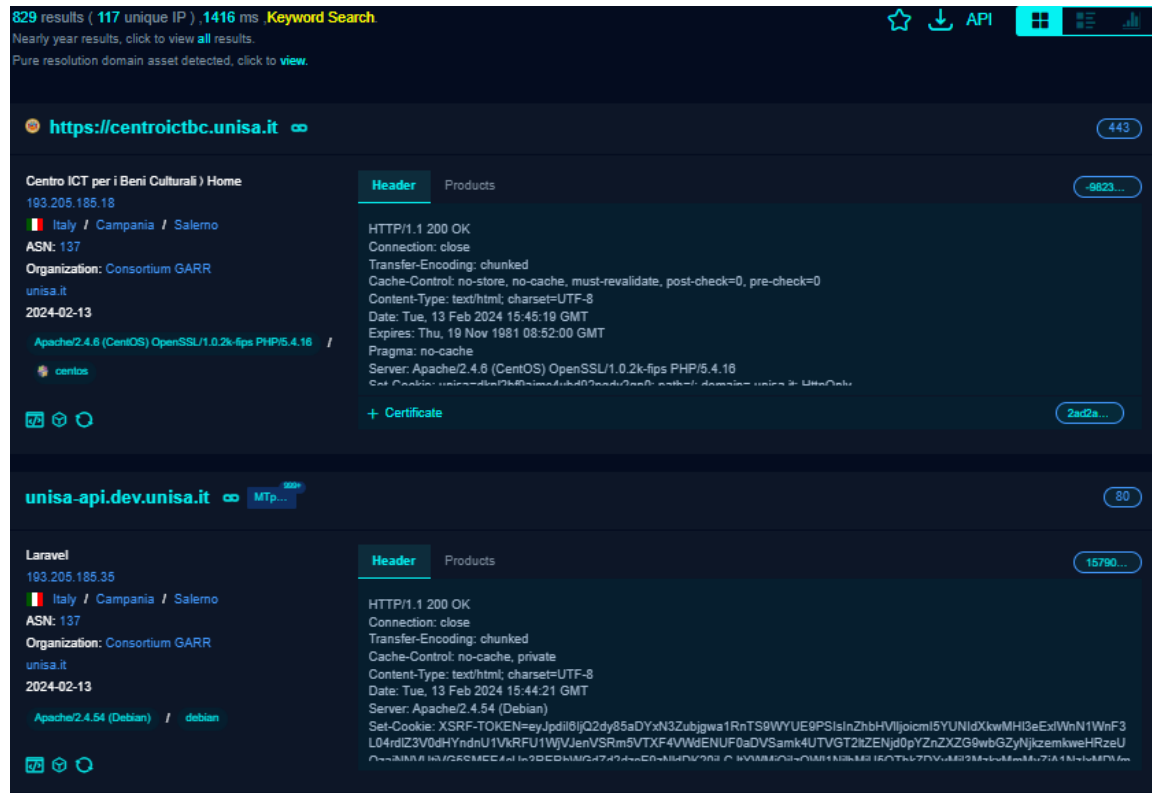
- Effettuiamo una ricerca per Autonomous System Network (**asn=as137**) e dominio **unisa.it**



FOFA

Esempio 2

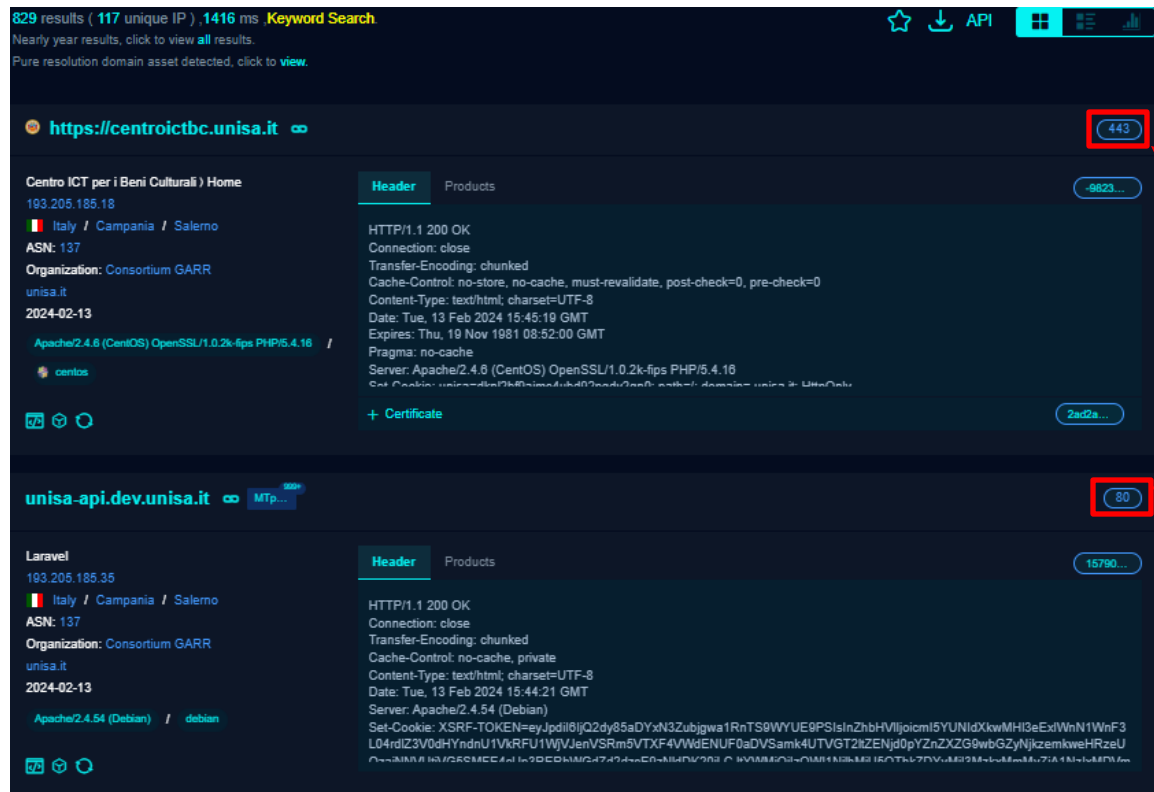
- Effettuiamo una ricerca per Autonomous System Network (**asn=as137**) e dominio **unisa.it**



FOFA

Esempio 2

- Effettuiamo una ricerca per Autonomous System Network (**asn=as137**) e dominio **unisa.it**



Porte Aperte

FOFA

Esempio 2

- Effettuiamo una ricerca per Autonomous System Network (**asn=as137**) e dominio **unisa.it**



TOP SERVERS

Apache	313
nginx/1.18.0 (Ubuntu)	72
Apache/2.4.7 (Ubuntu)	45
Apache/2.4.41 (Ubuntu)	30

TOP PROTOCOLS

http	58
https	32
ssh	7
dns	5
netbios-ssn	4

Outline

- Concetti Introduttivi
- Suite Protocollore TCP/IP
- Formato dei Messaggi TCP e UDP
- Active Enumeration
 - Network Scanner Nmap
 - Zenmap
 - Unicornscan
 - Masscan
- Passive Enumeration
 - Shodan
 - ZoomEye
 - FOFA
 - **Censys**

Censys

Caratteristiche

- Piattaforma open-source e commerciale per l'analisi della superficie d'attacco online
- Ricerca dispositivi, domini e certificati digitali esposti
- Individua anche vulnerabilità note, configurazioni errate, ed asset non registrati
- Scannerizza continuamente IPv4, IPv6, HTTP, HTTPS, SMTP, SSH, etc.
- Integra feed CVE, Shodan-like query, API, dashboard di analisi
- Supporta filtri avanzati, JSON search e tagging automatico

Censys

Progetto GitHub

➤ <https://github.com/censys>



Censys

Censys is a platform that helps information security practitioners discover, monitor, and analyze devices that are accessible from the Internet.

98 followers Ann Arbor <https://www.censys.com> @censysio support@censys.com

Overview Repositories 29 Packages People 7

Pinned

**censys-cloud-connector** Public
The Censys Unified Cloud Connector is a standalone connector that gathers assets from various cloud providers and stores them in Censys ASM. This Connector offers users the ability to supercharge o...
Python 8 6

**censys-python** Public
An easy-to-use and lightweight API wrapper for Censys APIs.
Python 435 97

Repositories

Find a repository... Type Language Sort

censys-sdk-python Public

People



Top lan

Go

TypeS

Most u

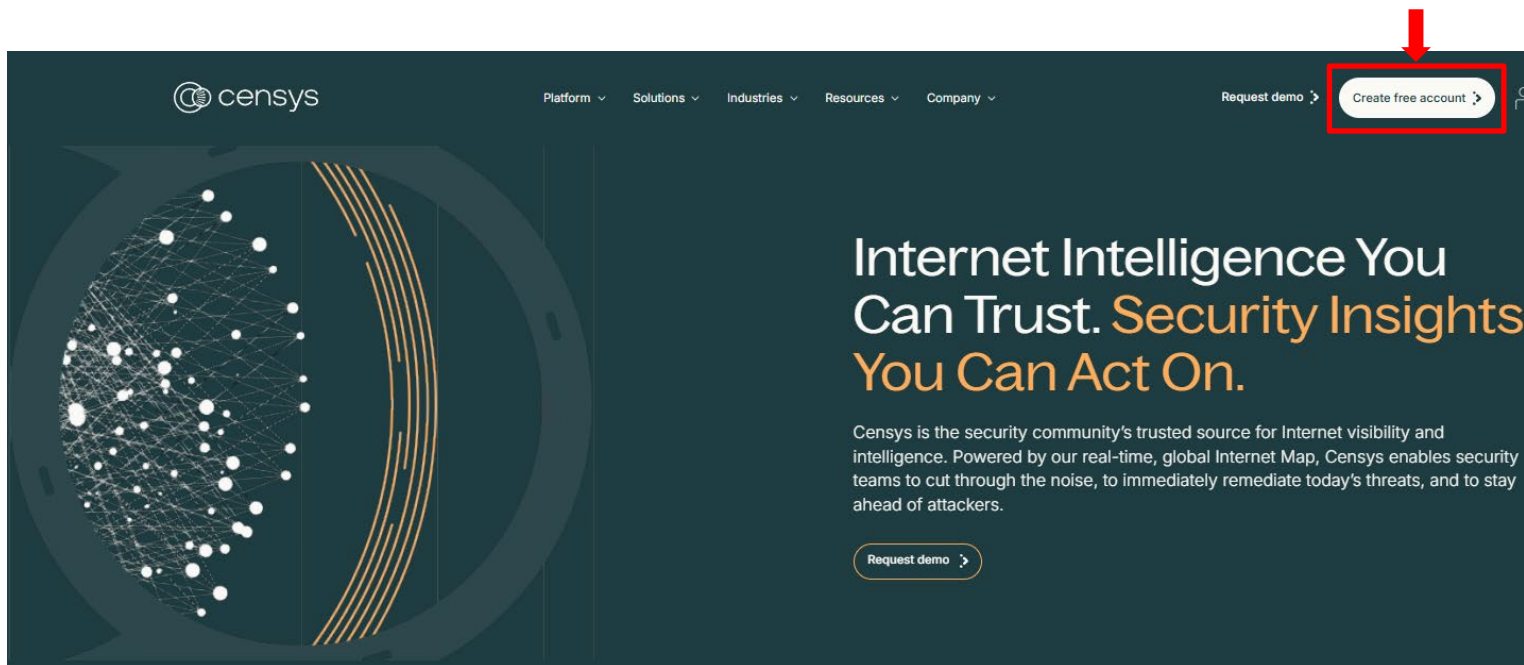
censys
python

Enumerating Target e Port Scanning

Censys

Interfaccia Web

➤ <https://censys.com/>

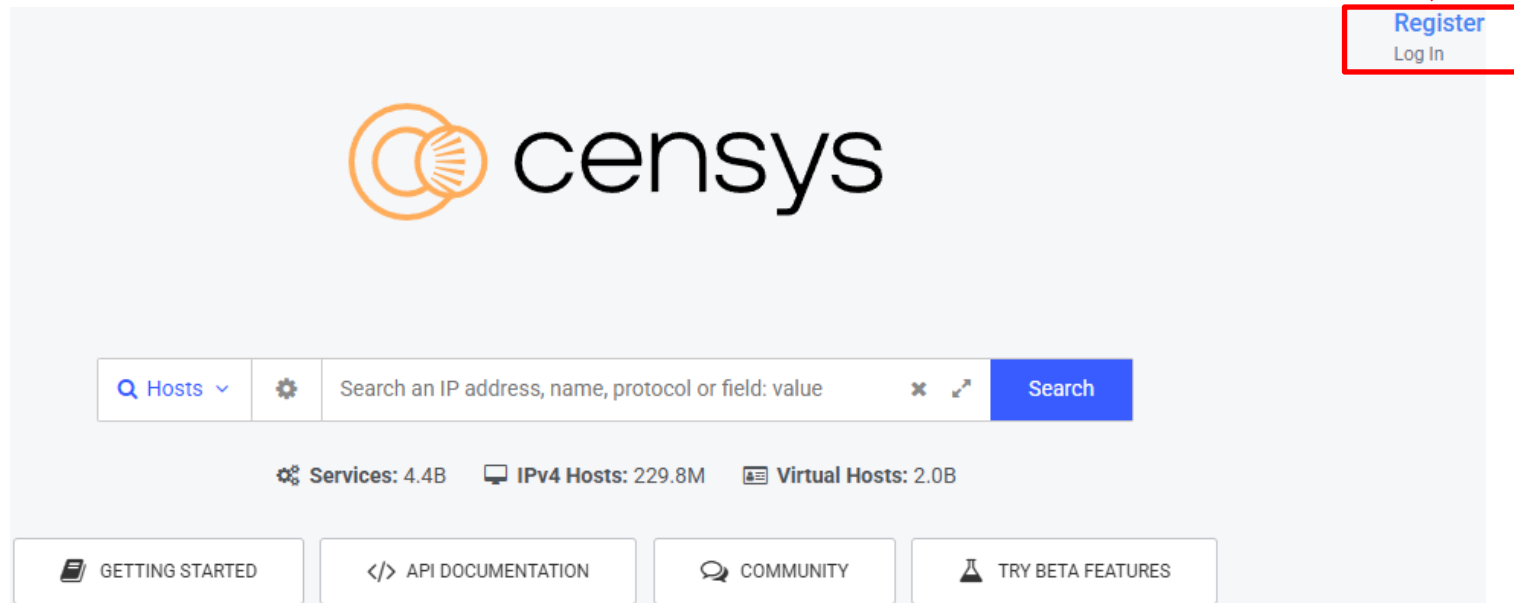


Per poter utilizzare Censys è necessaria la registrazione

Censys

Vecchia Interfaccia Web (ancora funzionante)

➤ <https://search.censys.io/>

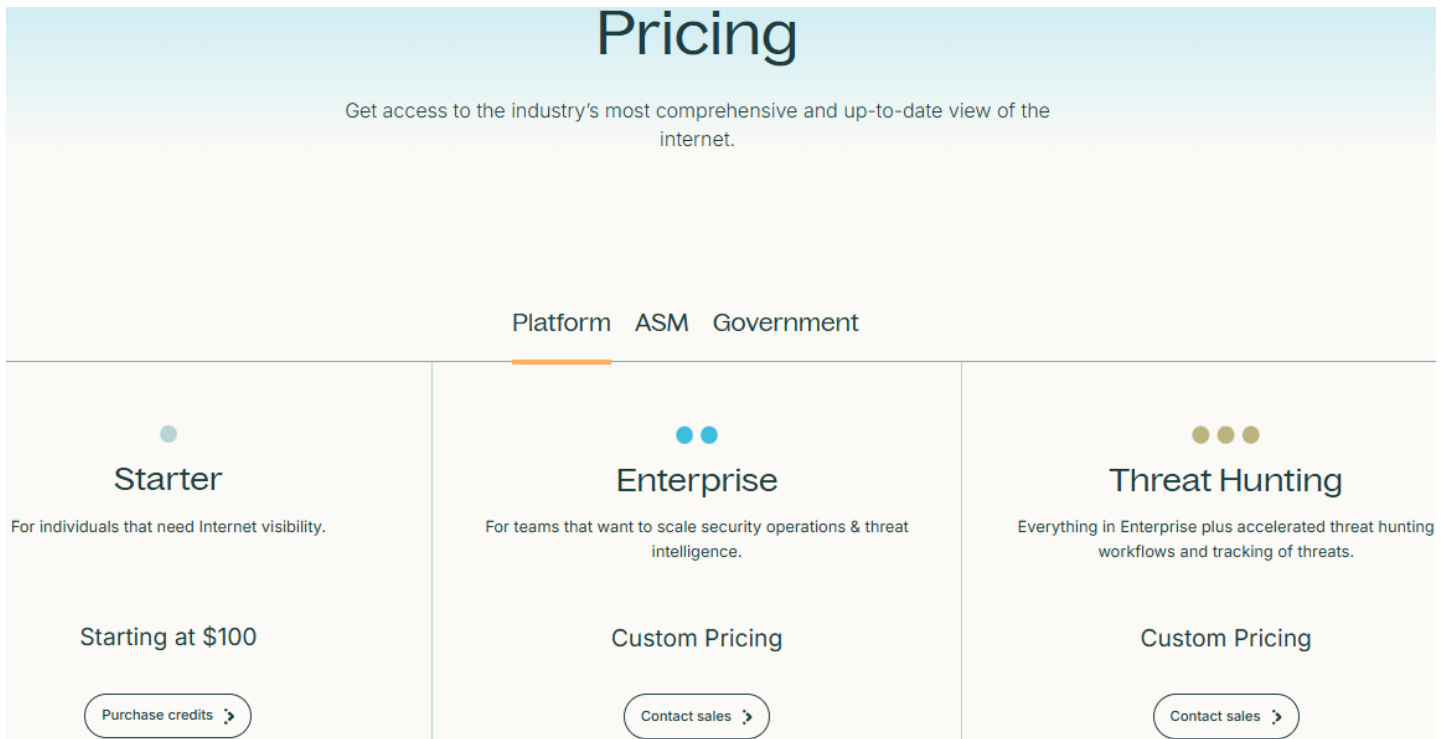


Per poter utilizzare Censys è necessaria la registrazione

Censys

Pricing

➤ <https://censys.com/resources/pricing>



The screenshot shows the Censys Pricing page. At the top, the word "Pricing" is centered in a large font. Below it, a subtitle reads: "Get access to the industry's most comprehensive and up-to-date view of the internet." Underneath the subtitle, there are three tabs: "Platform", "ASM", and "Government". The "Platform" tab is selected, indicated by an orange underline. Below the tabs, there are three pricing cards. The first card is for "Starter", described as "For individuals that need Internet visibility", with a starting price of "\$100" and a "Purchase credits" button. The second card is for "Enterprise", described as "For teams that want to scale security operations & threat intelligence", with "Custom Pricing" and a "Contact sales" button. The third card is for "Threat Hunting", described as "Everything in Enterprise plus accelerated threat hunting workflows and tracking of threats", with "Custom Pricing" and a "Contact sales" button. Each card has a set of three colored dots (blue, green, and yellow) above its title.

Pricing

Get access to the industry's most comprehensive and up-to-date view of the internet.

Platform ASM Government

Starter
For individuals that need Internet visibility.
Starting at \$100
Purchase credits ➤

Enterprise
For teams that want to scale security operations & threat intelligence.
Custom Pricing
Contact sales ➤

Threat Hunting
Everything in Enterprise plus accelerated threat hunting workflows and tracking of threats.
Custom Pricing
Contact sales ➤

Censys

Interfaccia di Ricerca

➤ <https://censys.io/>

The screenshot displays the Censys search interface. At the top, there is a search bar with the placeholder text "Search the Censys data for...". Below the search bar, a "Welcome," message is visible. The interface is divided into several sections:

- Getting Started:** Includes links for "Example Queries", "Data Definitions", and "Query Converter".
- QUERY:** A section with a "See More" link and four search filters:
 - GEOLOCATION:** Search for all hosts in Ann Arbor Michigan
 - LOGIN PAGES:** Find hosts with login pages
 - HTTP BODIES:** Search for web properties with default nginx bodies
 - SSH:** Show hosts with SSH services running on non-standard ports
- DISCOVER:** A section with three data statistics:
 - Hosts:** 416M
 - Certificates:** 16B
 - Web Properties:** 2.8B
- INVESTIGATE:** A section with three data statistics:
 - Protocols:** 193
 - ASNs in Chile:** 391
 - RDP Ports:** 2K +

Enumerating Target e Port Scanning

Censys

Interfaccia di Ricerca – Esempio

➤ <https://censys.io/>

The screenshot shows the Censys search interface. A red callout box highlights the query `"*.test.unisa.it"` entered in the search bar. The interface includes a sidebar with navigation links (Home, Getting Started, Example Queries, Data Definition), a search bar, and a main content area with various search filters and results. The results are categorized into DISCOVER and INVESTIGATE sections, showing metrics like Hosts (416M), Certificates (16B), Web Properties (2.8B), Protocols (193), ASNs in Chile (391), and RDP Ports (2K+).

Home

Search the Censys data for...

Welcome, Josh!

Getting Started Example Queries Data Definition

QUERY See More

GEOLOCATION Search for all hosts in Ann Arbor Michigan

LOGIN PAGES Find hosts with login pages

HTTP BODIES Search for web properties with default nginx bodies

SSH Show hosts with SSH services running on non-standard ports

DISCOVER

416M Hosts

16B Certificates

2.8B Web Properties

INVESTIGATE

193 Protocols

391 ASNs in Chile

2K+ RDP Ports

Enumerating Target e Port Scanning

Censys

Interfaccia di Ricerca – Esempio

Search

Search Results | Report Builder

▼ ASSET TYPES

- Hosts 6
- Certificates 15
- Web Properties 21

▼ SOFTWARE VENDORS

- apache 40
- php 30
- f5 10
- openwebui 4
- apple 2
- [More](#)

▼ SOFTWARE PRODUCTS

- httpd 40
- php 30
- nginx 10
- openwebui 4
- python 2
- [More](#)

RESULTS: 42 • DURATION: 1.61s

ugov-pth2.test.unisa.it: 443 • WEB PROPERTY

HTML Title	Open WebUI	1 Endpoint
Browser Trust	Trusted ✓	443 / HTTP
Software	Ngix Openwebui	

🔍 MATCHED FIELDS

ugov-pth2.test.unisa.it: 80 • WEB PROPERTY

HTML Title	301 Moved Permanently	1 Endpoint
Browser Trust	No Data	80 / HTTP
Software	Ngix	

🔍 MATCHED FIELDS

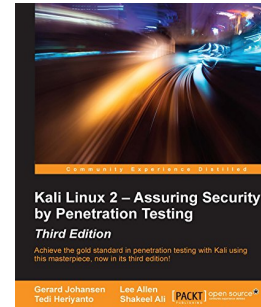
unisa.esse3.pp.cineca.it: 443 • WEB PROPERTY

HTML Title	Homepage area pubblica, Università di Salerno	2 Endpoints
Browser Trust	Trusted ✓	443 / HTTP 443 / HTTP /Root.do
Software	Httpd	

Bibliografia

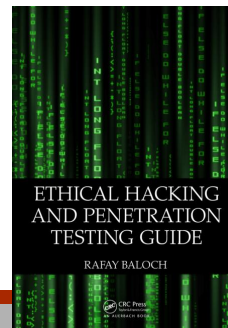
- **Kali Linux 2 - Assuring Security by Penetration Testing. Third Edition.** Gerard Johansen, Lee Allen, Tedi Heriyanto, Shakeel Ali. Packt Publishing. 2016

- Capitolo 6



- **Ethical Hacking and Penetration Testing Guide.** Rafay Baloch. CRC Press. 2014

- Capitolo 4



Bibliografia

➤ **Guida di riferimento di Nmap (pagina del manuale)**

➤ <https://nmap.org/man/it/>

➤ **Guida Rapida per Nmap**

➤ <https://www.stationx.net/nmap-cheat-sheet>

Bibliografia

➤ ZoomEye

➤ <https://www.zoomeye.org/doc>

➤ Shodan

➤ <https://danielmiessler.com/study/shodan/>

➤ <https://help.shodan.io/the-basics/search-query-fundamentals>

➤ <https://github.com/JavierOlmedo/shodan-filters>